

# Synthetic Underwriting Manual

Medical impairments, key questions and probable ratings for life insurance

A synthetic document, generated for a software demonstration.

No insurer or reinsurer wrote, uses or endorses this manual, and none of its wording comes from any insurer's or reinsurer's manual.

The debit percentages and the decisions to decline are invented for this synthetic manual. The thresholds follow the public source named in each rule, except the edges a rule marks as this manual's own.

It must not be used to assess any real person.

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# 1 Introduction

## 1.1 What this manual is

This is a synthetic underwriting manual. It was written for a software demonstration and for nothing else. No insurer or reinsurer uses it, none wrote it, and no part of its text was taken from any insurer's or reinsurer's manual.

It is laid out the way field guides to medical impairments usually are: for each impairment a short description, the questions an underwriter asks, and a table of probable ratings. That layout is common knowledge; every sentence here is original.

The clinical thresholds are real in one sense only: most band edges follow a published guideline from a public body, named in the rule. A few edges are this manual's own, drawn where a guideline's stage was split in two or a waiting period was needed; every rule with such an edge says which edge it is. The ratings attached to the bands are not real in any sense. Every debit percentage and every decision to decline was invented for this document, and none of them should be used to assess a real person.

## 1.2 How a section is laid out

Sections 2 onwards each cover one impairment, and each has the same five numbered parts. Part 1, the impairment, says what the condition is, why it matters to mortality and when the section applies at all. Part 2, key questions, lists what to find out and why each answer matters. Part 3, evidence and readings, says which document to trust, where that evidence commonly misleads and how to convert units.

Part 4, probable rating, holds the table of bands, the readings that meet no rule, the definition of each rule and what does not change the rating. Part 5 gives worked examples with invented applicants, the rule met by a reading exactly on each edge, the combinations seen most often and the rules of other sections that this section points to.

A section applies in one of two ways. Some apply only when the condition, event or treatment is on file: a reading alone is not enough. Others apply to any reading of their measure, diagnosis or not. Part 1 of each section says which, and names any other impairment whose diagnosis rules the section out.

Section numbers are printed with every heading, so a reference to section 10.4 always means the probable rating part of the tenth section.

### **1.3 Reading a rule**

Every rule has an identifier made of the letters UW, a short code for the impairment and a three-digit number. The identifier belongs to one rule only and never changes.

A rule is defined in exactly one place: in part 4 of its section, in a paragraph that starts with the word Rule, then the identifier, then a colon. That paragraph is complete in itself. It names the impairment and its section, states what is measured and the band, gives the rating, points to related rules, and names the public source of each edge of the band or says that the edge is this manual's own.

Everywhere else a rule is only mentioned, in words such as: see rule UW-HT-002. A mention is a pointer to the definition, not a second copy of it. If a mention and a definition ever seemed to disagree, the definition would be right.

A pointer always states the circumstance in which the other rule applies, in that rule's own terms: its impairment and its band. Most pointers lead to a rule that applies as well as the first. A few lead to a rule that applies in place of it, and say so.

## **1.4 Reading a rating**

A rating is one of two things. A debit is an addition to expected mortality, written as a percentage with a plus sign: a debit of +50 % means the applicant is expected to have one and a half times the mortality of a standard life of the same age and sex. Debits are always whole numbers.

Decline means that no terms are offered on the evidence in the file. Some rules that decline are postponements, and their rating reads decline as a postponement: the applicant may apply again once time has passed or a reading has improved.

A few rules carry no debit, written as no debit, +0 %. They exist so that a finding is recognised and named even though it adds nothing to the rating. A case that meets only such rules is a standard case.

## **1.5 Combining debits**

An applicant may meet several rules, in one section or in several. First find every rule that applies. Then combine them in this order.

If any rule that applies says decline, the outcome is decline, whatever the other rules say.

Otherwise add the debits of all the rules that apply. A debit of +50 % for raised blood pressure and a debit of +50 % for smoking give a loading of +100 %. Rules with no debit add nothing.

Only rules that can apply together are added. Where one rule applies in place of another, as a diabetes rule does in place of the prediabetes rule once the diagnosis is made, the first is dropped and the second stands alone.

If the total is +0 %, the case is standard. The sum is not capped in this manual, and no credits are given for favourable findings.

Within one measure the bands never overlap, so an applicant meets at most one rule for each thing measured. Where a section has two measures, such as present smoking status and lifetime pack-years, a rule for each can apply and both debits count.

## **1.6 Bands and their edges**

Each band says exactly which readings fall in it. The words from 7.0 to below 8.0 include 7.0 and exclude 8.0. The words above 40 and below 50 exclude both ends. The words of 180 or more include 180, and of 40 or less include 40.

Where the public guideline draws a line with a particular edge, the band keeps it. Otherwise the lower end of a band is included and the upper end is not.

Most sections rate only the abnormal range. The readings that meet no rule are listed under each rating table with what they mean, so that every reading a measure can take either falls in one band or is accounted for there.

## **1.7 Evidence and its age**

Rate on documents, not on recollection. A laboratory report outranks an attending physician's statement for a laboratory figure; the physician's statement outranks the application form for a diagnosis, a date or a habit.

Each section says how recent a reading must be. A reading older than that is not wrong, but it is not evidence of the present state, and a newer one should be asked for.

When two documents disagree, do not average them and do not choose the kinder one. Follow the section's instruction if it gives one; otherwise use the more recent document of the higher rank and record the disagreement.

When the reading a rule needs is missing, the rule cannot be applied. A rule is never applied on a guess, and a missing reading is never treated as a normal one.

## **1.8 When a person must decide**

This manual gives probable ratings. It does not decide cases. Many sections name circumstances that fall outside their bands: a recent hospital admission, an operation, a complication. Those cases go to a person, as does any case in which no rule fits the evidence, two rules of one measure seem to apply, or a deciding fact cannot be traced to a page of the file.

Software that reads this manual may suggest a rating and must show which rules and which facts it relied on. The suggestion is never the decision.

## **1.9 Sources**

The last section of this manual lists every public guideline cited, with the body that issued it, its year and the sections that rely on it. The citation beside each rule says where in the guideline the threshold is found.

Some bands have an edge that no guideline supplies, for example where a wide clinical stage is split in two. The definition of each rule and the note under each rating table say which edges come from the cited source and which are this manual's own.

## 2 Type 2 diabetes mellitus

### 2.1 The impairment

Type 2 diabetes is a long-term rise in blood glucose caused by resistance to insulin and, later, by falling insulin production. It usually starts in adult life and is often found on a routine blood test rather than through symptoms.

The extra mortality comes mostly from disease of the heart, the brain's blood supply and the kidneys. How high the glucose has run, and for how many years, is the best single guide to that risk, which is why this section rates on HbA1c.

An applicant treated with tablets alone and an applicant on insulin are rated by the same bands. Treatment tells the underwriter how far the disease has moved on, but the reading tells them how well it is held.

When this section applies. A diagnosis of type 2 diabetes is on file. A raised HbA1c with no diagnosis is read in the prediabetes section instead.

What is measured. The rating in this section depends on HbA1c (%). The thresholds are grounded in: American Diabetes Association, Standards of Care in Diabetes (2024).

### 2.2 Key questions

#### 1. When was diabetes diagnosed, and how old was the applicant?

Risk builds with the number of years of raised glucose, so a diagnosis at 35 weighs more than the same reading first found at 60.

#### 2. What is the most recent HbA1c, and what were the two before it?

One reading can flatter or mislead. Three readings show whether control is steady, improving or slipping.

**3. Which treatment is used: diet, tablets, injections other than insulin, or insulin?**

A move to insulin in type 2 diabetes usually means the disease has been present long enough to exhaust the pancreas.

**4. Has an eye, kidney or foot check found any damage?**

Damage to small vessels shows that past control was poor, whatever the latest reading says.

**5. Are blood pressure and cholesterol treated?**

Diabetes with untreated blood pressure or lipids carries more risk than the sum of its parts would suggest.

**2.3 Evidence and readings**

Use the HbA1c from a laboratory report or an attending physician's statement dated within the last twelve months. A figure the applicant recalls on the application form is a prompt to obtain evidence, not a reading to rate on.

Where the file holds several readings from the last twelve months, rate on the most recent one and mention in the notes any earlier reading that sits two bands higher. A fasting glucose on its own does not place the applicant in a band; ask for an HbA1c.

If the laboratory reports HbA1c in mmol/mol, convert it as shown below before reading the table, and record both figures.

**Pitfalls with this evidence.**

A recent fall in HbA1c of two points or more within six months is often the first effect of a new medicine and may not last. Rate on it, but record the earlier figure so that a person can see the trend.

Anaemia, recent blood loss, a transfusion and some inherited haemoglobin variants make HbA1c read falsely low or high. Where the full blood count shows a haemoglobin under 11.0 g/dL, say so beside the reading.

The words well controlled in a physician's statement are an opinion.  
The number is the evidence.

### **Converting units.**

HbA1c: A result in mmol/mol is multiplied by 0.0915 and 2.15 is added to give per cent: 53 mmol/mol is 7.0 %.

## **2.4 Probable rating**

The table lists every rule of this section; each is defined in full below it. The debit percentages and the decisions to decline are invented for this synthetic manual. The thresholds follow the public source named in each rule, except the edges a rule marks as this manual's own.

| <b>Rule id</b> | <b>Measure and band</b>        | <b>Probable rating</b>       |
|----------------|--------------------------------|------------------------------|
| UW-DM-001      | HbA1c below 7.0 %              | a debit of +25 %             |
| UW-DM-002      | HbA1c from 7.0 to below 8.0 %  | a debit of +50 %             |
| UW-DM-003      | HbA1c from 8.0 to below 9.0 %  | a debit of +100 %            |
| UW-DM-004      | HbA1c from 9.0 to below 10.0 % | a debit of +150 %            |
| UW-DM-005      | HbA1c of 10.0 % or more        | decline as a<br>postponement |

### **Readings that meet no rule.**

None: every reading these measures can take falls in one of the bands above.

Note on the bands. The edges at 7.0 % and 8.0 % follow the treatment goals in the cited standards. The steps at 9.0 % and 10.0 % are this manual's own, set so that each band is one percentage point wide.

Rule UW-DM-001: Type 2 diabetes mellitus (section 2.4). Threshold: HbA1c below 7.0 %. Probable rating: a debit of +25 %. Control at the usual treatment goal. If the applicant also has albumin in the urine with urine albumin-to-creatinine ratio from 30 to 300 mg/g, see rule UW-ALB-001. Source of the threshold: ADA, Standards of Care in Diabetes (2024), section 6, Glycemic Goals and Hypoglycemia (A1C goals for adults).

Rule UW-DM-002: Type 2 diabetes mellitus (section 2.4). Threshold: HbA1c from 7.0 to below 8.0 %. Probable rating: a debit of +50 %. Above the usual goal but inside the less stringent one. If the applicant also has hypertension with systolic blood pressure from 140 to below 160 mmHg, see rule UW-HT-002. Source of the threshold: ADA, Standards of Care in Diabetes (2024), section 6, Glycemic Goals and Hypoglycemia (A1C goals for adults).

Rule UW-DM-003: Type 2 diabetes mellitus (section 2.4). Threshold: HbA1c from 8.0 to below 9.0 %. Probable rating: a debit of +100 %. If the applicant also has chronic kidney disease with eGFR from 45 to below 60 mL/min/1.73 m<sup>2</sup>, see rule UW-CKD-001. Source of the edge at 8.0 %: ADA, Standards of Care in Diabetes (2024), section 6, Glycemic Goals and Hypoglycemia (A1C goals for adults). The edge at 9.0 % is this manual's own.

Rule UW-DM-004: Type 2 diabetes mellitus (section 2.4). Threshold: HbA1c from 9.0 to below 10.0 %. Probable rating: a debit of +150 %. If the applicant also has an unusual build with body mass index from 35.0 to below 40.0 kg/m<sup>2</sup>, see rule UW-BMI-003. The edges at 9.0 % and 10.0 % are this manual's own and no guideline supplies them. The source for the measure and the other edges of this section: ADA, Standards of Care in Diabetes (2024), section 6, Glycemic Goals and Hypoglycemia (A1C goals for adults).

Rule UW-DM-005: Type 2 diabetes mellitus (section 2.4). Threshold: HbA1c of 10.0 % or more. Probable rating: decline as a postponement. The applicant may apply again once two readings three months apart are below 10.0 %. The edge at 10.0 % is this manual's own and no guideline supplies it. The source for the measure and the other edges of this section: ADA, Standards of Care in Diabetes (2024), section 6, Glycemic Goals and Hypoglycemia (A1C goals for adults).

### **What does not change the rating.**

The type of treatment does not move an applicant between bands: diet alone, tablets, injected medicines other than insulin and insulin are all read on the same table.

A family history of diabetes adds nothing once the applicant has the diagnosis. A fasting glucose, however high or low, does not replace or adjust the HbA1c band.

## **2.5 Worked examples and related rules**

Example 1. The specialist's letter for a librarian aged 56 with type 2 diabetes records HbA1c 6.4 %. This is covered by rule UW-DM-001 (HbA1c below 7.0 %). Probable rating: a debit of +25 %. If the applicant also has albumin in the urine with urine albumin-to-creatinine ratio from 30 to 300 mg/g, rule UW-ALB-001 would apply as well; its rating is a debit of +25 %, and the two together would give +50 %.

Example 2. An applicant aged 37, a florist, has type 2 diabetes. The attending physician's statement records HbA1c 7.5 %. That meets rule UW-DM-002 (HbA1c from 7.0 to below 8.0 %), so the probable rating is a debit of +50 %. If the applicant also has hypertension with systolic blood pressure from 140 to below 160 mmHg, rule UW-HT-002 would apply as well; its rating is a debit of +50 %, and the two together would give +100 %.

Example 3. Take a market gardener of 48 with type 2 diabetes. On file, from the most recent clinic letter: HbA1c 8.5 %. The threshold met is that of rule UW-DM-003 (HbA1c from 8.0 to below 9.0 %), and the probable rating is a debit of +100 %. If the applicant also has chronic kidney disease with eGFR from 45 to below 60 mL/min/1.73 m<sup>2</sup>, rule UW-CKD-001 would apply as well; its rating is a debit of +50 %, and the two together would give +150 %.

Example 4. The report on file for a school caretaker aged 59 with type 2 diabetes records HbA1c 9.5 %. This is covered by rule UW-DM-004 (HbA1c from 9.0 to below 10.0 %). Probable rating: a debit of +150 %. If the applicant also has an unusual build with body mass index from 35.0 to below 40.0 kg/m<sup>2</sup>, rule UW-BMI-003 would apply as well; its rating is a debit of +50 %, and the two together would give +200 %.

Example 5. An applicant aged 40, a translator, has type 2 diabetes. The specialist's letter records HbA1c 10.8 %. That meets rule UW-DM-005 (HbA1c of 10.0 % or more), so the probable rating is decline as a postponement.

Readings on an edge: HbA1c 7.0 % exactly, rule UW-DM-002 (a debit of +50 %); HbA1c 8.0 % exactly, rule UW-DM-003 (a debit of +100 %); HbA1c 9.0 % exactly, rule UW-DM-004 (a debit of +150 %); HbA1c 10.0 % exactly, rule UW-DM-005 (decline as a postponement).

### **Common combinations.**

Diabetes rarely comes alone. Raised blood pressure, excess weight and raised LDL cholesterol are each rated in their own section and their debits are added to the diabetes debit; none of them is folded into it.

Kidney findings matter most. Albumin in the urine or a reduced eGFR in a person with diabetes shows damage to small vessels, and the kidney sections' debits are added in full. Diabetes together with a past heart attack or stroke is common and heavy: add the debits, and expect the total to pass +200 %.

**Related rules.**

Under rule UW-DM-001, if the applicant also has albumin in the urine with urine albumin-to-creatinine ratio from 30 to 300 mg/g, see rule UW-ALB-001 (section 28.4, Albuminuria).

Under rule UW-DM-002, if the applicant also has hypertension with systolic blood pressure from 140 to below 160 mmHg, see rule UW-HT-002 (section 10.4, Hypertension).

Under rule UW-DM-003, if the applicant also has chronic kidney disease with eGFR from 45 to below 60 mL/min/1.73 m<sup>2</sup>, see rule UW-CKD-001 (section 27.4, Chronic kidney disease).

Under rule UW-DM-004, if the applicant also has an unusual build with body mass index from 35.0 to below 40.0 kg/m<sup>2</sup>, see rule UW-BMI-003 (section 5.4, Build: underweight and obesity).

Rules of other sections that point to this one: rule UW-PD-001 (section 4.4, Prediabetes); rule UW-BMI-004 (section 5.4, Build: underweight and obesity); rule UW-PAD-002 (section 17.4, Peripheral artery disease); rule UW-ALB-002 (section 28.4, Albuminuria).

## 3 Type 1 diabetes mellitus

### 3.1 The impairment

Type 1 diabetes follows the loss of the insulin-producing cells of the pancreas. It usually begins in childhood or early adult life, and insulin is needed from the start and for life.

Because it starts early, an applicant of 40 may already have lived with raised glucose for 30 years. The bands are therefore stricter than for type 2 diabetes at the same HbA1c, and the highest band closes one step sooner.

When this section applies. A diagnosis of type 1 diabetes is on file, whatever the age at which it began.

What is measured. The rating in this section depends on HbA1c (%). The thresholds are grounded in: American Diabetes Association, Standards of Care in Diabetes (2024).

### 3.2 Key questions

#### 1. At what age did insulin treatment begin?

The years lived with the disease matter more here than the applicant's present age.

#### 2. How is insulin given, and is glucose monitored continuously?

A pump or a sensor does not earn a better rating by itself, but it explains a stable record.

#### 3. Has there been a hospital admission for ketoacidosis or a severe low in the last two years?

Either event points to unstable control that an average such as HbA1c can hide.

#### 4. What did the last kidney and eye screening show?

Protein in the urine is the earliest sign of the complication that shortens life most in type 1 diabetes.

### **3.3 Evidence and readings**

Rate on the most recent laboratory HbA1c from the last twelve months. A sensor's estimate of average glucose is useful background but is not a substitute for the laboratory figure.

A record of severe hypoglycaemia needing another person's help, or of ketoacidosis, within two years is outside these bands: send the case to a person with the hospital letter attached.

#### **Pitfalls with this evidence.**

Do not infer the type from the treatment. Many people with type 2 diabetes take insulin; the diagnosis written by the physician decides which section applies, and a file that leaves the type unclear goes to a person.

A very good HbA1c reached at the cost of frequent low glucose is not good control. If the clinic letter mentions hypoglycaemia needing help, the band alone understates the risk.

#### **Converting units.**

HbA1c: A result in mmol/mol is multiplied by 0.0915 and 2.15 is added to give per cent: 53 mmol/mol is 7.0 %.

### **3.4 Probable rating**

The table lists every rule of this section; each is defined in full below it. The debit percentages and the decisions to decline are invented for this synthetic manual. The thresholds follow the public source named in each rule, except the edges a rule marks as this manual's own.

| Rule id   | Measure and band              | Probable rating   |
|-----------|-------------------------------|-------------------|
| UW-DI-001 | HbA1c below 7.0 %             | a debit of +75 %  |
| UW-DI-002 | HbA1c from 7.0 to below 8.0 % | a debit of +100 % |
| UW-DI-003 | HbA1c from 8.0 to below 9.0 % | a debit of +150 % |
| UW-DI-004 | HbA1c of 9.0 % or more        | decline           |

### Readings that meet no rule.

None: every reading these measures can take falls in one of the bands above.

Note on the bands. The edges at 7.0 % and 8.0 % follow the goals in the cited standards; the edge at 9.0 % is this manual's own.

Rule UW-DI-001: Type 1 diabetes mellitus (section 3.4). Threshold: HbA1c below 7.0 %. Probable rating: a debit of +75 %. If the applicant also has albumin in the urine with urine albumin-to-creatinine ratio from 30 to 300 mg/g, see rule UW-ALB-001. Source of the threshold: ADA, Standards of Care in Diabetes (2024), section 6, Glycemic Goals and Hypoglycemia (A1C goals for adults).

Rule UW-DI-002: Type 1 diabetes mellitus (section 3.4). Threshold: HbA1c from 7.0 to below 8.0 %. Probable rating: a debit of +100 %. Source of the threshold: ADA, Standards of Care in Diabetes (2024), section 6, Glycemic Goals and Hypoglycemia (A1C goals for adults).

Rule UW-DI-003: Type 1 diabetes mellitus (section 3.4). Threshold: HbA1c from 8.0 to below 9.0 %. Probable rating: a debit of +150 %. If the applicant also has chronic kidney disease with eGFR from 45 to below 60 mL/min/1.73 m<sup>2</sup>, see rule UW-CKD-001. Source of the edge at 8.0 %: ADA, Standards of Care in Diabetes (2024), section 6, Glycemic Goals and Hypoglycemia (A1C goals for adults). The edge at 9.0 % is this manual's own.

Rule UW-DI-004: Type 1 diabetes mellitus (section 3.4). Threshold: HbA1c of 9.0 % or more. Probable rating: decline. The edge at 9.0 % is this manual's own and no guideline supplies it. The source for the measure and the other edges of this section: ADA, Standards of Care in Diabetes (2024), section 6, Glycemic Goals and Hypoglycemia (A1C goals for adults).

### **What does not change the rating.**

An insulin pump, a continuous sensor or a closed-loop system does not change the band. The age of the applicant does not change it either, though the duration of the disease is noted for the person who reviews the case.

A stable HbA1c for many years is reassuring but earns no reduction.

## **3.5 Worked examples and related rules**

Example 1. Take a surveyor of 33 with type 1 diabetes. On file, from the attending physician's statement: HbA1c 6.4 %. The threshold met is that of rule UW-DI-001 (HbA1c below 7.0 %), and the probable rating is a debit of +75 %. If the applicant also has albumin in the urine with urine albumin-to-creatinine ratio from 30 to 300 mg/g, rule UW-ALB-001 would apply as well; its rating is a debit of +25 %, and the two together would give +100 %.

Example 2. The most recent clinic letter for a dental technician aged 44 with type 1 diabetes records HbA1c 7.5 %. This is covered by rule UW-DI-002 (HbA1c from 7.0 to below 8.0 %). Probable rating: a debit of +100 %.

Example 3. An applicant aged 55, a pastry cook, has type 1 diabetes. The report on file records HbA1c 8.5 %. That meets rule UW-DI-003 (HbA1c from 8.0 to below 9.0 %), so the probable rating is a debit of +150 %. If the applicant also has chronic kidney disease with eGFR from 45 to below 60 mL/min/1.73 m<sup>2</sup>, rule UW-CKD-001 would apply as well; its rating is a debit of +50 %, and the two together would give +200 %.

Example 4. Take a joiner of 36 with type 1 diabetes. On file, from the specialist's letter: HbA1c 9.7 %. The threshold met is that of rule UW-DI-004 (HbA1c of 9.0 % or more), and the probable rating is decline.

Readings on an edge: HbA1c 7.0 % exactly, rule UW-DI-002 (a debit of +100 %); HbA1c 8.0 % exactly, rule UW-DI-003 (a debit of +150 %); HbA1c 9.0 % exactly, rule UW-DI-004 (decline).

### **Common combinations.**

The kidney is the usual second finding. Albumin in the urine after ten or more years of type 1 diabetes is rated in the albuminuria section and added; with a reduced eGFR as well, all three debits are added.

Thyroid disease and coeliac disease share an autoimmune cause with type 1 diabetes. A treated underactive thyroid with a settled TSH adds no debit. Raised blood pressure is added as it would be for anyone.

### **Related rules.**

Under rule UW-DI-001, if the applicant also has albumin in the urine with urine albumin-to-creatinine ratio from 30 to 300 mg/g, see rule UW-ALB-001 (section 28.4, Albuminuria).

Under rule UW-DI-003, if the applicant also has chronic kidney disease with eGFR from 45 to below 60 mL/min/1.73 m<sup>2</sup>, see rule UW-CKD-001 (section 27.4, Chronic kidney disease).

## 4 Prediabetes

### 4.1 The impairment

Prediabetes means glucose runs above the normal range without reaching the level that defines diabetes. Many people stay in this range for years; some return to normal with weight loss, and some go on to diabetes.

On its own the finding adds little mortality, so its single rule carries no debit. The section exists so that a reading in this range is recognised, named and followed to the right place if it later rises.

When this section applies. A laboratory HbA1c is on file and no diagnosis of diabetes of either type has been made.

What is measured. The rating in this section depends on HbA1c (%). The thresholds are grounded in: American Diabetes Association, Standards of Care in Diabetes (2024).

### 4.2 Key questions

#### 1. Was the reading repeated, and what was the second result?

A single borderline reading is often normal on repeat.

#### 2. Has a doctor ever used the word diabetes, or prescribed metformin?

Treatment with a glucose-lowering medicine suggests the diagnosis has already moved on.

#### 3. What are the weight, blood pressure and lipids?

Prediabetes matters mainly as part of a cluster; each of the others has its own section.

## 4.3 Evidence and readings

Use a laboratory HbA1c from the last twelve months. Where two readings from that period fall on either side of 6.5 %, treat the applicant as having diabetes until a third reading settles it.

A raised fasting glucose without an HbA1c is not enough to apply this rule; ask for the HbA1c.

### Pitfalls with this evidence.

The label matters. Notes that say impaired glucose tolerance, borderline diabetes or glucose intolerance all mean a reading in this range; notes that say diabetes, diet controlled, mean the diagnosis has been made and this section no longer applies.

Metformin prescribed for polycystic ovaries is not treatment for diabetes. Check the reason before treating a prescription as a diagnosis.

### Converting units.

HbA1c: A result in mmol/mol is multiplied by 0.0915 and 2.15 is added to give per cent: 53 mmol/mol is 7.0 %.

## 4.4 Probable rating

The table lists every rule of this section; each is defined in full below it. The debit percentages and the decisions to decline are invented for this synthetic manual. The thresholds follow the public source named in each rule, except the edges a rule marks as this manual's own.

| Rule id   | Measure and band              | Probable rating |
|-----------|-------------------------------|-----------------|
| UW-PD-001 | HbA1c from 5.7 to below 6.5 % | no debit, +0 %  |

### Readings that meet no rule.

HbA1c below 5.7 %: a normal reading; no rule and no debit.

HbA1c of 6.5 % or more: the level that defines diabetes; ask whether the diagnosis has been made, and read the diabetes section once it has.

Note on the bands. Both edges are the diagnostic criteria in the cited standards. Below 5.7 % no rule of this section applies.

Rule UW-PD-001: Prediabetes (section 4.4). Threshold: HbA1c from 5.7 to below 6.5 %. Probable rating: no debit, +0 %. If the applicant instead has type 2 diabetes with HbA1c below 7.0 %, see rule UW-DM-001, which applies in place of this rule. Source of the threshold: ADA, Standards of Care in Diabetes (2024), section 2, Diagnosis and Classification of Diabetes (A1C criteria for prediabetes and diabetes).

### **What does not change the rating.**

Advice to lose weight, a referral to a prevention programme or a repeat test booked for next year does not change the outcome: the single rule carries no debit.

A parent with diabetes does not add a debit to a reading in this range.

## **4.5 Worked examples and related rules**

Example 1. An applicant aged 40, a payroll clerk, has prediabetes. The most recent clinic letter records HbA1c 6.1 %. That meets rule UW-PD-001 (HbA1c from 5.7 to below 6.5 %), so the probable rating is no debit, +0 %. If the applicant instead has type 2 diabetes with HbA1c below 7.0 %, rule UW-DM-001 would apply in place of this one, and the probable rating would be a debit of +25 %.

Readings on an edge: HbA1c 5.7 % exactly, rule UW-PD-001 (no debit, +0 %); HbA1c 6.5 % exactly, no rule of this section.

**Common combinations.**

Prediabetes with a body mass index of 35 kg/m<sup>2</sup> or more, raised blood pressure and raised triglycerides is the familiar metabolic cluster. Each member is rated in its own section; this one contributes nothing, and the others are added as usual.

If a later reading reaches 6.5 %, or the physician records diabetes, the diabetes section takes over and its rule replaces this one. The two are never added together.

**Related rules.**

Under rule UW-PD-001, if the applicant instead has type 2 diabetes with HbA1c below 7.0 %, see rule UW-DM-001 (section 2.4, Type 2 diabetes mellitus).

## **5 Build: underweight and obesity**

### **5.1 The impairment**

Build is the relation of weight to height. Mortality is lowest across a broad middle range and rises at both ends: at the low end through frailty, undiagnosed illness and eating disorders, and at the high end through diabetes, heart disease and sleep apnoea.

This section rates the two ends only. A body mass index from 18.5 to below 35 kg/m<sup>2</sup> meets no rule here, although the conditions that travel with weight are rated in their own sections.

When this section applies. Every applicant: height and weight are on file for all of them.

What is measured. The rating in this section depends on body mass index (kg/m<sup>2</sup>). The thresholds are grounded in: World Health Organization, Obesity: preventing and managing the global epidemic, WHO Technical Report Series 894 (2000).

### **5.2 Key questions**

#### **1. Were height and weight measured by a clinician, or stated by the applicant?**

Stated weight tends to be lower, and stated height taller, than measured.

#### **2. Has weight changed by more than 5 kg in the last twelve months, and was the change intended?**

Unintended loss needs an explanation before any rating is offered.

#### **3. For a low weight: is there a history of an eating disorder, bowel disease or cancer?**

A low body mass index is a finding, not a diagnosis, and the cause decides the risk.

#### **4. For a high weight: has there been weight-loss surgery, and when?**

The first year after surgery has its own risks and its own unstable weight.

### **5.3 Evidence and readings**

Work out the body mass index from the measured height and weight in the attending physician's statement where there is one; otherwise from the application form. Round to one decimal place before reading the table.

Where measured and stated figures put the applicant in different bands, rate on the measured figure and say so in the notes.

#### **Pitfalls with this evidence.**

Weight on an application form is often rounded down and months old. A clinic weight in the attending physician's statement is preferred even when it is less flattering.

Body mass index overstates fatness in a very muscular person and understates it in a frail one. The manual does not adjust for either; where the physician comments on it, pass the comment to a person.

Weight during pregnancy is not rated. Use the last weight recorded before it.

#### **Converting units.**

body mass index: From pounds and inches: weight in pounds times 703, divided by the square of height in inches.

### **5.4 Probable rating**

The table lists every rule of this section; each is defined in full below it. The debit percentages and the decisions to decline are invented for this synthetic manual. The thresholds follow the public source named in each rule, except the edges a rule marks as this manual's own.

| Rule id    | Measure and band                              | Probable rating   |
|------------|-----------------------------------------------|-------------------|
| UW-BMI-001 | body mass index below 16.0 kg/m2              | a debit of +100 % |
| UW-BMI-002 | body mass index from 16.0 to below 18.5 kg/m2 | a debit of +25 %  |
| UW-BMI-003 | body mass index from 35.0 to below 40.0 kg/m2 | a debit of +50 %  |
| UW-BMI-004 | body mass index of 40.0 kg/m2 or more         | a debit of +100 % |

### Readings that meet no rule.

body mass index from 18.5 to below 35.0 kg/m2: no rule; no debit for build in this range.

Note on the bands. All four edges are those of the cited classification: severe thinness below 16, underweight below 18.5, obesity class II from 35 and class III from 40.

Rule UW-BMI-001: Build: underweight and obesity (section 5.4). Threshold: body mass index below 16.0 kg/m2. Probable rating: a debit of +100 %. If the applicant also has anaemia with haemoglobin from 8.0 to below 11.0 g/dL, see rule UW-ANA-001. Source of the threshold: WHO, Obesity: preventing and managing the global epidemic, WHO Technical Report Series 894 (2000), the classification of adults according to body mass index.

Rule UW-BMI-002: Build: underweight and obesity (section 5.4). Threshold: body mass index from 16.0 to below 18.5 kg/m2. Probable rating: a debit of +25 %. Source of the threshold: WHO, Obesity: preventing and managing the global epidemic, WHO Technical Report Series 894 (2000), the classification of adults according to body mass index.

Rule UW-BMI-003: Build: underweight and obesity (section 5.4).

Threshold: body mass index from 35.0 to below 40.0 kg/m<sup>2</sup>. Probable rating: a debit of +50 %. If the applicant also has obstructive sleep apnoea with apnoea-hypopnoea index from 15 to 30 events per hour, see rule UW-OSA-002. Source of the threshold: WHO, Obesity: preventing and managing the global epidemic, WHO Technical Report Series 894 (2000), the classification of adults according to body mass index.

Rule UW-BMI-004: Build: underweight and obesity (section 5.4).

Threshold: body mass index of 40.0 kg/m<sup>2</sup> or more. Probable rating: a debit of +100 %. If the applicant also has type 2 diabetes with HbA1c from 7.0 to below 8.0 %, see rule UW-DM-002. Source of the threshold: WHO, Obesity: preventing and managing the global epidemic, WHO Technical Report Series 894 (2000), the classification of adults according to body mass index.

### **What does not change the rating.**

Waist measurement, body fat percentage and the applicant's own account of fitness do not move the band. Neither does weight lost within the last twelve months: the present figure is rated, and the earlier one is noted.

A body mass index from 18.5 to below 35 kg/m<sup>2</sup> carries no debit, including the whole of the overweight range and obesity class I.

## **5.5 Worked examples and related rules**

Example 1. The report on file for a translator aged 47 with an unusual build records body mass index 14.7 kg/m<sup>2</sup>. This is covered by rule UW-BMI-001 (body mass index below 16.0 kg/m<sup>2</sup>). Probable rating: a debit of +100 %. If the applicant also has anaemia with haemoglobin from 8.0 to below 11.0 g/dL, rule UW-ANA-001 would apply as well; its rating is a debit of +50 %, and the two together would give +150 %.

Example 2. An applicant aged 58, a surveyor, has an unusual build. The specialist's letter records body mass index 17.2 kg/m<sup>2</sup>. That meets rule UW-BMI-002 (body mass index from 16.0 to below 18.5 kg/m<sup>2</sup>), so the probable rating is a debit of +25 %.

Example 3. Take a dental technician of 39 with an unusual build. On file, from the attending physician's statement: body mass index 37.5 kg/m<sup>2</sup>. The threshold met is that of rule UW-BMI-003 (body mass index from 35.0 to below 40.0 kg/m<sup>2</sup>), and the probable rating is a debit of +50 %. If the applicant also has obstructive sleep apnoea with apnoea-hypopnoea index from 15 to 30 events per hour, rule UW-OA-002 would apply as well; its rating is a debit of +25 %, and the two together would give +75 %.

Example 4. The most recent clinic letter for a pastry cook aged 50 with an unusual build records body mass index 43.2 kg/m<sup>2</sup>. This is covered by rule UW-BMI-004 (body mass index of 40.0 kg/m<sup>2</sup> or more). Probable rating: a debit of +100 %. If the applicant also has type 2 diabetes with HbA1c from 7.0 to below 8.0 %, rule UW-DM-002 would apply as well; its rating is a debit of +50 %, and the two together would give +150 %.

Readings on an edge: body mass index 16.0 kg/m<sup>2</sup> exactly, rule UW-BMI-002 (a debit of +25 %); body mass index 18.5 kg/m<sup>2</sup> exactly, no rule of this section; body mass index 35.0 kg/m<sup>2</sup> exactly, rule UW-BMI-003 (a debit of +50 %); body mass index 40.0 kg/m<sup>2</sup> exactly, rule UW-BMI-004 (a debit of +100 %).

### **Common combinations.**

High weight travels with diabetes, raised blood pressure, sleep apnoea and fatty liver disease. Each is rated in its own section and added to the build debit; the build debit is not reduced because the others are present.

Low weight with anaemia, an inflammatory bowel disease or chronic lung disease points to illness rather than constitution: add the debits, and send to a person where the weight loss is unexplained.

**Related rules.**

Under rule UW-BMI-001, if the applicant also has anaemia with haemoglobin from 8.0 to below 11.0 g/dL, see rule UW-ANA-001 (section 34.4, Anaemia).

Under rule UW-BMI-003, if the applicant also has obstructive sleep apnoea with apnoea-hypopnoea index from 15 to 30 events per hour, see rule UW-OSA-002 (section 26.4, Obstructive sleep apnoea).

Under rule UW-BMI-004, if the applicant also has type 2 diabetes with HbA1c from 7.0 to below 8.0 %, see rule UW-DM-002 (section 2.4, Type 2 diabetes mellitus).

Rules of other sections that point to this one: rule UW-DM-004 (section 2.4, Type 2 diabetes mellitus); rule UW-VTE-002 (section 20.4, Venous thromboembolism); rule UW-COPD-003 (section 25.4, Chronic obstructive pulmonary disease); rule UW-OSA-003 (section 26.4, Obstructive sleep apnoea); rule UW-FLD-002 (section 31.4, Fatty liver disease); rule UW-OST-001 (section 40.4, Osteoporosis).

## 6 Raised LDL cholesterol

### 6.1 The impairment

LDL cholesterol carries cholesterol into the artery wall. The higher it runs and the longer it stays high, the sooner fatty plaques narrow the coronary arteries.

Very high levels from early adult life usually have an inherited cause. This section rates the level itself; a heart attack or other event that has already happened is rated in its own section and the two are combined.

When this section applies. An LDL cholesterol result is on file, with or without a diagnosis or treatment.

What is measured. The rating in this section depends on LDL cholesterol (mg/dL). The thresholds are grounded in: American Heart Association and American College of Cardiology, Guideline on the Management of Blood Cholesterol (2018).

### 6.2 Key questions

#### 1. Was the sample taken fasting, and is the applicant on a statin?

The table is read with the level on treatment, if treatment is taken as prescribed.

#### 2. What was the highest level ever recorded before treatment?

An untreated level of 190 mg/dL or more suggests an inherited disorder, which matters for the family as well.

#### 3. Has a parent, brother or sister had a heart attack before the age of 60?

Early disease in the family strengthens the case for an inherited cause.

## 6.3 Evidence and readings

Use the LDL figure from a laboratory report in the last twelve months. If only total cholesterol is given, ask for the full lipid profile; do not estimate LDL from the total.

Where the report is in mmol/L, convert it as shown below and record the original figure beside the converted one.

### Pitfalls with this evidence.

An LDL figure worked out by the laboratory's formula is unreliable when triglycerides are above 400 mg/dL; the report usually says so. Ask for a directly measured LDL.

A result taken within eight weeks of a heart attack or a major illness reads falsely low. A result taken in the first weeks of statin treatment has not yet settled.

### Converting units.

LDL cholesterol: A result in mmol/L is multiplied by 38.67 to give mg/dL: 4.14 mmol/L is 160 mg/dL.

## 6.4 Probable rating

The table lists every rule of this section; each is defined in full below it. The debit percentages and the decisions to decline are invented for this synthetic manual. The thresholds follow the public source named in each rule, except the edges a rule marks as this manual's own.

| Rule id    | Measure and band                            | Probable rating  |
|------------|---------------------------------------------|------------------|
| UW-LDL-001 | LDL cholesterol from 160 to below 190 mg/dL | a debit of +25 % |
| UW-LDL-002 | LDL cholesterol of 190 mg/dL or more        | a debit of +50 % |

**Readings that meet no rule.**

LDL cholesterol below 160 mg/dL: no rule; no debit for LDL cholesterol.

Note on the bands. The edge at 190 mg/dL is the cited guideline's definition of severe primary hypercholesterolemia; 160 mg/dL is its risk-enhancing level.

Rule UW-LDL-001: Raised LDL cholesterol (section 6.4). Threshold: LDL cholesterol from 160 to below 190 mg/dL. Probable rating: a debit of +25 %. If the applicant also has hypertension with systolic blood pressure from 130 to below 140 mmHg, see rule UW-HT-001. Source of the threshold: AHA/ACC, Guideline on the Management of Blood Cholesterol (2018), severe primary hypercholesterolemia, the risk-enhancing factors and severe hypertriglyceridemia.

Rule UW-LDL-002: Raised LDL cholesterol (section 6.4). Threshold: LDL cholesterol of 190 mg/dL or more. Probable rating: a debit of +50 %. If the applicant also has a past heart attack with time since the heart attack from 12 to below 60 months, see rule UW-MI-002. Source of the threshold: AHA/ACC, Guideline on the Management of Blood Cholesterol (2018), severe primary hypercholesterolemia, the risk-enhancing factors and severe hypertriglyceridemia.

**What does not change the rating.**

Total cholesterol, HDL cholesterol and the ratio between them do not change the band; only LDL is rated. Taking a statin is not penalised and earns no credit beyond the lower reading it produces.

An LDL below 160 mg/dL meets no rule, however high it was before treatment.

## 6.5 Worked examples and related rules

Example 1. Take a bus driver of 54 with raised LDL cholesterol. On file, from the specialist's letter: LDL cholesterol 175 mg/dL. The threshold met is that of rule UW-LDL-001 (LDL cholesterol from 160 to below 190 mg/dL), and the probable rating is a debit of +25 %. If the applicant also has hypertension with systolic blood pressure from 130 to below 140 mmHg, rule UW-HT-001 would apply as well; its rating is a debit of +25 %, and the two together would give +50 %.

Example 2. The attending physician's statement for a payroll clerk aged 35 with raised LDL cholesterol records LDL cholesterol 205 mg/dL. This is covered by rule UW-LDL-002 (LDL cholesterol of 190 mg/dL or more). Probable rating: a debit of +50 %. If the applicant also has a past heart attack with time since the heart attack from 12 to below 60 months, rule UW-MI-002 would apply as well; its rating is a debit of +100 %, and the two together would give +150 %.

Readings on an edge: LDL cholesterol 160 mg/dL exactly, rule UW-LDL-001 (a debit of +25 %); LDL cholesterol 190 mg/dL exactly, rule UW-LDL-002 (a debit of +50 %).

### Common combinations.

Raised LDL with raised blood pressure, smoking or diabetes multiplies the chance of a heart attack; in this manual the debits are simply added, each from its own section.

After a heart attack the LDL debit is added to the debit for the event. An untreated LDL of 190 mg/dL or more with early heart disease in a parent suggests an inherited disorder: rate by the table and tell the person reviewing the case.

### Related rules.

Under rule UW-LDL-001, if the applicant also has hypertension with systolic blood pressure from 130 to below 140 mmHg, see rule UW-HT-001 (section 10.4, Hypertension).

Under rule UW-LDL-002, if the applicant also has a past heart attack with time since the heart attack from 12 to below 60 months, see rule UW-MI-002 (section 11.4, Myocardial infarction).

Rules of other sections that point to this one: rule UW-THY-002 (section 9.4, Hypothyroidism); rule UW-MI-003 (section 11.4, Myocardial infarction).

## 7 Raised triglycerides

### 7.1 The impairment

Triglycerides are the fat the body stores and burns. A moderate rise usually reflects weight, alcohol, diabetes or a medicine; a severe rise brings a real risk of inflammation of the pancreas.

The reading swings widely after food, so only a fasting sample is rated.

When this section applies. A fasting triglyceride result is on file.

What is measured. The rating in this section depends on fasting triglycerides (mg/dL). The thresholds are grounded in: American Heart Association and American College of Cardiology, Guideline on the Management of Blood Cholesterol (2018).

### 7.2 Key questions

#### 1. How long had the applicant fasted before the sample?

A sample taken after a meal can read twice as high as a fasting one.

#### 2. How much alcohol is taken in a usual week?

Alcohol is the commonest reversible cause of a high reading.

#### 3. Has there ever been pancreatitis?

A past attack with a level at or above 500 mg/dL changes the case from a laboratory finding to a disease.

### 7.3 Evidence and readings

Rate on a fasting laboratory result from the last twelve months. If the report does not say whether the sample was fasting, treat a level under 175 mg/dL as acceptable and ask for a fasting repeat for anything higher.

A history of pancreatitis is outside these bands and goes to a person.

### **Pitfalls with this evidence.**

The commonest error is rating a sample taken after food. Look for the word fasting on the report or in the clinic note.

Triglycerides fall quickly after a few days without alcohol, so a single good reading soon after a bad one proves little. Where two results a month apart differ by half or more, use the higher and say why.

### **Converting units.**

fasting triglycerides: A result in mmol/L is multiplied by 88.57 to give mg/dL: 2.0 mmol/L is about 177 mg/dL.

## **7.4 Probable rating**

The table lists every rule of this section; each is defined in full below it. The debit percentages and the decisions to decline are invented for this synthetic manual. The thresholds follow the public source named in each rule, except the edges a rule marks as this manual's own.

| <b>Rule id</b> | <b>Measure and band</b>                           | <b>Probable rating</b> |
|----------------|---------------------------------------------------|------------------------|
| UW-TG-001      | fasting triglycerides from 175 to below 500 mg/dL | a debit of +25 %       |
| UW-TG-002      | fasting triglycerides of 500 mg/dL or more        | a debit of +75 %       |

### **Readings that meet no rule.**

fasting triglycerides below 175 mg/dL: no rule; no debit for triglycerides.

Note on the bands. 175 mg/dL and 500 mg/dL are the cited guideline's levels for a persistent rise and for severe hypertriglyceridemia.

Rule UW-TG-001: Raised triglycerides (section 7.4). Threshold: fasting triglycerides from 175 to below 500 mg/dL. Probable rating: a debit of +25 %. Source of the threshold: AHA/ACC, Guideline on the Management of Blood Cholesterol (2018), severe primary hypercholesterolemia, the risk-enhancing factors and severe hypertriglyceridemia.

Rule UW-TG-002: Raised triglycerides (section 7.4). Threshold: fasting triglycerides of 500 mg/dL or more. Probable rating: a debit of +75 %. If the applicant also has a record of heavy drinking with AUDIT score from 8 to below 16 points, see rule UW-ALC-001. Source of the threshold: AHA/ACC, Guideline on the Management of Blood Cholesterol (2018), severe primary hypercholesterolemia, the risk-enhancing factors and severe hypertriglyceridemia.

### **What does not change the rating.**

Fish oil capsules, a fibrate or a statin on the prescription list does not change the band. A reading under 175 mg/dL meets no rule, even if earlier readings were far higher.

## **7.5 Worked examples and related rules**

Example 1. An applicant aged 31, a school caretaker, has raised triglycerides. The attending physician's statement records fasting triglycerides 338 mg/dL. That meets rule UW-TG-001 (fasting triglycerides from 175 to below 500 mg/dL), so the probable rating is a debit of +25 %.

Example 2. Take a translator of 42 with raised triglycerides. On file, from the most recent clinic letter: fasting triglycerides 540 mg/dL. The threshold met is that of rule UW-TG-002 (fasting triglycerides of 500 mg/dL or more), and the probable rating is a debit of +75 %. If the applicant also has a record of heavy drinking with AUDIT score from 8 to below 16 points, rule UW-ALC-001 would apply as well; its rating is a debit of +25 %, and the two together would give +100 %.

Readings on an edge: fasting triglycerides 175 mg/dL exactly, rule UW-TG-001 (a debit of +25 %); fasting triglycerides 500 mg/dL exactly, rule UW-TG-002 (a debit of +75 %).

**Common combinations.**

Raised triglycerides usually sit alongside excess weight, diabetes or heavy drinking, and the reading often falls when those are dealt with. Each has its own section; the debits are added.

A level of 500 mg/dL or more with an AUDIT score in the hazardous range is a combination to take seriously, because both lead to pancreatitis.

**Related rules.**

Under rule UW-TG-002, if the applicant also has a record of heavy drinking with AUDIT score from 8 to below 16 points, see rule UW-ALC-001 (section 22.4, Alcohol use).

## 8 Gout

### 8.1 The impairment

Gout is arthritis caused by urate crystals, classically a sudden, very painful swelling of the big toe. Attacks pass, but an untreated high urate brings more of them, joint damage and kidney stones.

Gout shortens life only modestly, and mostly through the company it keeps: high blood pressure, kidney disease and excess weight. The rating is small and looks to whether urate is at target.

When this section applies. Gout has been diagnosed. A raised urate in someone who has never had an attack is not gout and meets no rule here.

What is measured. The rating in this section depends on serum urate (mg/dL). The thresholds are grounded in: American College of Rheumatology, Guideline for the Management of Gout (2020).

### 8.2 Key questions

#### 1. How many attacks were there in the last twelve months?

Two or more attacks a year means the disease is not controlled, whatever medicine is taken.

#### 2. Is a urate-lowering medicine taken every day?

Treatment only during attacks leaves the cause untouched.

#### 3. Has kidney function been checked?

Poor kidneys both cause a high urate and are harmed by it.

### 8.3 Evidence and readings

Use a serum urate measured at least two weeks after an attack, since the level can fall misleadingly during one.

No urate on file and no attack for five years: no rule of this section applies, and the history is simply noted.

### **Pitfalls with this evidence.**

Urate falls during an acute attack, so a normal level taken that week does not show control. Check the date of the sample against the dates of attacks.

Colchicine and anti-inflammatory tablets treat attacks and do nothing to urate. Only allopurinol, febuxostat or a similar medicine counts as urate-lowering treatment.

### **Converting units.**

serum urate: A result in micromol/L is divided by 59.48 to give mg/dL:  
357 micromol/L is 6.0 mg/dL.

## **8.4 Probable rating**

The table lists every rule of this section; each is defined in full below it. The debit percentages and the decisions to decline are invented for this synthetic manual. The thresholds follow the public source named in each rule, except the edges a rule marks as this manual's own.

| <b>Rule id</b> | <b>Measure and band</b>          | <b>Probable rating</b> |
|----------------|----------------------------------|------------------------|
| UW-GOUT-001    | serum urate below 6.0 mg/dL      | no debit, +0 %         |
| UW-GOUT-002    | serum urate of 6.0 mg/dL or more | a debit of +25 %       |

### **Readings that meet no rule.**

None: every reading these measures can take falls in one of the bands above.

Note on the bands. 6 mg/dL is the treatment target in the cited guideline.

Rule UW-GOUT-001: Gout (section 8.4). Threshold: serum urate below 6.0 mg/dL. Probable rating: no debit, +0 %. Source of the threshold: ACR, Guideline for the Management of Gout (2020), the serum urate target for urate-lowering therapy.

Rule UW-GOUT-002: Gout (section 8.4). Threshold: serum urate of 6.0 mg/dL or more. Probable rating: a debit of +25 %. If the applicant also has chronic kidney disease with eGFR from 45 to below 60 mL/min/1.73 m<sup>2</sup>, see rule UW-CKD-001. Source of the threshold: ACR, Guideline for the Management of Gout (2020), the serum urate target for urate-lowering therapy.

### **What does not change the rating.**

The joint affected, visible deposits under the skin and the applicant's diet do not change the band. Nor does the number of years since diagnosis.

## **8.5 Worked examples and related rules**

Example 1. The most recent clinic letter for a joiner aged 38 with gout records serum urate 5.5 mg/dL. This is covered by rule UW-GOUT-001 (serum urate below 6.0 mg/dL). Probable rating: no debit, +0 %.

Example 2. An applicant aged 49, a bus driver, has gout. The report on file records serum urate 6.5 mg/dL. That meets rule UW-GOUT-002 (serum urate of 6.0 mg/dL or more), so the probable rating is a debit of +25 %. If the applicant also has chronic kidney disease with eGFR from 45 to below 60 mL/min/1.73 m<sup>2</sup>, rule UW-CKD-001 would apply as well; its rating is a debit of +50 %, and the two together would give +75 %.

Readings on an edge: serum urate 6.0 mg/dL exactly, rule UW-GOUT-002 (a debit of +25 %).

**Common combinations.**

Gout with reduced kidney function is the combination to look for: each worsens the other, and anti-inflammatory tablets taken for attacks harm the kidney further. Add the kidney section's debit.

Gout also keeps company with raised blood pressure and excess weight. Water tablets given for blood pressure raise urate; that explains a reading but does not excuse it.

**Related rules.**

Under rule UW-GOUT-002, if the applicant also has chronic kidney disease with eGFR from 45 to below 60 mL/min/1.73 m<sup>2</sup>, see rule UW-CKD-001 (section 27.4, Chronic kidney disease).

## 9 Hypothyroidism

### 9.1 The impairment

An underactive thyroid makes too little thyroid hormone. The usual causes are autoimmune inflammation of the gland and earlier treatment for an overactive one. Replacement with a daily tablet restores normal health.

A treated applicant with a settled TSH is a standard risk. A TSH that stays high shows the dose is too low or tablets are missed, and long neglect raises cholesterol and strains the heart.

When this section applies. An underactive thyroid has been diagnosed, treated or not.

What is measured. The rating in this section depends on TSH (mIU/L). The thresholds are grounded in: American Association of Clinical Endocrinologists and American Thyroid Association, Clinical Practice Guidelines for Hypothyroidism in Adults (2012).

### 9.2 Key questions

#### 1. What is the dose of levothyroxine, and when did it last change?

A dose changed within three months means the TSH on file may not yet reflect it.

#### 2. Why is the thyroid underactive?

After surgery for thyroid cancer the cancer, not the hormone level, is the impairment.

#### 3. When was TSH last measured?

Yearly testing is the sign of a followed-up patient.

## 9.3 Evidence and readings

Use the latest TSH from the last eighteen months. If the dose changed after that test, ask for a newer one.

A TSH below the laboratory's range on treatment suggests too high a dose; it meets no rule here, but mention it in the notes.

### Pitfalls with this evidence.

TSH takes six to eight weeks to settle after a change of dose. A result taken sooner than that reflects the old dose.

Biotin supplements interfere with some laboratory methods and can make TSH read falsely low. A surprising result in someone taking supplements should be repeated.

## 9.4 Probable rating

The table lists every rule of this section; each is defined in full below it. The debit percentages and the decisions to decline are invented for this synthetic manual. The thresholds follow the public source named in each rule, except the edges a rule marks as this manual's own.

| Rule id    | Measure and band          | Probable rating  |
|------------|---------------------------|------------------|
| UW-THY-001 | TSH below 10.0 mIU/L      | no debit, +0 %   |
| UW-THY-002 | TSH of 10.0 mIU/L or more | a debit of +25 % |

### Readings that meet no rule.

None: every reading these measures can take falls in one of the bands above.

Note on the bands. 10 mIU/L is the level above which the cited guidelines advise treatment even without symptoms.

Rule UW-THY-001: Hypothyroidism (section 9.4). Threshold: TSH below 10.0 mIU/L. Probable rating: no debit, +0 %. Source of the threshold: AACE/ATA, Clinical Practice Guidelines for Hypothyroidism in Adults (2012), the recommendation on treating a TSH above 10 mIU/L.

Rule UW-THY-002: Hypothyroidism (section 9.4). Threshold: TSH of 10.0 mIU/L or more. Probable rating: a debit of +25 %. If the applicant also has raised LDL cholesterol with LDL cholesterol from 160 to below 190 mg/dL, see rule UW-LDL-001. Source of the threshold: AACE/ATA, Clinical Practice Guidelines for Hypothyroidism in Adults (2012), the recommendation on treating a TSH above 10 mIU/L.

### **What does not change the rating.**

The dose of levothyroxine, the cause of the underactivity and the presence of thyroid antibodies do not change the band. A TSH below the laboratory's range meets the first rule and is noted.

## **9.5 Worked examples and related rules**

Example 1. Take a market gardener of 45 with an underactive thyroid. On file, from the report on file: TSH 9.2 mIU/L. The threshold met is that of rule UW-THY-001 (TSH below 10.0 mIU/L), and the probable rating is no debit, +0 %.

Example 2. The specialist's letter for a school caretaker aged 56 with an underactive thyroid records TSH 10.8 mIU/L. This is covered by rule UW-THY-002 (TSH of 10.0 mIU/L or more). Probable rating: a debit of +25 %. If the applicant also has raised LDL cholesterol with LDL cholesterol from 160 to below 190 mg/dL, rule UW-LDL-001 would apply as well; its rating is a debit of +25 %, and the two together would give +50 %.

Readings on an edge: TSH 10.0 mIU/L exactly, rule UW-THY-002 (a debit of +25 %).

**Common combinations.**

An underactive thyroid that is not replaced raises LDL cholesterol; the LDL section's debit is added, and both usually fall once the dose is right.

After treatment of an overactive thyroid with radioiodine or surgery, only the present replacement is rated. After surgery for thyroid cancer, this section still applies to the TSH, and the cancer goes to a person.

**Related rules.**

Under rule UW-THY-002, if the applicant also has raised LDL cholesterol with LDL cholesterol from 160 to below 190 mg/dL, see rule UW-LDL-001 (section 6.4, Raised LDL cholesterol).

# 10 Hypertension

## 10.1 The impairment

Hypertension is blood pressure that stays above the normal range. It rarely causes symptoms, and it is the largest single treatable cause of stroke, heart failure and kidney failure.

This section rates on the systolic reading, because in adults of insuring age it tracks risk more closely than the diastolic one. The diastolic reading has one rule of its own, for the very high level that marks a crisis.

Treatment is not penalised. An applicant whose readings are held below 130 mmHg by medicine meets no rule in this section.

When this section applies. Blood pressure readings from the last twelve months are on file, whether or not hypertension has been diagnosed or is treated.

What is measured. The rating in this section depends on systolic blood pressure (mmHg) and on diastolic blood pressure (mmHg). The thresholds are grounded in: American College of Cardiology and American Heart Association, Guideline for the Prevention, Detection, Evaluation, and Management of High Blood Pressure in Adults (2017).

## 10.2 Key questions

### 1. What were the last three readings, with their dates?

Blood pressure varies from visit to visit; the rating rests on the average of recent readings, not the best or the worst.

### 2. Which medicines are taken, and has the number risen in the last year?

A third or fourth medicine suggests pressure that resists treatment.

### **3. Has an ECG, echocardiogram or kidney test shown any effect on an organ?**

Thickening of the heart muscle or protein in the urine shows the pressure has been high for years.

### **4. Were the readings taken in a clinic, at home or over 24 hours?**

Home and 24-hour readings run a little lower than clinic readings, and a file should not mix them without saying so.

## **10.3 Evidence and readings**

Take the systolic readings recorded in the last twelve months, average them and round to a whole number. With one reading only, use it, and say in the notes that it stands alone.

If the readings fall in different bands, the average decides. A reading taken during acute pain or illness is left out when the record says so.

The diastolic rule is read on the single highest diastolic figure in the last twelve months, not on an average, because it is there to catch a crisis.

### **Pitfalls with this evidence.**

A single reading taken at an insurance examination tends to run high. Where the physician's own readings are lower and there are three or more of them, the average of the physician's readings is used.

Check that the two numbers have not been transposed or the systolic mistaken for a pulse rate. A reading such as 82/130 is a transcription error; ask for the original.

Readings taken with a cuff too small for a large arm overstate the pressure. The manual does not adjust for this.

### **Converting units.**

systolic blood pressure: A reading in kPa is multiplied by 7.5 to give mmHg.

diastolic blood pressure: A reading in kPa is multiplied by 7.5 to give mmHg.

## 10.4 Probable rating

The table lists every rule of this section; each is defined in full below it. The debit percentages and the decisions to decline are invented for this synthetic manual. The thresholds follow the public source named in each rule, except the edges a rule marks as this manual's own.

| Rule id   | Measure and band                                   | Probable rating   |
|-----------|----------------------------------------------------|-------------------|
| UW-HT-001 | systolic blood pressure from 130 to below 140 mmHg | a debit of +25 %  |
| UW-HT-002 | systolic blood pressure from 140 to below 160 mmHg | a debit of +50 %  |
| UW-HT-003 | systolic blood pressure from 160 to below 180 mmHg | a debit of +100 % |
| UW-HT-004 | systolic blood pressure of 180 mmHg or more        | decline           |
| UW-HT-005 | diastolic blood pressure of 120 mmHg or more       | decline           |

### Readings that meet no rule.

systolic blood pressure below 130 mmHg: no rule; normal or elevated pressure, or hypertension held at goal, carries no debit.

diastolic blood pressure below 120 mmHg: no rule of its own; the systolic rules decide the rating.

Note on the bands. 130, 140 and 180 mmHg systolic and 120 mmHg diastolic are the cited guideline's stage 1, stage 2 and crisis levels. The edge at 160 mmHg is this manual's own, to split stage 2. Rating on the systolic reading alone departs from the cited classification, which places a person in the higher of the stages given by the systolic and the diastolic reading; here a diastolic reading from 80 to below 120 mmHg does not raise the rating.

Rule UW-HT-001: Hypertension (section 10.4). Threshold: systolic blood pressure from 130 to below 140 mmHg. Probable rating: a debit of +25 %. Stage 1 by the systolic reading. Source of the threshold: ACC/AHA, Guideline for the Prevention, Detection, Evaluation, and Management of High Blood Pressure in Adults (2017), the table of blood pressure categories and the section on hypertensive crises.

Rule UW-HT-002: Hypertension (section 10.4). Threshold: systolic blood pressure from 140 to below 160 mmHg. Probable rating: a debit of +50 %. The lower part of stage 2. If the applicant also has a history of smoking with smoking status: current smoker, see rule UW-TOB-001. Source of the edge at 140 mmHg: ACC/AHA, Guideline for the Prevention, Detection, Evaluation, and Management of High Blood Pressure in Adults (2017), the table of blood pressure categories and the section on hypertensive crises. The edge at 160 mmHg is this manual's own.

Rule UW-HT-003: Hypertension (section 10.4). Threshold: systolic blood pressure from 160 to below 180 mmHg. Probable rating: a debit of +100 %. The upper part of stage 2. If the applicant also has chronic kidney disease with eGFR from 45 to below 60 mL/min/1.73 m<sup>2</sup>, see rule UW-CKD-001. Source of the edge at 180 mmHg: ACC/AHA, Guideline for the Prevention, Detection, Evaluation, and Management of High Blood Pressure in Adults (2017), the table of blood pressure categories and the section on hypertensive crises. The edge at 160 mmHg is this manual's own.

Rule UW-HT-004: Hypertension (section 10.4). Threshold: systolic blood pressure of 180 mmHg or more. Probable rating: decline. If the record also shows diastolic blood pressure of 120 mmHg or more, see rule UW-HT-005. Source of the threshold: ACC/AHA, Guideline for the Prevention, Detection, Evaluation, and Management of High Blood Pressure in Adults (2017), the table of blood pressure categories and the section on hypertensive crises.

Rule UW-HT-005: Hypertension (section 10.4). Threshold: diastolic blood pressure of 120 mmHg or more. Probable rating: decline. A diastolic reading below 120 mmHg does not change the rating given by the systolic rules. If the record also shows systolic blood pressure of 180 mmHg or more, see rule UW-HT-004. Source of the threshold: ACC/AHA, Guideline for the Prevention, Detection, Evaluation, and Management of High Blood Pressure in Adults (2017), the table of blood pressure categories and the section on hypertensive crises.

**What does not change the rating.**

The number and names of the medicines do not change the band, and neither does the number of years since diagnosis. A diagnosis of hypertension with readings held below 130 mmHg meets no rule.

White-coat hypertension confirmed by 24-hour monitoring is rated on the monitored average.

**10.5 Worked examples and related rules**

Example 1. An applicant aged 52, a pastry cook, has hypertension. The specialist's letter records systolic blood pressure 135 mmHg. That meets rule UW-HT-001 (systolic blood pressure from 130 to below 140 mmHg), so the probable rating is a debit of +25 %.

Example 2. Take a joiner of 33 with hypertension. On file, from the attending physician's statement: systolic blood pressure 150 mmHg. The threshold met is that of rule UW-HT-002 (systolic blood pressure from 140 to below 160 mmHg), and the probable rating is a debit of +50 %. If the applicant also has a history of smoking with smoking status: current smoker, rule UW-TOB-001 would apply as well; its rating is a debit of +50 %, and the two together would give +100 %.

Example 3. The most recent clinic letter for a bus driver aged 44 with hypertension records systolic blood pressure 170 mmHg. This is covered by rule UW-HT-003 (systolic blood pressure from 160 to below 180 mmHg). Probable rating: a debit of +100 %. If the applicant also has chronic kidney disease with eGFR from 45 to below 60 mL/min/1.73 m<sup>2</sup>, rule UW-CKD-001 would apply as well; its rating is a debit of +50 %, and the two together would give +150 %.

Example 4. An applicant aged 55, a payroll clerk, has hypertension. The report on file records systolic blood pressure 194 mmHg. That meets rule UW-HT-004 (systolic blood pressure of 180 mmHg or more), so the probable rating is decline. If the record also shows diastolic blood pressure of 120 mmHg or more, rule UW-HT-005 would apply as well; its rating is decline, and the outcome would be decline.

Example 5. Take a librarian of 36 with hypertension. On file, from the specialist's letter: diastolic blood pressure 130 mmHg. The threshold met is that of rule UW-HT-005 (diastolic blood pressure of 120 mmHg or more), and the probable rating is decline. If the record also shows systolic blood pressure of 180 mmHg or more, rule UW-HT-004 would apply as well; its rating is decline, and the outcome would be decline.

Readings on an edge: systolic blood pressure 130 mmHg exactly, rule UW-HT-001 (a debit of +25 %); systolic blood pressure 140 mmHg exactly, rule UW-HT-002 (a debit of +50 %); systolic blood pressure 160 mmHg exactly, rule UW-HT-003 (a debit of +100 %); systolic blood pressure 180 mmHg exactly, rule UW-HT-004 (decline); diastolic blood pressure 120 mmHg exactly, rule UW-HT-005 (decline).

### **Common combinations.**

Hypertension is the commonest partner of every other cardiovascular and kidney impairment. Its debit is added to the debit for diabetes, smoking, kidney disease or a past stroke; it is never treated as already included in them.

With reduced kidney function, pressure that is not controlled speeds the loss of the kidney. With sleep apnoea, pressure often resists treatment until the apnoea is treated.

**Related rules.**

Under rule UW-HT-002, if the applicant also has a history of smoking with smoking status: current smoker, see rule UW-TOB-001 (section 21.4, Tobacco use).

Under rule UW-HT-003, if the applicant also has chronic kidney disease with eGFR from 45 to below 60 mL/min/1.73 m<sup>2</sup>, see rule UW-CKD-001 (section 27.4, Chronic kidney disease).

Under rule UW-HT-004, if the record also shows diastolic blood pressure of 120 mmHg or more, see rule UW-HT-005 (section 10.4, Hypertension).

Under rule UW-HT-005, if the record also shows systolic blood pressure of 180 mmHg or more, see rule UW-HT-004 (section 10.4, Hypertension).

Rules of other sections that point to this one: rule UW-DM-002 (section 2.4, Type 2 diabetes mellitus); rule UW-LDL-001 (section 6.4, Raised LDL cholesterol); rule UW-HF-003 (section 12.4, Heart failure); rule UW-AAA-002 (section 16.4, Abdominal aortic aneurysm); rule UW-OSA-002 (section 26.4, Obstructive sleep apnoea); rule UW-CKD-002 (section 27.4, Chronic kidney disease).

# 11 Myocardial infarction

## 11.1 The impairment

A myocardial infarction, or heart attack, is the death of heart muscle after a coronary artery is blocked. Survival has improved greatly, but the risk of another event and of sudden death is highest in the first year and never returns fully to normal.

This section rates by the time that has passed. How much muscle was lost is rated separately through the ejection fraction.

When this section applies. A heart attack is recorded in the history.

What is measured. The rating in this section depends on time since the heart attack (months). The thresholds are grounded in: American College of Cardiology and American Heart Association, Guideline Focused Update on Duration of Dual Antiplatelet Therapy in Patients With Coronary Artery Disease (2016).

## 11.2 Key questions

### 1. On what date did the heart attack happen, and was there more than one?

The clock in the table starts at the most recent event.

### 2. Was a stent placed or bypass surgery done?

Restored blood flow improves the outlook, and the procedure note usually states how many arteries are diseased.

### 3. What was the ejection fraction at the latest echocardiogram?

It measures the damage left behind and is the strongest guide to long-term survival.

### 4. Has chest pain returned, and is the applicant still smoking?

Either one marks a course that is not settled.

## 11.3 Evidence and readings

Take the date of the event from the hospital discharge letter or the attending physician's statement, and count whole months to the date of the application.

If there is no echocardiogram after the event, ask for one before offering terms beyond twelve months.

### Pitfalls with this evidence.

The words angina, acute coronary syndrome and unstable angina are not the same as a heart attack. Look for the diagnosis in the discharge letter; a raised troponin with the word infarction settles it.

A stent placed for stable narrowing without any heart attack is not rated here and goes to a person.

## 11.4 Probable rating

The table lists every rule of this section; each is defined in full below it. The debit percentages and the decisions to decline are invented for this synthetic manual. The thresholds follow the public source named in each rule, except the edges a rule marks as this manual's own.

| Rule id   | Measure and band                                       | Probable rating           |
|-----------|--------------------------------------------------------|---------------------------|
| UW-MI-001 | time since the heart attack below 12 months            | decline as a postponement |
| UW-MI-002 | time since the heart attack from 12 to below 60 months | a debit of +100 %         |
| UW-MI-003 | time since the heart attack of 60 months or more       | a debit of +50 %          |

### Readings that meet no rule.

None: every reading these measures can take falls in one of the bands above.

Note on the bands. Twelve months matches the period for which the cited guideline advises two antiplatelet medicines after an acute coronary syndrome. The edge at 60 months is this manual's own.

Rule UW-MI-001: Myocardial infarction (section 11.4). Threshold: time since the heart attack below 12 months. Probable rating: decline as a postponement. This is a postponement: the applicant may apply again after twelve months. Source of the threshold: ACC/AHA, Guideline Focused Update on Duration of Dual Antiplatelet Therapy in Patients With Coronary Artery Disease (2016), the recommendations for at least 12 months of treatment after an acute coronary syndrome.

Rule UW-MI-002: Myocardial infarction (section 11.4). Threshold: time since the heart attack from 12 to below 60 months. Probable rating: a debit of +100 %. If the applicant also has heart failure with left ventricular ejection fraction above 40 and below 50 %, see rule UW-HF-002. Source of the edge at 12 months: ACC/AHA, Guideline Focused Update on Duration of Dual Antiplatelet Therapy in Patients With Coronary Artery Disease (2016), the recommendations for at least 12 months of treatment after an acute coronary syndrome. The edge at 60 months is this manual's own.

Rule UW-MI-003: Myocardial infarction (section 11.4). Threshold: time since the heart attack of 60 months or more. Probable rating: a debit of +50 %. If the applicant also has raised LDL cholesterol with LDL cholesterol from 160 to below 190 mg/dL, see rule UW-LDL-001. The edge at 60 months is this manual's own and no guideline supplies it. The source for the measure and the other edges of this section: ACC/AHA, Guideline Focused Update on Duration of Dual Antiplatelet Therapy in Patients With Coronary Artery Disease (2016), the recommendations for at least 12 months of treatment after an acute coronary syndrome.

**What does not change the rating.**

Whether the artery was opened with a stent, bypassed or treated with medicines alone does not change the band. Nor does the part of the heart affected, or a normal exercise test afterwards.

**11.5 Worked examples and related rules**

Example 1. The attending physician's statement for a florist aged 59 with a past heart attack records time since the heart attack 11 months. This is covered by rule UW-MI-001 (time since the heart attack below 12 months). Probable rating: decline as a postponement.

Example 2. An applicant aged 40, a market gardener, has a past heart attack. The most recent clinic letter records time since the heart attack 36 months. That meets rule UW-MI-002 (time since the heart attack from 12 to below 60 months), so the probable rating is a debit of +100 %. If the applicant also has heart failure with left ventricular ejection fraction above 40 and below 50 %, rule UW-HF-002 would apply as well; its rating is a debit of +150 %, and the two together would give +250 %.

Example 3. Take a school caretaker of 51 with a past heart attack. On file, from the report on file: time since the heart attack 65 months. The threshold met is that of rule UW-MI-003 (time since the heart attack of 60 months or more), and the probable rating is a debit of +50 %. If the applicant also has raised LDL cholesterol with LDL cholesterol from 160 to below 190 mg/dL, rule UW-LDL-001 would apply as well; its rating is a debit of +25 %, and the two together would give +75 %.

Readings on an edge: time since the heart attack 12 months exactly, rule UW-MI-002 (a debit of +100 %); time since the heart attack 60 months exactly, rule UW-MI-003 (a debit of +50 %).

**Common combinations.**

The ejection fraction measured after the event decides most of the long-term outlook. Below 50 %, the heart failure section's debit is added to the debit here.

Continued smoking after a heart attack, an LDL cholesterol still at 160 mg/dL or more and diabetes are each added from their own sections. A second heart attack restarts the clock and goes to a person.

**Related rules.**

Under rule UW-MI-002, if the applicant also has heart failure with left ventricular ejection fraction above 40 and below 50 %, see rule UW-HF-002 (section 12.4, Heart failure).

Under rule UW-MI-003, if the applicant also has raised LDL cholesterol with LDL cholesterol from 160 to below 190 mg/dL, see rule UW-LDL-001 (section 6.4, Raised LDL cholesterol).

Rules of other sections that point to this one: rule UW-LDL-002 (section 6.4, Raised LDL cholesterol).

# 12 Heart failure

## 12.1 The impairment

Heart failure means the heart cannot pump enough blood for the body's needs without raised filling pressures. Breathlessness, tiredness and swollen ankles are the usual complaints.

It is always a serious impairment. The ejection fraction separates the forms with the worst outlook from those that can be considered at a heavy rating.

When this section applies. Heart failure has been diagnosed.

What is measured. The rating in this section depends on left ventricular ejection fraction (%). The thresholds are grounded in: American Heart Association, American College of Cardiology and Heart Failure Society of America, Guideline for the Management of Heart Failure (2022).

## 12.2 Key questions

### 1. What caused the heart failure?

Failure after a heart attack, from a valve, from alcohol and from high blood pressure have different courses.

### 2. What was the ejection fraction on the latest echocardiogram, and on the one before?

A fraction that has recovered on treatment is better than the same figure reached on the way down.

### 3. Has there been a hospital admission for heart failure in the last twelve months?

An admission is the strongest short-term marker of death in this condition.

#### 4. How far can the applicant walk on the flat without stopping?

Everyday capacity adds to what the echocardiogram shows.

### 12.3 Evidence and readings

Use the ejection fraction from an echocardiogram or cardiac MRI in the last twelve months. Where the report gives a range, such as 45 to 50 %, take the lower figure.

An admission for heart failure in the last twelve months is outside these bands and goes to a person, whatever the ejection fraction.

#### Pitfalls with this evidence.

Ejection fraction varies by a few points between scans and between methods. A change from 48 to 51 % is within that noise; where two recent scans straddle an edge, use the lower.

A low ejection fraction found by chance, with no diagnosis of heart failure, is not rated here and goes to a person.

### 12.4 Probable rating

The table lists every rule of this section; each is defined in full below it. The debit percentages and the decisions to decline are invented for this synthetic manual. The thresholds follow the public source named in each rule, except the edges a rule marks as this manual's own.

| Rule id   | Measure and band                                           | Probable rating   |
|-----------|------------------------------------------------------------|-------------------|
| UW-HF-001 | left ventricular ejection fraction of 40 % or less         | decline           |
| UW-HF-002 | left ventricular ejection fraction above 40 and below 50 % | a debit of +150 % |
| UW-HF-003 | left ventricular ejection fraction of 50 % or more         | a debit of +75 %  |

**Readings that meet no rule.**

None: every reading these measures can take falls in one of the bands above.

Note on the bands. The edges at 40 % and 50 % are the cited guideline's classes: reduced, mildly reduced and preserved ejection fraction.

Rule UW-HF-001: Heart failure (section 12.4). Threshold: left ventricular ejection fraction of 40 % or less. Probable rating: decline. Source of the threshold: AHA/ACC/HFSA, Guideline for the Management of Heart Failure (2022), the classification of heart failure by left ventricular ejection fraction.

Rule UW-HF-002: Heart failure (section 12.4). Threshold: left ventricular ejection fraction above 40 and below 50 %. Probable rating: a debit of +150 %. If the applicant also has atrial fibrillation with CHA2DS2-VASc score from 2 to below 4 points, see rule UW-AF-002. Source of the threshold: AHA/ACC/HFSA, Guideline for the Management of Heart Failure (2022), the classification of heart failure by left ventricular ejection fraction.

Rule UW-HF-003: Heart failure (section 12.4). Threshold: left ventricular ejection fraction of 50 % or more. Probable rating: a debit of +75 %. If the applicant also has hypertension with systolic blood pressure from 140 to below 160 mmHg, see rule UW-HT-002. Source of the threshold: AHA/ACC/HFSA, Guideline for the Management of Heart Failure (2022), the classification of heart failure by left ventricular ejection fraction.

**What does not change the rating.**

The names and doses of the medicines do not change the band. An implanted defibrillator or pacemaker does not change it either, but must be mentioned to the person reviewing the case.

## 12.5 Worked examples and related rules

Example 1. Take a dental technician of 36 with heart failure. On file, from the most recent clinic letter: left ventricular ejection fraction 37 %. The threshold met is that of rule UW-HF-001 (left ventricular ejection fraction of 40 % or less), and the probable rating is decline.

Example 2. The report on file for a pastry cook aged 47 with heart failure records left ventricular ejection fraction 45 %. This is covered by rule UW-HF-002 (left ventricular ejection fraction above 40 and below 50 %). Probable rating: a debit of +150 %. If the applicant also has atrial fibrillation with CHA2DS2-VASc score from 2 to below 4 points, rule UW-AF-002 would apply as well; its rating is a debit of +50 %, and the two together would give +200 %.

Example 3. An applicant aged 58, a joiner, has heart failure. The specialist's letter records left ventricular ejection fraction 54 %. That meets rule UW-HF-003 (left ventricular ejection fraction of 50 % or more), so the probable rating is a debit of +75 %. If the applicant also has hypertension with systolic blood pressure from 140 to below 160 mmHg, rule UW-HT-002 would apply as well; its rating is a debit of +50 %, and the two together would give +125 %.

Readings on an edge: left ventricular ejection fraction 40 % exactly, rule UW-HF-001 (decline); left ventricular ejection fraction 50 % exactly, rule UW-HF-003 (a debit of +75 %).

### Common combinations.

Atrial fibrillation appears in a large share of people with heart failure and worsens the outlook: add its debit.

Heart failure after a heart attack is rated in both sections. Raised blood pressure and reduced kidney function are added as usual; a falling eGFR in heart failure is a poor sign.

**Related rules.**

Under rule UW-HF-002, if the applicant also has atrial fibrillation with CHA2DS2-VASc score from 2 to below 4 points, see rule UW-AF-002 (section 13.4, Atrial fibrillation).

Under rule UW-HF-003, if the applicant also has hypertension with systolic blood pressure from 140 to below 160 mmHg, see rule UW-HT-002 (section 10.4, Hypertension).

Rules of other sections that point to this one: rule UW-MI-002 (section 11.4, Myocardial infarction); rule UW-AF-003 (section 13.4, Atrial fibrillation); rule UW-AS-002 (section 14.4, Aortic stenosis).

## 13 Atrial fibrillation

### 13.1 The impairment

In atrial fibrillation the upper chambers of the heart quiver instead of beating. The pulse is irregular, and blood pooling in the atria can clot and travel to the brain.

Stroke is the main danger, so the section rates on the standard stroke risk score rather than on how often the rhythm occurs.

When this section applies. Atrial fibrillation or atrial flutter has been recorded on an ECG.

What is measured. The rating in this section depends on CHA<sub>2</sub>DS<sub>2</sub>-VASc score (points). The thresholds are grounded in: American College of Cardiology, American Heart Association, American College of Clinical Pharmacy and Heart Rhythm Society, Guideline for the Diagnosis and Management of Atrial Fibrillation (2023).

### 13.2 Key questions

#### 1. Is the rhythm occasional, persistent or permanent?

The pattern shapes treatment, though the stroke risk is much the same for all three.

#### 2. Is an anticoagulant taken, and which?

An applicant with a score of 2 or more who takes none is not following usual advice, and the reason should be on file.

#### 3. Which items of the score apply: heart failure, hypertension, age, diabetes, stroke, vascular disease, sex?

The underwriter should be able to rebuild the score from the file rather than accept a bare number.

#### **4. Has an ablation been done, and has the rhythm returned since?**

A successful ablation eases symptoms but does not remove the stroke score.

### **13.3 Evidence and readings**

Use the score the cardiologist recorded if the file shows how it was built; otherwise work it out from the history. Count one point each for heart failure, hypertension, diabetes, vascular disease, age 65 to 74 and female sex, and two each for age 75 or more and for a previous stroke.

A single episode with a clear, removed cause, such as an overactive thyroid since treated, and no recurrence for two years meets no rule here.

#### **Pitfalls with this evidence.**

Clinic letters often quote a score without its parts, and some quote an older version of the score. Rebuild it from the history.

Female sex counts one point in the score. A woman with no other item has a score of 1 and meets the lowest rule.

### **13.4 Probable rating**

The table lists every rule of this section; each is defined in full below it. The debit percentages and the decisions to decline are invented for this synthetic manual. The thresholds follow the public source named in each rule, except the edges a rule marks as this manual's own.

| Rule id   | Measure and band                            | Probable rating   |
|-----------|---------------------------------------------|-------------------|
| UW-AF-001 | CHA2DS2-VASc score below 2 points           | a debit of +25 %  |
| UW-AF-002 | CHA2DS2-VASc score from 2 to below 4 points | a debit of +50 %  |
| UW-AF-003 | CHA2DS2-VASc score of 4 points or more      | a debit of +100 % |

**Readings that meet no rule.**

None: every reading these measures can take falls in one of the bands above.

Note on the bands. A score of 2 is the level at which the cited guideline recommends an anticoagulant for men; the edge at 4 is this manual's own.

Rule UW-AF-001: Atrial fibrillation (section 13.4). Threshold: CHA2DS2-VASc score below 2 points. Probable rating: a debit of +25 %. Source of the threshold: ACC/AHA/ACCP/HRS, Guideline for the Diagnosis and Management of Atrial Fibrillation (2023), the recommendations on stroke risk scores and anticoagulation.

Rule UW-AF-002: Atrial fibrillation (section 13.4). Threshold: CHA2DS2-VASc score from 2 to below 4 points. Probable rating: a debit of +50 %. If the applicant also has a past stroke with NIHSS score at admission below 6 points, see rule UW-CVA-001. Source of the edge at 2 points: ACC/AHA/ACCP/HRS, Guideline for the Diagnosis and Management of Atrial Fibrillation (2023), the recommendations on stroke risk scores and anticoagulation. The edge at 4 points is this manual's own.

Rule UW-AF-003: Atrial fibrillation (section 13.4). Threshold: CHA2DS2-VASc score of 4 points or more. Probable rating: a debit of +100 %. If the applicant also has heart failure with left ventricular ejection fraction above 40 and below 50 %, see rule UW-HF-002. The edge at 4 points is this manual's own and no guideline supplies it. The source for the measure and the other edges of this section: ACC/AHA/ACCP/HRS, Guideline for the Diagnosis and Management of Atrial Fibrillation (2023), the recommendations on stroke risk scores and anticoagulation.

### **What does not change the rating.**

Whether the rhythm is occasional, persistent or permanent does not change the band, and neither does the choice between rate control and rhythm control. Taking an anticoagulant does not lower the score.

## **13.5 Worked examples and related rules**

Example 1. An applicant aged 43, a librarian, has atrial fibrillation. The report on file records CHA2DS2-VASc score 1 points. That meets rule UW-AF-001 (CHA2DS2-VASc score below 2 points), so the probable rating is a debit of +25 %.

Example 2. Take a florist of 54 with atrial fibrillation. On file, from the specialist's letter: CHA2DS2-VASc score 3 points. The threshold met is that of rule UW-AF-002 (CHA2DS2-VASc score from 2 to below 4 points), and the probable rating is a debit of +50 %. If the applicant also has a past stroke with NIHSS score at admission below 6 points, rule UW-CVA-001 would apply as well; its rating is a debit of +75 %, and the two together would give +125 %.

Example 3. The attending physician's statement for a market gardener aged 35 with atrial fibrillation records CHA2DS2-VASc score 5 points. This is covered by rule UW-AF-003 (CHA2DS2-VASc score of 4 points or more). Probable rating: a debit of +100 %. If the applicant also has heart failure with left ventricular ejection fraction above 40 and below 50 %, rule UW-HF-002 would apply as well; its rating is a debit of +150 %, and the two together would give +250 %.

Readings on an edge: CHA2DS2-VASc score 2 points exactly, rule UW-AF-002 (a debit of +50 %); CHA2DS2-VASc score 4 points exactly, rule UW-AF-003 (a debit of +100 %).

### **Common combinations.**

The items of the score are themselves impairments. Hypertension, diabetes, heart failure and a past stroke each raise the score here and are also rated in their own sections; both effects stand, and the debits are added.

Atrial fibrillation with a narrowed mitral valve or a mechanical heart valve is outside this section and goes to a person.

### **Related rules.**

Under rule UW-AF-002, if the applicant also has a past stroke with NIHSS score at admission below 6 points, see rule UW-CVA-001 (section 19.4, Stroke).

Under rule UW-AF-003, if the applicant also has heart failure with left ventricular ejection fraction above 40 and below 50 %, see rule UW-HF-002 (section 12.4, Heart failure).

Rules of other sections that point to this one: rule UW-HF-002 (section 12.4, Heart failure); rule UW-HCM-001 (section 15.4, Hypertrophic cardiomyopathy); rule UW-CVA-001 (section 19.4, Stroke).

# 14 Aortic stenosis

## 14.1 The impairment

Aortic stenosis is a narrowing of the valve through which the heart empties into the aorta. It progresses slowly for years, then quickly once chest pain, fainting or breathlessness appear.

Echocardiography grades the narrowing by how fast blood must travel to get through the valve.

When this section applies. Aortic stenosis has been reported on an echocardiogram.

What is measured. The rating in this section depends on peak aortic jet velocity (m/s). The thresholds are grounded in: American College of Cardiology and American Heart Association, Guideline for the Management of Patients With Valvular Heart Disease (2020).

## 14.2 Key questions

### 1. What did the latest echocardiogram report as the peak velocity?

It is the figure this section rates on, and it should be less than twelve months old.

### 2. Does the applicant have chest pain, fainting or breathlessness on effort?

Symptoms with severe narrowing mean the valve needs replacing now.

### 3. Is the valve bicuspid?

A two-leaflet valve narrows earlier in life and may come with a widened aorta.

## 14.3 Evidence and readings

Take the peak aortic jet velocity from the echocardiogram report. If only the valve area or mean gradient is reported, ask the cardiologist for the velocity.

A replaced valve is outside these bands and goes to a person with the operation note.

### Pitfalls with this evidence.

Velocity understates the narrowing when the heart is pumping weakly. Where the ejection fraction is below 50 % and the valve area is reported as 1.0 cm<sup>2</sup> or less, send the case to a person whatever the velocity.

A murmur described as aortic sclerosis, with a velocity under 2.0 m/s, is not stenosis.

## 14.4 Probable rating

The table lists every rule of this section; each is defined in full below it. The debit percentages and the decisions to decline are invented for this synthetic manual. The thresholds follow the public source named in each rule, except the edges a rule marks as this manual's own.

| Rule id   | Measure and band                                   | Probable rating           |
|-----------|----------------------------------------------------|---------------------------|
| UW-AS-001 | peak aortic jet velocity from 2.0 to below 3.0 m/s | a debit of +25 %          |
| UW-AS-002 | peak aortic jet velocity from 3.0 to below 4.0 m/s | a debit of +100 %         |
| UW-AS-003 | peak aortic jet velocity of 4.0 m/s or more        | decline as a postponement |

### Readings that meet no rule.

peak aortic jet velocity below 2.0 m/s: no rule; thickening of the valve without narrowing.

Note on the bands. The edges at 2.0, 3.0 and 4.0 m/s are the cited guideline's mild, moderate and severe grades.

Rule UW-AS-001: Aortic stenosis (section 14.4). Threshold: peak aortic jet velocity from 2.0 to below 3.0 m/s. Probable rating: a debit of +25 %. Source of the threshold: ACC/AHA, Guideline for the Management of Patients With Valvular Heart Disease (2020), the stages of aortic stenosis by peak aortic jet velocity.

Rule UW-AS-002: Aortic stenosis (section 14.4). Threshold: peak aortic jet velocity from 3.0 to below 4.0 m/s. Probable rating: a debit of +100 %. If the applicant also has heart failure with left ventricular ejection fraction above 40 and below 50 %, see rule UW-HF-002. Source of the threshold: ACC/AHA, Guideline for the Management of Patients With Valvular Heart Disease (2020), the stages of aortic stenosis by peak aortic jet velocity.

Rule UW-AS-003: Aortic stenosis (section 14.4). Threshold: peak aortic jet velocity of 4.0 m/s or more. Probable rating: decline as a postponement. The applicant may apply again twelve months after a valve replacement. Source of the threshold: ACC/AHA, Guideline for the Management of Patients With Valvular Heart Disease (2020), the stages of aortic stenosis by peak aortic jet velocity.

### **What does not change the rating.**

The cause of the narrowing and the applicant's age do not change the band. An absence of symptoms does not lower it.

## **14.5 Worked examples and related rules**

Example 1. The specialist's letter for a surveyor aged 50 with aortic stenosis records peak aortic jet velocity 2.5 m/s. This is covered by rule UW-AS-001 (peak aortic jet velocity from 2.0 to below 3.0 m/s). Probable rating: a debit of +25 %.

Example 2. An applicant aged 31, a dental technician, has aortic stenosis. The attending physician's statement records peak aortic jet velocity 3.5 m/s. That meets rule UW-AS-002 (peak aortic jet velocity from 3.0 to below 4.0 m/s), so the probable rating is a debit of +100 %. If the applicant also has heart failure with left ventricular ejection fraction above 40 and below 50 %, rule UW-HF-002 would apply as well; its rating is a debit of +150 %, and the two together would give +250 %.

Example 3. Take a pastry cook of 42 with aortic stenosis. On file, from the most recent clinic letter: peak aortic jet velocity 4.3 m/s. The threshold met is that of rule UW-AS-003 (peak aortic jet velocity of 4.0 m/s or more), and the probable rating is decline as a postponement.

Readings on an edge: peak aortic jet velocity 2.0 m/s exactly, rule UW-AS-001 (a debit of +25 %); peak aortic jet velocity 3.0 m/s exactly, rule UW-AS-002 (a debit of +100 %); peak aortic jet velocity 4.0 m/s exactly, rule UW-AS-003 (decline as a postponement).

### **Common combinations.**

A narrowed valve makes the heart muscle thicken and, late on, weaken. An ejection fraction below 50 % with aortic stenosis is rated in the heart failure section as well, and the debits are added.

Coronary artery disease often coexists because the two share causes. A past heart attack is added from its own section.

### **Related rules.**

Under rule UW-AS-002, if the applicant also has heart failure with left ventricular ejection fraction above 40 and below 50 %, see rule UW-HF-002 (section 12.4, Heart failure).

# 15 Hypertrophic cardiomyopathy

## 15.1 The impairment

Hypertrophic cardiomyopathy is an inherited thickening of the heart muscle that is not explained by blood pressure or a valve. Many people with it live a normal span; a few die suddenly from a rhythm disturbance.

The thickness of the wall is one of the recognised markers of that sudden risk and is the measure used here.

When this section applies. Hypertrophic cardiomyopathy has been diagnosed by a cardiologist.

What is measured. The rating in this section depends on maximum left ventricular wall thickness (mm). The thresholds are grounded in: American Heart Association and American College of Cardiology, Guideline for the Diagnosis and Treatment of Patients With Hypertrophic Cardiomyopathy (2020).

## 15.2 Key questions

### 1. Has a close relative died suddenly before the age of 50?

Family history of sudden death is an independent risk marker.

### 2. Has the applicant fainted without explanation?

An unexplained faint may have been a rhythm disturbance that stopped by itself.

### 3. Has a defibrillator been implanted or advised?

Advice to implant one tells the underwriter how the cardiologist judges the risk.

## 15.3 Evidence and readings

Use the maximum wall thickness from the latest echocardiogram or cardiac MRI; where both exist, the MRI figure is preferred.

An implanted defibrillator, or one advised and refused, takes the case to a person.

### Pitfalls with this evidence.

Thickening from long-standing high blood pressure or from athletic training can look similar on a scan. Only the cardiologist's diagnosis brings a case into this section.

A gene found on family screening, with a normal scan, is not the disease. It meets no rule and goes to a person.

### Converting units.

maximum left ventricular wall thickness: A thickness in cm is multiplied by ten to give mm.

## 15.4 Probable rating

The table lists every rule of this section; each is defined in full below it. The debit percentages and the decisions to decline are invented for this synthetic manual. The thresholds follow the public source named in each rule, except the edges a rule marks as this manual's own.

| Rule id    | Measure and band                                                  | Probable rating   |
|------------|-------------------------------------------------------------------|-------------------|
| UW-HCM-001 | maximum left ventricular wall thickness<br>from 15 to below 30 mm | a debit of +100 % |
| UW-HCM-002 | maximum left ventricular wall thickness<br>of 30 mm or more       | decline           |

### Readings that meet no rule.

maximum left ventricular wall thickness below 15 mm: below the diagnostic thickness; no rule; send to a person if the diagnosis is stated all the same.

Note on the bands. 15 mm is the diagnostic thickness and 30 mm the marker of high sudden-death risk in the cited guideline.

Rule UW-HCM-001: Hypertrophic cardiomyopathy (section 15.4). Threshold: maximum left ventricular wall thickness from 15 to below 30 mm. Probable rating: a debit of +100 %. If the applicant also has atrial fibrillation with CHA2DS2-VASc score below 2 points, see rule UW-AF-001. Source of the threshold: AHA/ACC, Guideline for the Diagnosis and Treatment of Patients With Hypertrophic Cardiomyopathy (2020), the diagnostic wall thickness and the risk markers for sudden cardiac death.

Rule UW-HCM-002: Hypertrophic cardiomyopathy (section 15.4). Threshold: maximum left ventricular wall thickness of 30 mm or more. Probable rating: decline. Source of the threshold: AHA/ACC, Guideline for the Diagnosis and Treatment of Patients With Hypertrophic Cardiomyopathy (2020), the diagnostic wall thickness and the risk markers for sudden cardiac death.

### **What does not change the rating.**

Obstruction to outflow, the gene involved and the medicines taken do not change the band.

## **15.5 Worked examples and related rules**

Example 1. Take a payroll clerk of 57 with hypertrophic cardiomyopathy. On file, from the attending physician's statement: maximum left ventricular wall thickness 22 mm. The threshold met is that of rule UW-HCM-001 (maximum left ventricular wall thickness from 15 to below 30 mm), and the probable rating is a debit of +100 %. If the applicant also has atrial fibrillation with CHA2DS2-VASc score below 2 points, rule UW-AF-001 would apply as well; its rating is a debit of +25 %, and the two together would give +125 %.

Example 2. The most recent clinic letter for a librarian aged 38 with hypertrophic cardiomyopathy records maximum left ventricular wall thickness 32 mm. This is covered by rule UW-HCM-002 (maximum left ventricular wall thickness of 30 mm or more). Probable rating: decline.

Readings on an edge: maximum left ventricular wall thickness 15 mm exactly, rule UW-HCM-001 (a debit of +100 %); maximum left ventricular wall thickness 30 mm exactly, rule UW-HCM-002 (decline).

**Common combinations.**

Atrial fibrillation is common in this condition and carries a high stroke risk whatever the score: its debit is added.

Heart failure late in the disease is rated in its own section and added.

**Related rules.**

Under rule UW-HCM-001, if the applicant also has atrial fibrillation with CHA2DS2-VASc score below 2 points, see rule UW-AF-001 (section 13.4, Atrial fibrillation).

# 16 Abdominal aortic aneurysm

## 16.1 The impairment

An abdominal aortic aneurysm is a ballooning of the body's main artery below the kidneys. It gives no warning. The risk of rupture, which is usually fatal, climbs steeply with diameter.

Small aneurysms are watched with ultrasound and large ones are repaired. The rating follows the same sizes that set the watching interval.

When this section applies. An abdominal aortic aneurysm has been found on imaging.

What is measured. The rating in this section depends on abdominal aortic diameter (cm). The thresholds are grounded in: Society for Vascular Surgery, Practice guidelines on the care of patients with an abdominal aortic aneurysm (2018).

## 16.2 Key questions

### 1. What was the diameter at the last scan, and how much had it grown in a year?

Growth of more than 1 cm in a year is itself a reason for repair.

### 2. Is the applicant enrolled in regular surveillance?

An aneurysm found once and never rescanned is an unknown size today.

### 3. Does the applicant smoke?

Smoking speeds growth and raises the chance of rupture.

## 16.3 Evidence and readings

Use the diameter from an ultrasound or CT in the last twelve months, or in the last six months once the diameter has reached 5.0 cm.

A repaired aneurysm is outside these bands and goes to a person with the operation note and the latest follow-up scan.

### Pitfalls with this evidence.

Ultrasound and CT differ by a few millimetres, and CT usually reads larger. Compare like with like when judging growth.

Reports sometimes give the diameter of the channel that carries blood rather than the outer wall. The outer diameter is the one that is rated.

### Converting units.

abdominal aortic diameter: A diameter in mm is divided by ten to give cm.

## 16.4 Probable rating

The table lists every rule of this section; each is defined in full below it. The debit percentages and the decisions to decline are invented for this synthetic manual. The thresholds follow the public source named in each rule, except the edges a rule marks as this manual's own.

| Rule id    | Measure and band                                   | Probable rating   |
|------------|----------------------------------------------------|-------------------|
| UW-AAA-001 | abdominal aortic diameter from 3.0 to below 4.0 cm | a debit of +50 %  |
| UW-AAA-002 | abdominal aortic diameter from 4.0 to below 5.5 cm | a debit of +150 % |
| UW-AAA-003 | abdominal aortic diameter of 5.5 cm or more        | decline           |

**Readings that meet no rule.**

abdominal aortic diameter below 3.0 cm: no rule; not an aneurysm.

Note on the bands. 3.0 cm defines an aneurysm and 5.5 cm is the size for planned repair in the cited guidelines; 4.0 cm is where they shorten the surveillance interval.

Rule UW-AAA-001: Abdominal aortic aneurysm (section 16.4).

Threshold: abdominal aortic diameter from 3.0 to below 4.0 cm.

Probable rating: a debit of +50 %. If the applicant also has a history of smoking with smoking status: current smoker, see rule UW-TOB-001.

Source of the threshold: SVS, Practice guidelines on the care of patients with an abdominal aortic aneurysm (2018), the surveillance intervals by diameter and the threshold for elective repair.

Rule UW-AAA-002: Abdominal aortic aneurysm (section 16.4).

Threshold: abdominal aortic diameter from 4.0 to below 5.5 cm.

Probable rating: a debit of +150 %. If the applicant also has hypertension with systolic blood pressure from 140 to below 160 mmHg, see rule UW-HT-002. Source of the threshold: SVS, Practice guidelines on the care of patients with an abdominal aortic aneurysm (2018), the surveillance intervals by diameter and the threshold for elective repair.

Rule UW-AAA-003: Abdominal aortic aneurysm (section 16.4).

Threshold: abdominal aortic diameter of 5.5 cm or more. Probable rating: decline. Source of the threshold: SVS, Practice guidelines on the care of patients with an abdominal aortic aneurysm (2018), the surveillance intervals by diameter and the threshold for elective repair.

**What does not change the rating.**

The applicant's age and sex do not change the band, although the cited guidelines advise repair at a smaller size in women; such a case goes to a person.

## 16.5 Worked examples and related rules

Example 1. An applicant aged 34, a translator, has an abdominal aortic aneurysm. The most recent clinic letter records abdominal aortic diameter 3.5 cm. That meets rule UW-AAA-001 (abdominal aortic diameter from 3.0 to below 4.0 cm), so the probable rating is a debit of +50 %. If the applicant also has a history of smoking with smoking status: current smoker, rule UW-TOB-001 would apply as well; its rating is a debit of +50 %, and the two together would give +100 %.

Example 2. Take a surveyor of 45 with an abdominal aortic aneurysm. On file, from the report on file: abdominal aortic diameter 4.8 cm. The threshold met is that of rule UW-AAA-002 (abdominal aortic diameter from 4.0 to below 5.5 cm), and the probable rating is a debit of +150 %. If the applicant also has hypertension with systolic blood pressure from 140 to below 160 mmHg, rule UW-HT-002 would apply as well; its rating is a debit of +50 %, and the two together would give +200 %.

Example 3. The specialist's letter for a dental technician aged 56 with an abdominal aortic aneurysm records abdominal aortic diameter 5.9 cm. This is covered by rule UW-AAA-003 (abdominal aortic diameter of 5.5 cm or more). Probable rating: decline.

Readings on an edge: abdominal aortic diameter 3.0 cm exactly, rule UW-AAA-001 (a debit of +50 %); abdominal aortic diameter 4.0 cm exactly, rule UW-AAA-002 (a debit of +150 %); abdominal aortic diameter 5.5 cm exactly, rule UW-AAA-003 (decline).

### Common combinations.

Smoking is the strongest driver of growth, and its debit is added. Raised blood pressure is added too.

People with an aneurysm very often have coronary or leg artery disease; each is rated in its own section and added.

**Related rules.**

Under rule UW-AAA-001, if the applicant also has a history of smoking with smoking status: current smoker, see rule UW-TOB-001 (section 21.4, Tobacco use).

Under rule UW-AAA-002, if the applicant also has hypertension with systolic blood pressure from 140 to below 160 mmHg, see rule UW-HT-002 (section 10.4, Hypertension).

# 17 Peripheral artery disease

## 17.1 The impairment

Peripheral artery disease is narrowing of the arteries of the legs by the same fatty plaque that affects the heart. Calf pain on walking is the typical symptom, but many people have none.

Its importance for mortality lies less in the legs than in what it says about the coronary and brain arteries. A simple pressure ratio at the ankle confirms it.

When this section applies. Peripheral artery disease has been diagnosed or an ankle-brachial index has been measured for leg symptoms.

What is measured. The rating in this section depends on resting ankle-brachial index. The thresholds are grounded in: American Heart Association and American College of Cardiology, Guideline on the Management of Patients With Lower Extremity Peripheral Artery Disease (2016).

## 17.2 Key questions

### 1. How far can the applicant walk before calf pain stops them?

A shrinking walking distance shows the disease is advancing.

### 2. Has there been an angioplasty, a bypass or an amputation?

Each marks more severe disease than the index alone conveys.

### 3. Does the applicant smoke or have diabetes?

These two drive both the leg disease and the events that shorten life.

## 17.3 Evidence and readings

Use the lower of the two legs' resting ankle-brachial index from a vascular laboratory report.

An index above 1.40 means the arteries are too stiff to compress and the test cannot show narrowing; it has its own rule. Values above 0.90 and up to 1.40 meet no rule of this section.

### Pitfalls with this evidence.

An index measured after exercise is lower than a resting one; only the resting figure is rated.

In diabetes and in kidney failure the arteries may be too stiff to compress, giving a normal or high index in a diseased leg. A toe pressure is then the better test.

## 17.4 Probable rating

The table lists every rule of this section; each is defined in full below it. The debit percentages and the decisions to decline are invented for this synthetic manual. The thresholds follow the public source named in each rule, except the edges a rule marks as this manual's own.

| Rule id    | Measure and band                             | Probable rating  |
|------------|----------------------------------------------|------------------|
| UW-PAD-001 | resting ankle-brachial index of 0.90 or less | a debit of +75 % |
| UW-PAD-002 | resting ankle-brachial index above 1.40      | a debit of +50 % |

### Readings that meet no rule.

resting ankle-brachial index above 0.90 and up to 1.40: no rule; a normal or borderline index.

Note on the bands. 0.90 and 1.40 are the cited guideline's limits for an abnormal index and for arteries that cannot be compressed.

Rule UW-PAD-001: Peripheral artery disease (section 17.4).

Threshold: resting ankle-brachial index of 0.90 or less. Probable rating: a debit of +75 %. If the applicant also has a history of smoking with smoking status: current smoker, see rule UW-TOB-001. Source of the threshold: AHA/ACC, Guideline on the Management of Patients With Lower Extremity Peripheral Artery Disease (2016), the interpretation of the resting ankle-brachial index.

Rule UW-PAD-002: Peripheral artery disease (section 17.4).

Threshold: resting ankle-brachial index above 1.40. Probable rating: a debit of +50 %. If the applicant also has type 2 diabetes with HbA1c from 7.0 to below 8.0 %, see rule UW-DM-002. Source of the threshold: AHA/ACC, Guideline on the Management of Patients With Lower Extremity Peripheral Artery Disease (2016), the interpretation of the resting ankle-brachial index.

### **What does not change the rating.**

Walking distance, the side affected and treatment with a supervised exercise programme do not change the band.

## **17.5 Worked examples and related rules**

Example 1. The report on file for a bus driver aged 41 with peripheral artery disease records resting ankle-brachial index 0.83. This is covered by rule UW-PAD-001 (resting ankle-brachial index of 0.90 or less). Probable rating: a debit of +75 %. If the applicant also has a history of smoking with smoking status: current smoker, rule UW-TOB-001 would apply as well; its rating is a debit of +50 %, and the two together would give +125 %.

Example 2. An applicant aged 52, a payroll clerk, has peripheral artery disease. The specialist's letter records resting ankle-brachial index 1.51. That meets rule UW-PAD-002 (resting ankle-brachial index above 1.40), so the probable rating is a debit of +50 %. If the applicant also has type 2 diabetes with HbA1c from 7.0 to below 8.0 %, rule UW-DM-002 would apply as well; its rating is a debit of +50 %, and the two together would give +100 %.

Readings on an edge: resting ankle-brachial index 0.90 exactly, rule UW-PAD-001 (a debit of +75 %); resting ankle-brachial index 1.40 exactly, no rule of this section.

### **Common combinations.**

This disease is a marker for the whole arterial tree. A past heart attack, a past stroke and carotid narrowing are each rated in their own sections and added.

Smoking and diabetes are the two great causes; both are added. An ulcer that will not heal, or any amputation, goes to a person.

### **Related rules.**

Under rule UW-PAD-001, if the applicant also has a history of smoking with smoking status: current smoker, see rule UW-TOB-001 (section 21.4, Tobacco use).

Under rule UW-PAD-002, if the applicant also has type 2 diabetes with HbA1c from 7.0 to below 8.0 %, see rule UW-DM-002 (section 2.4, Type 2 diabetes mellitus).

Rules of other sections that point to this one: rule UW-CAS-002 (section 18.4, Carotid artery stenosis).

# 18 Carotid artery stenosis

## 18.1 The impairment

The carotid arteries in the neck supply most of the brain. Plaque at the point where each one divides can shed fragments that cause a stroke or a passing loss of vision or speech.

The degree of narrowing decides both the stroke risk and whether surgery is advised.

When this section applies. Narrowing of a carotid artery has been reported on imaging.

What is measured. The rating in this section depends on carotid artery stenosis (%). The thresholds are grounded in: American Heart Association and American Stroke Association, Guideline for the Prevention of Stroke in Patients With Stroke and Transient Ischemic Attack (2021).

## 18.2 Key questions

### 1. Was the narrowing found after symptoms, or by chance?

A narrowing that has already caused symptoms is far more likely to cause a stroke.

### 2. What percentage did the latest scan report, and for which side?

The table is read on the worse side.

### 3. Has surgery or a stent been done?

A treated artery is rated on its present state and on the history that led to treatment.

## 18.3 Evidence and readings

Use the percentage from a duplex ultrasound, CT or MR angiogram in the last twelve months. Where a range is given, such as 50 to 69 %, the band is clear without a single figure; record the range.

Narrowing under 50 % meets no rule of this section.

### Pitfalls with this evidence.

Different scanning methods and different ways of measuring give different percentages for the same artery. Use the figure the report gives as its conclusion.

A complete blockage is reported as occlusion, not as 100 % stenosis, and is outside this section.

## 18.4 Probable rating

The table lists every rule of this section; each is defined in full below it. The debit percentages and the decisions to decline are invented for this synthetic manual. The thresholds follow the public source named in each rule, except the edges a rule marks as this manual's own.

| Rule id    | Measure and band                              | Probable rating   |
|------------|-----------------------------------------------|-------------------|
| UW-CAS-001 | carotid artery stenosis from 50 to below 70 % | a debit of +50 %  |
| UW-CAS-002 | carotid artery stenosis of 70 % or more       | a debit of +100 % |

### Readings that meet no rule.

carotid artery stenosis below 50 %: no rule; mild narrowing.

Note on the bands. 50 % and 70 % are the cited guideline's limits for moderate and severe stenosis.

Rule UW-CAS-001: Carotid artery stenosis (section 18.4). Threshold: carotid artery stenosis from 50 to below 70 %. Probable rating: a debit of +50 %. If the applicant also has a past stroke with NIHSS score at admission below 6 points, see rule UW-CVA-001. Source of the threshold: AHA/ASA, Guideline for the Prevention of Stroke in Patients With Stroke and Transient Ischemic Attack (2021), the recommendations on extracranial carotid stenosis.

Rule UW-CAS-002: Carotid artery stenosis (section 18.4). Threshold: carotid artery stenosis of 70 % or more. Probable rating: a debit of +100 %. If the applicant also has peripheral artery disease with resting ankle-brachial index of 0.90 or less, see rule UW-PAD-001. Source of the threshold: AHA/ASA, Guideline for the Prevention of Stroke in Patients With Stroke and Transient Ischemic Attack (2021), the recommendations on extracranial carotid stenosis.

### **What does not change the rating.**

A noise heard over the artery with a stethoscope, the appearance of the plaque and which side is affected do not change the band.

## **18.5 Worked examples and related rules**

Example 1. Take a school caretaker of 48 with carotid artery stenosis. On file, from the specialist's letter: carotid artery stenosis 60 %. The threshold met is that of rule UW-CAS-001 (carotid artery stenosis from 50 to below 70 %), and the probable rating is a debit of +50 %. If the applicant also has a past stroke with NIHSS score at admission below 6 points, rule UW-CVA-001 would apply as well; its rating is a debit of +75 %, and the two together would give +125 %.

Example 2. The attending physician's statement for a translator aged 59 with carotid artery stenosis records carotid artery stenosis 76 %. This is covered by rule UW-CAS-002 (carotid artery stenosis of 70 % or more). Probable rating: a debit of +100 %. If the applicant also has peripheral artery disease with resting ankle-brachial index of 0.90 or less, rule UW-PAD-001 would apply as well; its rating is a debit of +75 %, and the two together would give +175 %.

Readings on an edge: carotid artery stenosis 50 % exactly, rule UW-CAS-001 (a debit of +50 %); carotid artery stenosis 70 % exactly, rule UW-CAS-002 (a debit of +100 %).

### **Common combinations.**

Where the narrowing has already caused a stroke, the stroke section's debit is added.

Carotid narrowing with coronary or leg artery disease is one disease in three places: each is added. Smoking and raised LDL cholesterol are added as well.

### **Related rules.**

Under rule UW-CAS-001, if the applicant also has a past stroke with NIHSS score at admission below 6 points, see rule UW-CVA-001 (section 19.4, Stroke).

Under rule UW-CAS-002, if the applicant also has peripheral artery disease with resting ankle-brachial index of 0.90 or less, see rule UW-PAD-001 (section 17.4, Peripheral artery disease).

Rules of other sections that point to this one: rule UW-CVA-002 (section 19.4, Stroke).

# 19 Stroke

## 19.1 The impairment

A stroke is damage to part of the brain from a blocked or burst blood vessel. What is left afterwards ranges from nothing noticeable to dependence on others for daily life.

Survivors face a raised risk of another stroke and of heart disease. How severe the stroke was when the applicant reached hospital is recorded on a standard scale in nearly every stroke unit, and that score is the measure used here.

When this section applies. A stroke is recorded in the history, with damage shown on brain imaging.

What is measured. The rating in this section depends on NIHSS score at admission (points). The thresholds are grounded in: American Heart Association and American Stroke Association, Guidelines for the Early Management of Patients With Acute Ischemic Stroke, 2019 update (2019).

## 19.2 Key questions

### 1. When did the stroke happen, and was it a blockage or a bleed?

The two types have different causes and different chances of recurring.

### 2. What can the applicant not do now that they could do before?

Lasting disability is not scored in this section, but heavy dependence on others takes the case to a person.

### 3. Was a cause found, such as atrial fibrillation or a narrowed carotid artery?

A cause that has been found and treated lowers the chance of a second stroke.

#### 4. Has the applicant returned to work or to driving?

Both are practical checks on how complete the recovery has been.

### 19.3 Evidence and readings

Take the NIHSS score at admission from the hospital discharge letter. Within six months of a stroke, send the case to a person whatever the score.

If the letter gives no score, ask the stroke unit for it. Do not estimate one from a description of the symptoms. A transient ischaemic attack, with symptoms gone within a day and no damage on imaging, meets no rule of this section and goes to a person.

#### Pitfalls with this evidence.

The score falls quickly in the first days as symptoms improve; the score at admission is the one that is rated, not the score at discharge.

A bleed into the brain and a blocked artery are both strokes for this section. A bleed under the lining of the brain from a burst aneurysm is not, and goes to a person.

### 19.4 Probable rating

The table lists every rule of this section; each is defined in full below it. The debit percentages and the decisions to decline are invented for this synthetic manual. The thresholds follow the public source named in each rule, except the edges a rule marks as this manual's own.

| Rule id    | Measure and band                             | Probable rating   |
|------------|----------------------------------------------|-------------------|
| UW-CVA-001 | NIHSS score at admission below 6 points      | a debit of +75 %  |
| UW-CVA-002 | NIHSS score at admission of 6 points or more | a debit of +200 % |

**Readings that meet no rule.**

None: every reading these measures can take falls in one of the bands above.

Note on the bands. A score of 6 or more is among the cited guidelines' criteria for removing the clot mechanically, which marks the stroke as more than minor.

Rule UW-CVA-001: Stroke (section 19.4). Threshold: NIHSS score at admission below 6 points. Probable rating: a debit of +75 %. If the applicant also has atrial fibrillation with CHA2DS2-VASc score from 2 to below 4 points, see rule UW-AF-002. Source of the threshold: AHA/ASA, Guidelines for the Early Management of Patients With Acute Ischemic Stroke, 2019 update (2019), the criteria for mechanical thrombectomy, among them an NIHSS score of 6 or more.

Rule UW-CVA-002: Stroke (section 19.4). Threshold: NIHSS score at admission of 6 points or more. Probable rating: a debit of +200 %. If the applicant also has carotid artery stenosis with carotid artery stenosis of 70 % or more, see rule UW-CAS-002. Source of the threshold: AHA/ASA, Guidelines for the Early Management of Patients With Acute Ischemic Stroke, 2019 update (2019), the criteria for mechanical thrombectomy, among them an NIHSS score of 6 or more.

**What does not change the rating.**

Clot-dissolving treatment, mechanical removal of the clot and the length of the hospital stay do not change the band. A full recovery does not lower it.

**19.5 Worked examples and related rules**

Example 1. An applicant aged 55, a joiner, has a past stroke. The attending physician's statement records NIHSS score at admission 5 points. That meets rule UW-CVA-001 (NIHSS score at admission below 6 points), so the probable rating is a debit of +75 %. If the applicant also has atrial fibrillation with CHA2DS2-VASc score from 2 to below 4 points, rule UW-AF-002 would apply as well; its rating is a debit of +50 %, and the two together would give +125 %.

Example 2. Take a bus driver of 36 with a past stroke. On file, from the most recent clinic letter: NIHSS score at admission 7 points. The threshold met is that of rule UW-CVA-002 (NIHSS score at admission of 6 points or more), and the probable rating is a debit of +200 %. If the applicant also has carotid artery stenosis with carotid artery stenosis of 70 % or more, rule UW-CAS-002 would apply as well; its rating is a debit of +100 %, and the two together would give +300 %.

Readings on an edge: NIHSS score at admission 6 points exactly, rule UW-CVA-002 (a debit of +200 %).

### **Common combinations.**

Atrial fibrillation is found in about a quarter of strokes caused by a blocked artery; its debit is added, and its score already counts two points for the stroke.

Carotid narrowing, raised blood pressure, diabetes and smoking are added from their sections. Depression after stroke is common and is rated in its own section.

### **Related rules.**

Under rule UW-CVA-001, if the applicant also has atrial fibrillation with CHA2DS2-VASc score from 2 to below 4 points, see rule UW-AF-002 (section 13.4, Atrial fibrillation).

Under rule UW-CVA-002, if the applicant also has carotid artery stenosis with carotid artery stenosis of 70 % or more, see rule UW-CAS-002 (section 18.4, Carotid artery stenosis).

Rules of other sections that point to this one: rule UW-AF-002 (section 13.4, Atrial fibrillation); rule UW-CAS-001 (section 18.4, Carotid artery stenosis).

## 20 Venous thromboembolism

### 20.1 The impairment

Venous thromboembolism covers a clot in a deep vein, usually of the leg, and a clot that has travelled to the lungs. Anticoagulant treatment for the first months prevents the clot from growing or returning.

After that treatment phase the outlook depends on why the clot formed. A clot after surgery or a long-haul flight seldom returns; one with no cause often does.

When this section applies. A deep vein thrombosis or a pulmonary embolism is recorded in the history.

What is measured. The rating in this section depends on time since the clot (months). The thresholds are grounded in: American College of Chest Physicians, Antithrombotic Therapy for VTE Disease, second update of the CHEST guideline and expert panel report (2021).

### 20.2 Key questions

#### 1. What was the date of the clot, and was it in a leg, the lungs or both?

A clot in the lungs carries more early risk than one confined to the calf.

#### 2. Was there a trigger: surgery, a plaster cast, pregnancy, a hormone medicine, cancer?

A removable trigger is the best predictor that it will not happen again.

#### 3. Is an anticoagulant still taken, and for how long is it planned?

Lifelong treatment tells the underwriter the clinic judges the risk of recurrence to be high.

## 20.3 Evidence and readings

Take the date from the hospital letter and count whole months to the application. With more than one clot, count from the latest and send the case to a person if there have been three or more.

A clot found during investigation for cancer is rated under the cancer, not here.

### Pitfalls with this evidence.

A clot in a surface vein, often called phlebitis, is not a deep vein thrombosis and is not rated here.

The date that matters is the date of diagnosis, not the date treatment stopped.

## 20.4 Probable rating

The table lists every rule of this section; each is defined in full below it. The debit percentages and the decisions to decline are invented for this synthetic manual. The thresholds follow the public source named in each rule, except the edges a rule marks as this manual's own.

| Rule id    | Measure and band                              | Probable rating           |
|------------|-----------------------------------------------|---------------------------|
| UW-VTE-001 | time since the clot below 3 months            | decline as a postponement |
| UW-VTE-002 | time since the clot from 3 to below 12 months | a debit of +50 %          |
| UW-VTE-003 | time since the clot of 12 months or more      | no debit, +0 %            |

### Readings that meet no rule.

None: every reading these measures can take falls in one of the bands above.

Note on the bands. Three months is the treatment phase in the cited guideline. The edge at twelve months is this manual's own.

Rule UW-VTE-001: Venous thromboembolism (section 20.4).

Threshold: time since the clot below 3 months. Probable rating: decline as a postponement. This is a postponement until the treatment phase is complete. Source of the threshold: CHEST, Antithrombotic Therapy for VTE Disease, second update of the CHEST guideline and expert panel report (2021), the three-month treatment phase of anticoagulation.

Rule UW-VTE-002: Venous thromboembolism (section 20.4).

Threshold: time since the clot from 3 to below 12 months. Probable rating: a debit of +50 %. If the applicant also has an unusual build with body mass index from 35.0 to below 40.0 kg/m<sup>2</sup>, see rule UW-BMI-003. Source of the edge at 3 months: CHEST, Antithrombotic Therapy for VTE Disease, second update of the CHEST guideline and expert panel report (2021), the three-month treatment phase of anticoagulation. The edge at 12 months is this manual's own.

Rule UW-VTE-003: Venous thromboembolism (section 20.4).

Threshold: time since the clot of 12 months or more. Probable rating: no debit, +0 %. The edge at 12 months is this manual's own and no guideline supplies it. The source for the measure and the other edges of this section: CHEST, Antithrombotic Therapy for VTE Disease, second update of the CHEST guideline and expert panel report (2021), the three-month treatment phase of anticoagulation.

### **What does not change the rating.**

The anticoagulant chosen, a filter placed in the main vein and an inherited clotting tendency found on testing do not change the band; the last of these is passed to a person.

## **20.5 Worked examples and related rules**

Example 1. The most recent clinic letter for a market gardener aged 32 with a past blood clot in the veins records time since the clot 2 months. This is covered by rule UW-VTE-001 (time since the clot below 3 months). Probable rating: decline as a postponement.

Example 2. An applicant aged 43, a school caretaker, has a past blood clot in the veins. The report on file records time since the clot 8 months. That meets rule UW-VTE-002 (time since the clot from 3 to below 12 months), so the probable rating is a debit of +50 %. If the applicant also has an unusual build with body mass index from 35.0 to below 40.0 kg/m<sup>2</sup>, rule UW-BMI-003 would apply as well; its rating is a debit of +50 %, and the two together would give +100 %.

Example 3. Take a translator of 54 with a past blood clot in the veins. On file, from the specialist's letter: time since the clot 13 months. The threshold met is that of rule UW-VTE-003 (time since the clot of 12 months or more), and the probable rating is no debit, +0 %.

Readings on an edge: time since the clot 3 months exactly, rule UW-VTE-002 (a debit of +50 %); time since the clot 12 months exactly, rule UW-VTE-003 (no debit, +0 %).

### **Common combinations.**

Obesity makes a second clot more likely, and the build debit is added. A clot during treatment with an oestrogen medicine that has since been stopped is rated here like any other.

A clot in someone with active cancer is rated under the cancer. Raised pressure in the lung arteries after an embolism goes to a person.

### **Related rules.**

Under rule UW-VTE-002, if the applicant also has an unusual build with body mass index from 35.0 to below 40.0 kg/m<sup>2</sup>, see rule UW-BMI-003 (section 5.4, Build: underweight and obesity).

## 21 Tobacco use

### 21.1 The impairment

Cigarette smoking is the largest avoidable cause of early death. It damages the arteries and the lungs and causes cancer in many organs, and the harm grows with the amount smoked and the years of smoking.

This section has a rule for present status and a second, separate rule for the lifetime total. A long-term heavy smoker meets both, and the two debits are added.

Stopping helps at any age. The former smoker's rule carries no debit, but a large lifetime total still counts after stopping.

When this section applies. Every applicant: smoking status is recorded for all of them.

What is measured. The rating in this section depends on smoking status and on lifetime smoking (pack-years). The thresholds are grounded in: Centers for Disease Control and Prevention, National Center for Health Statistics, National Health Interview Survey, adult tobacco use glossary (2017); US Preventive Services Task Force, Lung Cancer: Screening, recommendation statement (2021).

### 21.2 Key questions

#### **1. Does the applicant smoke now, and how many cigarettes a day?**

Status is the first rule of this section, and the daily number is needed for the lifetime total.

#### **2. At what age did smoking start, and were there years without it?**

Pack-years are packs a day multiplied by years actually smoked.

**3. If stopped, on what date, and has there been any relapse?**

A former smoker is one who has stopped; an applicant who stopped last month and one who stopped ten years ago are the same in status but not in notes.

**4. Does the medical record agree with the application form?**

Smoking is under-declared more often than any other habit, and the physician's record is the better witness.

**21.3 Evidence and readings**

Take status from the attending physician's statement where it differs from the application form. A current smoker is someone who has smoked at least 100 cigarettes in their life and smokes now, every day or on some days; a former smoker has smoked that many and has stopped.

Work out pack-years from the record: 10 cigarettes a day for 30 years is half a pack for 30 years, or 15 pack-years. Round down to a whole number.

Someone who has smoked fewer than 100 cigarettes in total has never smoked for the purposes of this section and meets no rule.

**Pitfalls with this evidence.**

An applicant who stopped within the last few weeks is a former smoker by the survey definition this section uses. Record the date of stopping so that a person can weigh a very recent change.

Cotinine in a urine or saliva test shows nicotine from any source, including patches and gum. A positive test with a credible account of nicotine replacement goes to a person.

Cigars, pipes and electronic cigarettes are not counted in pack-years.

**Converting units.**

lifetime smoking: Cigarettes a day divided by 20, times the years smoked: 10 a day for 30 years is 15 pack-years.

**21.4 Probable rating**

The table lists every rule of this section; each is defined in full below it. The debit percentages and the decisions to decline are invented for this synthetic manual. The thresholds follow the public source named in each rule, except the edges a rule marks as this manual's own.

| Rule id    | Measure and band                          | Probable rating  |
|------------|-------------------------------------------|------------------|
| UW-TOB-001 | smoking status: current smoker            | a debit of +50 % |
| UW-TOB-002 | lifetime smoking of 20 pack-years or more | a debit of +25 % |
| UW-TOB-003 | smoking status: former smoker             | no debit, +0 %   |

**Readings that meet no rule.**

smoking status: never smoked: no rule; fewer than 100 cigarettes in a lifetime.

lifetime smoking below 20 pack-years: no rule for the lifetime total; the status rules still apply.

Note on the bands. The status definitions and the 20 pack-year level are those of the cited survey glossary and screening recommendation.

Rule UW-TOB-001: Tobacco use (section 21.4). Threshold: smoking status: current smoker. Probable rating: a debit of +50 %. If the record also shows lifetime smoking of 20 pack-years or more, see rule UW-TOB-002. Source of the threshold: CDC, National Health Interview Survey, adult tobacco use glossary (2017), the definitions of current smoker and former smoker.

Rule UW-TOB-002: Tobacco use (section 21.4). Threshold: lifetime smoking of 20 pack-years or more. Probable rating: a debit of +25 %. This debit is added to the one for present status, whether the applicant smokes now or has stopped. If the applicant also has chronic obstructive pulmonary disease with FEV1 after a bronchodilator from 50 to below 80 % of predicted, see rule UW-COPD-002. Source of the threshold: USPSTF, Lung Cancer: Screening, recommendation statement (2021), the 20 pack-year smoking history that defines the screened group.

Rule UW-TOB-003: Tobacco use (section 21.4). Threshold: smoking status: former smoker. Probable rating: no debit, +0 %. If the record also shows lifetime smoking of 20 pack-years or more, see rule UW-TOB-002. Source of the threshold: CDC, National Health Interview Survey, adult tobacco use glossary (2017), the definitions of current smoker and former smoker.

### **What does not change the rating.**

The brand, the tar content and whether the smoke is inhaled do not change either rule. Smoking fewer than 20 cigarettes a day does not lower the status debit.

## **21.5 Worked examples and related rules**

Example 1. Take a pastry cook of 39 with a history of smoking. On file, from the report on file: smoking status: current smoker. The threshold met is that of rule UW-TOB-001 (smoking status: current smoker), and the probable rating is a debit of +50 %. If the record also shows lifetime smoking of 20 pack-years or more, rule UW-TOB-002 would apply as well; its rating is a debit of +25 %, and the two together would give +75 %.

Example 2. The specialist's letter for a joiner aged 50 with a history of smoking records lifetime smoking 22 pack-years. This is covered by rule UW-TOB-002 (lifetime smoking of 20 pack-years or more). Probable rating: a debit of +25 %. If the applicant also has chronic obstructive pulmonary disease with FEV1 after a bronchodilator from 50 to below 80 % of predicted, rule UW-COPD-002 would apply as well; its rating is a debit of +50 %, and the two together would give +75 %.

Example 3. An applicant aged 31, a bus driver, has a history of smoking. The attending physician's statement records smoking status: former smoker. That meets rule UW-TOB-003 (smoking status: former smoker), so the probable rating is no debit, +0 %. If the record also shows lifetime smoking of 20 pack-years or more, rule UW-TOB-002 would apply as well; its rating is a debit of +25 %, and the two together would give +25 %.

Readings on an edge: lifetime smoking 20 pack-years exactly, rule UW-TOB-002 (a debit of +25 %).

### **Common combinations.**

Smoking makes every arterial and lung impairment worse, and its debit is always added to theirs: raised blood pressure, a past heart attack, an aneurysm, leg artery disease and chronic obstructive pulmonary disease above all.

A smoker with diabetes or with raised LDL cholesterol carries all the debits together.

### **Related rules.**

Under rule UW-TOB-001, if the record also shows lifetime smoking of 20 pack-years or more, see rule UW-TOB-002 (section 21.4, Tobacco use).

Under rule UW-TOB-002, if the applicant also has chronic obstructive pulmonary disease with FEV1 after a bronchodilator from 50 to below 80 % of predicted, see rule UW-COPD-002 (section 25.4, Chronic obstructive pulmonary disease).

Under rule UW-TOB-003, if the record also shows lifetime smoking of 20 pack-years or more, see rule UW-TOB-002 (section 21.4, Tobacco use).

Rules of other sections that point to this one: rule UW-HT-002 (section 10.4, Hypertension); rule UW-AAA-001 (section 16.4, Abdominal aortic aneurysm); rule UW-PAD-001 (section 17.4, Peripheral artery disease); rule UW-COPD-002 (section 25.4, Chronic obstructive pulmonary disease).

## 22 Alcohol use

### 22.1 The impairment

Heavy drinking shortens life through liver disease, accidents, several cancers, heart muscle damage and suicide. The amount an applicant states is an unreliable guide, since most heavy drinkers understate it.

A structured screening questionnaire gives a steadier picture than a single figure for weekly units, because it also asks about loss of control and harm.

When this section applies. An AUDIT score recorded by a clinician is on file.

What is measured. The rating in this section depends on AUDIT score (points). The thresholds are grounded in: World Health Organization, AUDIT, the Alcohol Use Disorders Identification Test: guidelines for use in primary health care, second edition (2001).

### 22.2 Key questions

#### 1. Has a doctor ever advised the applicant to cut down?

Advice recorded in the notes means the drinking was enough to worry a clinician.

#### 2. Has there been treatment for alcohol dependence, or an admission for detoxification?

Either takes the case out of the lower bands, whatever today's score.

#### 3. Are the liver enzymes or the red cell size raised?

Blood tests are indirect witnesses that do not depend on what the applicant says.

#### 4. Has there been a drink-driving offence?

It is one of the few events that reliably marks drinking beyond control.

## 22.3 Evidence and readings

Use an AUDIT total recorded by a clinician in the last twelve months. Do not derive a score from weekly units on the application form: the questionnaire measures more than quantity.

With no score on file and nothing in the record suggesting harm, no rule of this section applies. Where the record suggests harm and there is no score, send the case to a person.

### Pitfalls with this evidence.

A score taken during treatment for dependence reflects abstinence, not the drinking that led to treatment. The history decides such a case, and it goes to a person.

Some clinics record only the first three questions, which score from 0 to 12. That short form is not the score this section uses.

## 22.4 Probable rating

The table lists every rule of this section; each is defined in full below it. The debit percentages and the decisions to decline are invented for this synthetic manual. The thresholds follow the public source named in each rule, except the edges a rule marks as this manual's own.

| Rule id    | Measure and band                       | Probable rating  |
|------------|----------------------------------------|------------------|
| UW-ALC-001 | AUDIT score from 8 to below 16 points  | a debit of +25 % |
| UW-ALC-002 | AUDIT score from 16 to below 20 points | a debit of +75 % |
| UW-ALC-003 | AUDIT score of 20 points or more       | decline          |

### Readings that meet no rule.

AUDIT score below 8 points: no rule; the low-risk zone.

Note on the bands. 8, 16 and 20 are the lower limits of the hazardous, harmful and possible-dependence zones in the cited manual.

Rule UW-ALC-001: Alcohol use (section 22.4). Threshold: AUDIT score from 8 to below 16 points. Probable rating: a debit of +25 %. If the applicant also has fatty liver disease with FIB-4 index from 1.3 to 2.67, see rule UW-FLD-002. Source of the threshold: WHO, AUDIT, the Alcohol Use Disorders Identification Test: guidelines for use in primary health care, second edition (2001), the risk zones by total AUDIT score.

Rule UW-ALC-002: Alcohol use (section 22.4). Threshold: AUDIT score from 16 to below 20 points. Probable rating: a debit of +75 %. If the applicant also has depression with PHQ-9 score below 16 points, see rule UW-DEP-001. Source of the threshold: WHO, AUDIT, the Alcohol Use Disorders Identification Test: guidelines for use in primary health care, second edition (2001), the risk zones by total AUDIT score.

Rule UW-ALC-003: Alcohol use (section 22.4). Threshold: AUDIT score of 20 points or more. Probable rating: decline. If the applicant also has cirrhosis with MELD score below 15 points, see rule UW-CIR-001. Source of the threshold: WHO, AUDIT, the Alcohol Use Disorders Identification Test: guidelines for use in primary health care, second edition (2001), the risk zones by total AUDIT score.

### **What does not change the rating.**

The type of drink and whether drinking is daily or at weekends do not change the band. Normal liver enzymes do not lower it.

## **22.5 Worked examples and related rules**

Example 1. An applicant aged 46, a florist, has a record of heavy drinking. The specialist's letter records AUDIT score 12 points. That meets rule UW-ALC-001 (AUDIT score from 8 to below 16 points), so the probable rating is a debit of +25 %. If the applicant also has fatty liver disease with FIB-4 index from 1.3 to 2.67, rule UW-FLD-002 would apply as well; its rating is a debit of +50 %, and the two together would give +75 %.

Example 2. Take a market gardener of 57 with a record of heavy drinking. On file, from the attending physician's statement: AUDIT score 18 points. The threshold met is that of rule UW-ALC-002 (AUDIT score from 16 to below 20 points), and the probable rating is a debit of +75 %. If the applicant also has depression with PHQ-9 score below 16 points, rule UW-DEP-001 would apply as well; its rating is a debit of +25 %, and the two together would give +100 %.

Example 3. The most recent clinic letter for a school caretaker aged 38 with a record of heavy drinking records AUDIT score 22 points. This is covered by rule UW-ALC-003 (AUDIT score of 20 points or more). Probable rating: decline. If the applicant also has cirrhosis with MELD score below 15 points, rule UW-CIR-001 would apply as well; its rating is a debit of +200 %, and the outcome would be decline.

Readings on an edge: AUDIT score 8 points exactly, rule UW-ALC-001 (a debit of +25 %); AUDIT score 16 points exactly, rule UW-ALC-002 (a debit of +75 %); AUDIT score 20 points exactly, rule UW-ALC-003 (decline).

### **Common combinations.**

Alcohol and the liver: where fatty liver disease has been diagnosed in a drinker, its FIB-4 band is rated in that section and added, and cirrhosis is rated in its own section.

Alcohol and mood travel together, each making the other worse; the depression debit is added. Raised triglycerides and raised blood pressure caused by drinking are still rated in their own sections.

**Related rules.**

Under rule UW-ALC-001, if the applicant also has fatty liver disease with FIB-4 index from 1.3 to 2.67, see rule UW-FLD-002 (section 31.4, Fatty liver disease).

Under rule UW-ALC-002, if the applicant also has depression with PHQ-9 score below 16 points, see rule UW-DEP-001 (section 37.4, Depression).

Under rule UW-ALC-003, if the applicant also has cirrhosis with MELD score below 15 points, see rule UW-CIR-001 (section 32.4, Cirrhosis of the liver).

Rules of other sections that point to this one: rule UW-TG-002 (section 7.4, Raised triglycerides); rule UW-CIR-001 (section 32.4, Cirrhosis of the liver); rule UW-DEP-001 (section 37.4, Depression); rule UW-EPI-001 (section 38.4, Epilepsy).

## 23 Long-term opioid therapy

### 23.1 The impairment

Opioid medicines prescribed for months or years against chronic pain carry a risk of fatal overdose that rises with the daily dose. The risk is higher still when sedatives or alcohol are taken as well.

This section covers prescribed treatment for pain that is not due to cancer. Illicit use and treatment for opioid dependence are not rated here and go to a person.

When this section applies. An opioid medicine has been prescribed for pain, other than cancer pain, for three months or longer.

What is measured. The rating in this section depends on daily opioid dose (MME a day). The thresholds are grounded in: Centers for Disease Control and Prevention, Guideline for Prescribing Opioids for Chronic Pain, United States (2016).

### 23.2 Key questions

#### 1. Which opioids are prescribed, at what dose, and for how long?

The daily total across all products is the figure the table needs.

#### 2. Is a benzodiazepine or another sedative prescribed too?

The combination multiplies the risk of breathing stopping during sleep.

#### 3. What is the painful condition?

The underlying disease may be an impairment in its own right.

#### 4. Has the dose risen over the last year?

A climbing dose suggests tolerance and predicts further rises.

## 23.3 Evidence and readings

Add up the daily dose of every opioid on the current prescription list and convert it to morphine milligram equivalents with the factors shown below, which are those of the cited guideline. Record the working.

Treatment for less than three months after surgery or injury meets no rule of this section.

### Pitfalls with this evidence.

Prescription lists often show an old dose alongside the present one after a change. Use the dose in the most recent clinic letter.

Medicines taken as needed are counted at the most the prescription allows in a day, unless the record shows what is actually taken.

### Converting units.

daily opioid dose: Multiply each medicine's daily milligrams by its factor and add the results: morphine 1, hydrocodone 1, oxycodone 1.5, hydromorphone 4, codeine 0.15, tramadol 0.1. A fentanyl patch counts 2.4 for each microgram an hour.

## 23.4 Probable rating

The table lists every rule of this section; each is defined in full below it. The debit percentages and the decisions to decline are invented for this synthetic manual. The thresholds follow the public source named in each rule, except the edges a rule marks as this manual's own.

| Rule id    | Measure and band                                | Probable rating   |
|------------|-------------------------------------------------|-------------------|
| UW-OPI-001 | daily opioid dose below 50 MME a day            | a debit of +25 %  |
| UW-OPI-002 | daily opioid dose from 50 to below 90 MME a day | a debit of +75 %  |
| UW-OPI-003 | daily opioid dose of 90 MME a day or more       | a debit of +150 % |

**Readings that meet no rule.**

None: every reading these measures can take falls in one of the bands above.

Note on the bands. 50 and 90 morphine milligram equivalents a day are the cited guideline's levels for added caution and for avoidance.

Rule UW-OPI-001: Long-term opioid therapy (section 23.4).

Threshold: daily opioid dose below 50 MME a day. Probable rating: a debit of +25 %. Source of the threshold: CDC, Guideline for Prescribing Opioids for Chronic Pain, United States (2016), the recommendation on dosage: reassess at 50, avoid 90 morphine milligram equivalents a day or more.

Rule UW-OPI-002: Long-term opioid therapy (section 23.4).

Threshold: daily opioid dose from 50 to below 90 MME a day.

Probable rating: a debit of +75 %. If the applicant also has depression with PHQ-9 score of 16 points or more, see rule UW-DEP-002. Source of the threshold: CDC, Guideline for Prescribing Opioids for Chronic Pain, United States (2016), the recommendation on dosage: reassess at 50, avoid 90 morphine milligram equivalents a day or more.

Rule UW-OPI-003: Long-term opioid therapy (section 23.4).

Threshold: daily opioid dose of 90 MME a day or more. Probable rating: a debit of +150 %. If the applicant also has obstructive sleep apnoea with apnoea-hypopnoea index above 30 events per hour, see rule UW-OSA-003. Source of the threshold: CDC, Guideline for Prescribing Opioids for Chronic Pain, United States (2016), the recommendation on dosage: reassess at 50, avoid 90 morphine milligram equivalents a day or more.

**What does not change the rating.**

A signed treatment agreement, urine screening and the form of the medicine, tablet or patch, do not change the band.

## 23.5 Worked examples and related rules

Example 1. The attending physician's statement for a dental technician aged 53 with long-term opioid treatment for pain records daily opioid dose 46 MME a day. This is covered by rule UW-OPI-001 (daily opioid dose below 50 MME a day). Probable rating: a debit of +25 %.

Example 2. An applicant aged 34, a pastry cook, has long-term opioid treatment for pain. The most recent clinic letter records daily opioid dose 70 MME a day. That meets rule UW-OPI-002 (daily opioid dose from 50 to below 90 MME a day), so the probable rating is a debit of +75 %. If the applicant also has depression with PHQ-9 score of 16 points or more, rule UW-DEP-002 would apply as well; its rating is a debit of +100 %, and the two together would give +175 %.

Example 3. Take a joiner of 45 with long-term opioid treatment for pain. On file, from the report on file: daily opioid dose 97 MME a day. The threshold met is that of rule UW-OPI-003 (daily opioid dose of 90 MME a day or more), and the probable rating is a debit of +150 %. If the applicant also has obstructive sleep apnoea with apnoea-hypopnoea index above 30 events per hour, rule UW-OSA-003 would apply as well; its rating is a debit of +75 %, and the two together would give +225 %.

Readings on an edge: daily opioid dose 50 MME a day exactly, rule UW-OPI-002 (a debit of +75 %); daily opioid dose 90 MME a day exactly, rule UW-OPI-003 (a debit of +150 %).

### Common combinations.

Sedatives, sleeping tablets and alcohol each add to the danger of breathing stopping during sleep. Severe sleep apnoea does the same from another direction. Add the debits from those sections, and say plainly in the notes that the combination is present.

Depression is common in chronic pain and raises the risk of overdose; its debit is added.

**Related rules.**

Under rule UW-OPI-002, if the applicant also has depression with PHQ-9 score of 16 points or more, see rule UW-DEP-002 (section 37.4, Depression).

Under rule UW-OPI-003, if the applicant also has obstructive sleep apnoea with apnoea-hypopnoea index above 30 events per hour, see rule UW-OSA-003 (section 26.4, Obstructive sleep apnoea).

Rules of other sections that point to this one: rule UW-DEP-002 (section 37.4, Depression).

## 24 Asthma

### 24.1 The impairment

Asthma is long-term inflammation of the airways that makes them narrow in bouts, with wheeze, cough and breathlessness. Between bouts most people with asthma breathe normally.

Deaths from asthma are uncommon and nearly all follow severe attacks. Low lung function between attacks is one of the recognised warnings that such an attack is more likely.

When this section applies. Asthma has been diagnosed.

What is measured. The rating in this section depends on FEV1 (% of predicted). The thresholds are grounded in: Global Initiative for Asthma, Global Strategy for Asthma Management and Prevention (2023).

### 24.2 Key questions

#### 1. How many courses of steroid tablets were needed in the last twelve months?

Each course marks an attack that inhalers could not control.

#### 2. Has there ever been an admission to intensive care for asthma?

A previous near-fatal attack is the strongest predictor of a fatal one.

#### 3. Is a preventer inhaler used every day?

Reliance on the reliever alone is a known risk factor.

#### 4. Does the applicant smoke?

Smoking blunts the effect of preventer inhalers and may mean the diagnosis is partly something else.

## 24.3 Evidence and readings

Use the FEV1 as a percentage of predicted from spirometry done when the applicant was well, in the last two years.

An admission to hospital for asthma in the last twelve months, or any past admission to intensive care, takes the case to a person.

### Pitfalls with this evidence.

Spirometry during or just after an attack reads low and does not show the usual state. Check the date against the dates of steroid courses.

Peak flow is not FEV1 and cannot be read on this table.

## 24.4 Probable rating

The table lists every rule of this section; each is defined in full below it. The debit percentages and the decisions to decline are invented for this synthetic manual. The thresholds follow the public source named in each rule, except the edges a rule marks as this manual's own.

| Rule id    | Measure and band                  | Probable rating  |
|------------|-----------------------------------|------------------|
| UW-AST-001 | FEV1 of 60 % of predicted or more | no debit, +0 %   |
| UW-AST-002 | FEV1 below 60 % of predicted      | a debit of +75 % |

### Readings that meet no rule.

None: every reading these measures can take falls in one of the bands above.

Note on the bands. 60 % of predicted is the level the cited strategy lists as a risk factor for exacerbations.

Rule UW-AST-001: Asthma (section 24.4). Threshold: FEV1 of 60 % of predicted or more. Probable rating: no debit, +0 %. Source of the threshold: GINA, Global Strategy for Asthma Management and Prevention (2023), the risk factors for exacerbations, among them an FEV1 below 60 % of predicted.

Rule UW-AST-002: Asthma (section 24.4). Threshold: FEV1 below 60 % of predicted. Probable rating: a debit of +75 %. If the applicant also has chronic obstructive pulmonary disease with FEV1 after a bronchodilator from 50 to below 80 % of predicted, see rule UW-COPD-002. Source of the threshold: GINA, Global Strategy for Asthma Management and Prevention (2023), the risk factors for exacerbations, among them an FEV1 below 60 % of predicted.

### **What does not change the rating.**

Allergies, hay fever, eczema and the brand of inhaler do not change the band. Childhood asthma that ended before the age of 16 with no treatment since meets no rule and is simply noted.

## **24.5 Worked examples and related rules**

Example 1. Take a librarian of 60 with asthma. On file, from the most recent clinic letter: FEV1 65 % of predicted. The threshold met is that of rule UW-AST-001 (FEV1 of 60 % of predicted or more), and the probable rating is no debit, +0 %.

Example 2. The report on file for a florist aged 41 with asthma records FEV1 55 % of predicted. This is covered by rule UW-AST-002 (FEV1 below 60 % of predicted). Probable rating: a debit of +75 %. If the applicant also has chronic obstructive pulmonary disease with FEV1 after a bronchodilator from 50 to below 80 % of predicted, rule UW-COPD-002 would apply as well; its rating is a debit of +50 %, and the two together would give +125 %.

Readings on an edge: FEV1 60 % of predicted exactly, rule UW-AST-001 (no debit, +0 %).

### **Common combinations.**

In a smoker over 40, asthma and chronic obstructive pulmonary disease overlap. If narrowing persists after a bronchodilator, the other section's rule is read as well and the debits are added.

Frequent courses of steroid tablets thin the bones and raise glucose; those effects are rated in their own sections.

**Related rules.**

Under rule UW-AST-002, if the applicant also has chronic obstructive pulmonary disease with FEV1 after a bronchodilator from 50 to below 80 % of predicted, see rule UW-COPD-002 (section 25.4, Chronic obstructive pulmonary disease).

## 25 Chronic obstructive pulmonary disease

### 25.1 The impairment

Chronic obstructive pulmonary disease is permanent narrowing of the airways and destruction of the air sacs, nearly always from smoking. Breathlessness worsens over the years, with flare-ups that may need hospital care.

Lung function after a bronchodilator grades the fixed obstruction and is the basis of the rating. Unlike in asthma, the figure does not return to normal.

When this section applies. Chronic obstructive pulmonary disease, chronic bronchitis with obstruction or emphysema has been diagnosed.

What is measured. The rating in this section depends on FEV1 after a bronchodilator (% of predicted). The thresholds are grounded in: Global Initiative for Chronic Obstructive Lung Disease, Global Strategy for Prevention, Diagnosis and Management of COPD (2024).

### 25.2 Key questions

#### 1. How many flare-ups needed antibiotics or steroids last year, and did any need hospital?

Two flare-ups, or one admission, in a year marks a faster decline.

#### 2. Does the applicant still smoke?

Stopping is the only measure that slows the loss of lung function.

#### 3. Is oxygen used at home?

Home oxygen means the disease is at its last stage.

#### 4. Has weight been lost without trying?

Weight loss in this disease is a sign of advanced illness.

## 25.3 Evidence and readings

Use the FEV1 after a bronchodilator from spirometry in the last two years. A figure taken without a bronchodilator overstates the obstruction; ask whether one was given.

Home oxygen takes the case to a person whatever the spirometry shows.

### Pitfalls with this evidence.

The diagnosis itself needs a ratio of FEV1 to total exhaled volume below 0.7 after a bronchodilator. A report with a normal ratio does not support the diagnosis, whatever the FEV1.

Spirometry within six weeks of a flare-up understates lung function.

## 25.4 Probable rating

The table lists every rule of this section; each is defined in full below it. The debit percentages and the decisions to decline are invented for this synthetic manual. The thresholds follow the public source named in each rule, except the edges a rule marks as this manual's own.

| Rule id     | Measure and band                                               | Probable rating   |
|-------------|----------------------------------------------------------------|-------------------|
| UW-COPD-001 | FEV1 after a bronchodilator of 80 % of predicted or more       | a debit of +25 %  |
| UW-COPD-002 | FEV1 after a bronchodilator from 50 to below 80 % of predicted | a debit of +50 %  |
| UW-COPD-003 | FEV1 after a bronchodilator from 30 to below 50 % of predicted | a debit of +150 % |
| UW-COPD-004 | FEV1 after a bronchodilator below 30 % of predicted            | decline           |

### Readings that meet no rule.

None: every reading these measures can take falls in one of the bands above.

Note on the bands. 80, 50 and 30 % of predicted are the limits of grades 1 to 4 in the cited strategy.

Rule UW-COPD-001: Chronic obstructive pulmonary disease (section 25.4). Threshold: FEV1 after a bronchodilator of 80 % of predicted or more. Probable rating: a debit of +25 %. Source of the threshold: GOLD, Global Strategy for Prevention, Diagnosis and Management of COPD (2024), the grades of airflow obstruction by FEV1 after a bronchodilator.

Rule UW-COPD-002: Chronic obstructive pulmonary disease (section 25.4). Threshold: FEV1 after a bronchodilator from 50 to below 80 % of predicted. Probable rating: a debit of +50 %. If the applicant also has a history of smoking with smoking status: current smoker, see rule UW-TOB-001. Source of the threshold: GOLD, Global Strategy for Prevention, Diagnosis and Management of COPD (2024), the grades of airflow obstruction by FEV1 after a bronchodilator.

Rule UW-COPD-003: Chronic obstructive pulmonary disease (section 25.4). Threshold: FEV1 after a bronchodilator from 30 to below 50 % of predicted. Probable rating: a debit of +150 %. If the applicant also has an unusual build with body mass index from 16.0 to below 18.5 kg/m<sup>2</sup>, see rule UW-BMI-002. Source of the threshold: GOLD, Global Strategy for Prevention, Diagnosis and Management of COPD (2024), the grades of airflow obstruction by FEV1 after a bronchodilator.

Rule UW-COPD-004: Chronic obstructive pulmonary disease (section 25.4). Threshold: FEV1 after a bronchodilator below 30 % of predicted. Probable rating: decline. Source of the threshold: GOLD, Global Strategy for Prevention, Diagnosis and Management of COPD (2024), the grades of airflow obstruction by FEV1 after a bronchodilator.

**What does not change the rating.**

The inhalers prescribed and the applicant's account of breathlessness do not change the band. Having stopped smoking does not lower it, though it is the best sign for the future.

**25.5 Worked examples and related rules**

Example 1. An applicant aged 37, a surveyor, has chronic obstructive pulmonary disease. The report on file records FEV1 after a bronchodilator 86 % of predicted. That meets rule UW-COPD-001 (FEV1 after a bronchodilator of 80 % of predicted or more), so the probable rating is a debit of +25 %.

Example 2. Take a dental technician of 48 with chronic obstructive pulmonary disease. On file, from the specialist's letter: FEV1 after a bronchodilator 65 % of predicted. The threshold met is that of rule UW-COPD-002 (FEV1 after a bronchodilator from 50 to below 80 % of predicted), and the probable rating is a debit of +50 %. If the applicant also has a history of smoking with smoking status: current smoker, rule UW-TOB-001 would apply as well; its rating is a debit of +50 %, and the two together would give +100 %.

Example 3. The attending physician's statement for a pastry cook aged 59 with chronic obstructive pulmonary disease records FEV1 after a bronchodilator 40 % of predicted. This is covered by rule UW-COPD-003 (FEV1 after a bronchodilator from 30 to below 50 % of predicted). Probable rating: a debit of +150 %. If the applicant also has an unusual build with body mass index from 16.0 to below 18.5 kg/m<sup>2</sup>, rule UW-BMI-002 would apply as well; its rating is a debit of +25 %, and the two together would give +175 %.

Example 4. An applicant aged 40, a joiner, has chronic obstructive pulmonary disease. The most recent clinic letter records FEV1 after a bronchodilator 28 % of predicted. That meets rule UW-COPD-004 (FEV1 after a bronchodilator below 30 % of predicted), so the probable rating is decline.

Readings on an edge: FEV1 after a bronchodilator 30 % of predicted exactly, rule UW-COPD-003 (a debit of +150 %); FEV1 after a bronchodilator 50 % of predicted exactly, rule UW-COPD-002 (a debit of +50 %); FEV1 after a bronchodilator 80 % of predicted exactly, rule UW-COPD-001 (a debit of +25 %).

**Common combinations.**

Continued smoking is added from the tobacco section, as is the lifetime total. Lung cancer and heart disease, not breathing failure, cause many of the deaths in this group, so a past heart attack is added in full.

Unintended weight loss into the underweight range is a marker of advanced disease and is added from the build section.

**Related rules.**

Under rule UW-COPD-002, if the applicant also has a history of smoking with smoking status: current smoker, see rule UW-TOB-001 (section 21.4, Tobacco use).

Under rule UW-COPD-003, if the applicant also has an unusual build with body mass index from 16.0 to below 18.5 kg/m<sup>2</sup>, see rule UW-BMI-002 (section 5.4, Build: underweight and obesity).

Rules of other sections that point to this one: rule UW-TOB-002 (section 21.4, Tobacco use); rule UW-AST-002 (section 24.4, Asthma).

## 26 Obstructive sleep apnoea

### 26.1 The impairment

In obstructive sleep apnoea the throat closes repeatedly during sleep. Each closure drops the blood oxygen and briefly wakes the sleeper, who is then sleepy by day.

Untreated severe disease raises blood pressure and the risk of heart rhythm problems and road accidents. Treatment with a pressure mask at night works well when it is used.

When this section applies. Obstructive sleep apnoea has been diagnosed by a sleep study.

What is measured. The rating in this section depends on apnoea-hypopnoea index (events per hour). The thresholds are grounded in: American Academy of Sleep Medicine, Clinical Guideline for the Evaluation, Management and Long-term Care of Obstructive Sleep Apnea in Adults (2009).

### 26.2 Key questions

#### 1. What index did the sleep study report before treatment?

The table is read on the untreated figure, since that is the severity of the disease.

#### 2. Is a pressure mask used, for how many hours a night?

Use for four hours or more on most nights is the usual test of adequate treatment, and the machine records it.

#### 3. Has the applicant fallen asleep while driving?

Sleepiness at the wheel is the most immediate danger.

### 26.3 Evidence and readings

Use the apnoea-hypopnoea index from the diagnostic sleep study, not from the treatment machine's nightly report.

Good recorded use of a pressure mask does not move the applicant to a lower band, but it should be noted, and a person may take it into account.

**Pitfalls with this evidence.**

Home studies that do not record sleep itself tend to give a lower index than a laboratory study. The manual does not adjust for this.

The oxygen desaturation index is a different number. Do not read it on this table.

### 26.4 Probable rating

The table lists every rule of this section; each is defined in full below it. The debit percentages and the decisions to decline are invented for this synthetic manual. The thresholds follow the public source named in each rule, except the edges a rule marks as this manual's own.

| Rule id    | Measure and band                                          | Probable rating  |
|------------|-----------------------------------------------------------|------------------|
| UW-OSA-001 | apnoea-hypopnoea index from 5 to below 15 events per hour | no debit, +0 %   |
| UW-OSA-002 | apnoea-hypopnoea index from 15 to 30 events per hour      | a debit of +25 % |
| UW-OSA-003 | apnoea-hypopnoea index above 30 events per hour           | a debit of +75 % |

**Readings that meet no rule.**

apnoea-hypopnoea index below 5 events per hour: no rule; not sleep apnoea.

Note on the bands. 5, 15 and 30 events per hour are the limits of mild, moderate and severe disease in the cited guideline.

Rule UW-OSA-001: Obstructive sleep apnoea (section 26.4).

Threshold: apnoea-hypopnoea index from 5 to below 15 events per hour. Probable rating: no debit, +0 %. Source of the threshold: AASM, Clinical Guideline for the Evaluation, Management and Long-term Care of Obstructive Sleep Apnea in Adults (2009), the severity grades by apnoea-hypopnoea index.

Rule UW-OSA-002: Obstructive sleep apnoea (section 26.4).

Threshold: apnoea-hypopnoea index from 15 to 30 events per hour. Probable rating: a debit of +25 %. If the applicant also has hypertension with systolic blood pressure from 140 to below 160 mmHg, see rule UW-HT-002. Source of the threshold: AASM, Clinical Guideline for the Evaluation, Management and Long-term Care of Obstructive Sleep Apnea in Adults (2009), the severity grades by apnoea-hypopnoea index.

Rule UW-OSA-003: Obstructive sleep apnoea (section 26.4).

Threshold: apnoea-hypopnoea index above 30 events per hour. Probable rating: a debit of +75 %. If the applicant also has an unusual build with body mass index of 40.0 kg/m<sup>2</sup> or more, see rule UW-BMI-004. Source of the threshold: AASM, Clinical Guideline for the Evaluation, Management and Long-term Care of Obstructive Sleep Apnea in Adults (2009), the severity grades by apnoea-hypopnoea index.

### **What does not change the rating.**

Use of a pressure mask, a jaw splint or surgery to the throat does not change the band, which is read on the untreated index. Snoring without a sleep study meets no rule.

## **26.5 Worked examples and related rules**

Example 1. The specialist's letter for a payroll clerk aged 44 with obstructive sleep apnoea records apnoea-hypopnoea index 10 events per hour. This is covered by rule UW-OSA-001 (apnoea-hypopnoea index from 5 to below 15 events per hour). Probable rating: no debit, +0 %.

Example 2. An applicant aged 55, a librarian, has obstructive sleep apnoea. The attending physician's statement records apnoea-hypopnoea index 22 events per hour. That meets rule UW-OSA-002 (apnoea-hypopnoea index from 15 to 30 events per hour), so the probable rating is a debit of +25 %. If the applicant also has hypertension with systolic blood pressure from 140 to below 160 mmHg, rule UW-HT-002 would apply as well; its rating is a debit of +50 %, and the two together would give +75 %.

Example 3. Take a florist of 36 with obstructive sleep apnoea. On file, from the most recent clinic letter: apnoea-hypopnoea index 32 events per hour. The threshold met is that of rule UW-OSA-003 (apnoea-hypopnoea index above 30 events per hour), and the probable rating is a debit of +75 %. If the applicant also has an unusual build with body mass index of 40.0 kg/m<sup>2</sup> or more, rule UW-BMI-004 would apply as well; its rating is a debit of +100 %, and the two together would give +175 %.

Readings on an edge: apnoea-hypopnoea index 5 events per hour exactly, rule UW-OSA-001 (no debit, +0 %); apnoea-hypopnoea index 15 events per hour exactly, rule UW-OSA-002 (a debit of +25 %); apnoea-hypopnoea index 30 events per hour exactly, rule UW-OSA-002 (a debit of +25 %).

### **Common combinations.**

Sleep apnoea and obesity come as a pair in most cases: both debits are added. Blood pressure that resists three medicines is typical and is rated in its section.

Opioid medicines and sedatives deepen the pauses in breathing; the opioid section's debit is added.

**Related rules.**

Under rule UW-OSA-002, if the applicant also has hypertension with systolic blood pressure from 140 to below 160 mmHg, see rule UW-HT-002 (section 10.4, Hypertension).

Under rule UW-OSA-003, if the applicant also has an unusual build with body mass index of 40.0 kg/m<sup>2</sup> or more, see rule UW-BMI-004 (section 5.4, Build: underweight and obesity).

Rules of other sections that point to this one: rule UW-BMI-003 (section 5.4, Build: underweight and obesity); rule UW-OPI-003 (section 23.4, Long-term opioid therapy).

## 27 Chronic kidney disease

### 27.1 The impairment

Chronic kidney disease is a loss of kidney function, or a sign of kidney damage, that has lasted three months or more. Diabetes and high blood pressure cause most of it.

Few people with reduced kidney function reach dialysis; far more die early of heart disease, and that risk rises step by step as filtration falls. The rating follows the standard filtration categories.

When this section applies. Two eGFR results at least three months apart are on file.

What is measured. The rating in this section depends on eGFR (mL/min/1.73 m<sup>2</sup>). The thresholds are grounded in: Kidney Disease: Improving Global Outcomes, Clinical Practice Guideline for the Evaluation and Management of Chronic Kidney Disease (2024).

### 27.2 Key questions

#### 1. What were the last three eGFR results, and over what period?

A stable figure over years is a different risk from the same figure reached by a fall of 10 in twelve months.

#### 2. What is the cause of the kidney disease?

Inherited cysts, inflammation of the filters and diabetes each run a different course.

#### 3. Is there albumin in the urine?

Albumin and filtration together predict the outcome better than either alone; albumin has its own section.

#### 4. Is the applicant under a kidney clinic?

Referral usually happens once the eGFR is under 30 or falling quickly.

## 27.3 Evidence and readings

Use the most recent eGFR, provided a second result at least three months earlier was also below 60; one low reading does not establish chronic disease.

An eGFR of 60 or more meets no rule of this section, even when a diagnosis of early kidney disease is recorded on the strength of albumin alone.

### Pitfalls with this evidence.

eGFR is worked out from creatinine, which depends on muscle. It reads low in a very muscular person or after a large meat meal and high in a frail one. A result that does not fit the person should be repeated.

A single low result during an acute illness, dehydration or a course of anti-inflammatory tablets is not chronic disease.

## 27.4 Probable rating

The table lists every rule of this section; each is defined in full below it. The debit percentages and the decisions to decline are invented for this synthetic manual. The thresholds follow the public source named in each rule, except the edges a rule marks as this manual's own.

| Rule id    | Measure and band                        | Probable rating   |
|------------|-----------------------------------------|-------------------|
| UW-CKD-001 | eGFR from 45 to below 60 mL/min/1.73 m2 | a debit of +50 %  |
| UW-CKD-002 | eGFR from 30 to below 45 mL/min/1.73 m2 | a debit of +100 % |
| UW-CKD-003 | eGFR from 15 to below 30 mL/min/1.73 m2 | a debit of +200 % |
| UW-CKD-004 | eGFR below 15 mL/min/1.73 m2            | decline           |

**Readings that meet no rule.**

eGFR of 60 mL/min/1.73 m<sup>2</sup> or more: no rule of this section; albumin in the urine is read in its own section.

Note on the bands. 60, 45, 30 and 15 are the limits of categories G3a, G3b, G4 and G5 in the cited guideline.

Rule UW-CKD-001: Chronic kidney disease (section 27.4). Threshold: eGFR from 45 to below 60 mL/min/1.73 m<sup>2</sup>. Probable rating: a debit of +50 %. If the applicant also has albumin in the urine with urine albumin-to-creatinine ratio from 30 to 300 mg/g, see rule UW-ALB-001. Source of the threshold: KDIGO, Clinical Practice Guideline for the Evaluation and Management of Chronic Kidney Disease (2024), the GFR categories G1 to G5 and the albuminuria categories A1 to A3.

Rule UW-CKD-002: Chronic kidney disease (section 27.4). Threshold: eGFR from 30 to below 45 mL/min/1.73 m<sup>2</sup>. Probable rating: a debit of +100 %. If the applicant also has hypertension with systolic blood pressure from 140 to below 160 mmHg, see rule UW-HT-002. Source of the threshold: KDIGO, Clinical Practice Guideline for the Evaluation and Management of Chronic Kidney Disease (2024), the GFR categories G1 to G5 and the albuminuria categories A1 to A3.

Rule UW-CKD-003: Chronic kidney disease (section 27.4). Threshold: eGFR from 15 to below 30 mL/min/1.73 m<sup>2</sup>. Probable rating: a debit of +200 %. If the applicant also has anaemia with haemoglobin from 8.0 to below 11.0 g/dL, see rule UW-ANA-001. Source of the threshold: KDIGO, Clinical Practice Guideline for the Evaluation and Management of Chronic Kidney Disease (2024), the GFR categories G1 to G5 and the albuminuria categories A1 to A3.

Rule UW-CKD-004: Chronic kidney disease (section 27.4). Threshold: eGFR below 15 mL/min/1.73 m<sup>2</sup>. Probable rating: decline. Source of the threshold: KDIGO, Clinical Practice Guideline for the Evaluation and Management of Chronic Kidney Disease (2024), the GFR categories G1 to G5 and the albuminuria categories A1 to A3.

### **What does not change the rating.**

The cause of the kidney disease does not change the band, although inherited cystic disease and inflammation of the filters go to a person. A single kidney with an eGFR of 60 or more meets no rule.

## **27.5 Worked examples and related rules**

Example 1. Take a translator of 51 with chronic kidney disease. On file, from the attending physician's statement: eGFR 52 mL/min/1.73 m<sup>2</sup>. The threshold met is that of rule UW-CKD-001 (eGFR from 45 to below 60 mL/min/1.73 m<sup>2</sup>), and the probable rating is a debit of +50 %. If the applicant also has albumin in the urine with urine albumin-to-creatinine ratio from 30 to 300 mg/g, rule UW-ALB-001 would apply as well; its rating is a debit of +25 %, and the two together would give +75 %.

Example 2. The most recent clinic letter for a surveyor aged 32 with chronic kidney disease records eGFR 38 mL/min/1.73 m<sup>2</sup>. This is covered by rule UW-CKD-002 (eGFR from 30 to below 45 mL/min/1.73 m<sup>2</sup>). Probable rating: a debit of +100 %. If the applicant also has hypertension with systolic blood pressure from 140 to below 160 mmHg, rule UW-HT-002 would apply as well; its rating is a debit of +50 %, and the two together would give +150 %.

Example 3. An applicant aged 43, a dental technician, has chronic kidney disease. The report on file records eGFR 22 mL/min/1.73 m<sup>2</sup>. That meets rule UW-CKD-003 (eGFR from 15 to below 30 mL/min/1.73 m<sup>2</sup>), so the probable rating is a debit of +200 %. If the applicant also has anaemia with haemoglobin from 8.0 to below 11.0 g/dL, rule UW-ANA-001 would apply as well; its rating is a debit of +50 %, and the two together would give +250 %.

Example 4. Take a pastry cook of 54 with chronic kidney disease. On file, from the specialist's letter: eGFR 14 mL/min/1.73 m<sup>2</sup>. The threshold met is that of rule UW-CKD-004 (eGFR below 15 mL/min/1.73 m<sup>2</sup>), and the probable rating is decline.

Readings on an edge: eGFR 15 mL/min/1.73 m<sup>2</sup> exactly, rule UW-CKD-003 (a debit of +200 %); eGFR 30 mL/min/1.73 m<sup>2</sup> exactly, rule UW-CKD-002 (a debit of +100 %); eGFR 45 mL/min/1.73 m<sup>2</sup> exactly, rule UW-CKD-001 (a debit of +50 %); eGFR 60 mL/min/1.73 m<sup>2</sup> exactly, no rule of this section.

### **Common combinations.**

Albumin in the urine is rated in its own section and always added: a low eGFR with heavy albumin loss is far worse than either alone.

Diabetes and raised blood pressure cause most kidney disease and are added. Anaemia appears as filtration falls below 30 and is rated in its section.

### **Related rules.**

Under rule UW-CKD-001, if the applicant also has albumin in the urine with urine albumin-to-creatinine ratio from 30 to 300 mg/g, see rule UW-ALB-001 (section 28.4, Albuminuria).

Under rule UW-CKD-002, if the applicant also has hypertension with systolic blood pressure from 140 to below 160 mmHg, see rule UW-HT-002 (section 10.4, Hypertension).

Under rule UW-CKD-003, if the applicant also has anaemia with haemoglobin from 8.0 to below 11.0 g/dL, see rule UW-ANA-001 (section 34.4, Anaemia).

Rules of other sections that point to this one: rule UW-DM-003 (section 2.4, Type 2 diabetes mellitus); rule UW-DI-003 (section 3.4, Type 1 diabetes mellitus); rule UW-GOUT-002 (section 8.4, Gout); rule UW-HT-003 (section 10.4, Hypertension); rule UW-ALB-001 (section 28.4, Albuminuria); rule UW-ANA-001 (section 34.4, Anaemia).

## 28 Albuminuria

### 28.1 The impairment

Healthy kidneys keep albumin in the blood. Albumin in the urine means the filters are leaking, and it is often the first evidence of damage from diabetes or high blood pressure.

Even with normal filtration, albumin in the urine marks a higher risk of heart disease and of later kidney failure. It is rated here and combined with any rule the filtration rate meets.

When this section applies. A laboratory urine albumin-to-creatinine ratio is on file.

What is measured. The rating in this section depends on urine albumin-to-creatinine ratio (mg/g). The thresholds are grounded in: Kidney Disease: Improving Global Outcomes, Clinical Practice Guideline for the Evaluation and Management of Chronic Kidney Disease (2024).

### 28.2 Key questions

#### 1. Was the finding confirmed on a second sample?

Exercise, fever and infection can all cause a passing rise.

#### 2. Is a medicine that protects the kidney taken?

Treatment with one of the blood pressure medicines that lower albumin loss shows the finding is being acted on.

#### 3. Does the applicant have diabetes or high blood pressure?

Without either, the cause needs to be found before terms are offered.

## 28.3 Evidence and readings

Use a urine albumin-to-creatinine ratio from the last twelve months that was confirmed by a second sample within three months of it.

A dipstick result is not a ratio. Where only a dipstick is on file, ask for a laboratory ratio.

### Pitfalls with this evidence.

A sample taken during a urine infection, a fever or a period, or within a day of hard exercise, can read high. That is why a second sample is required.

A protein-to-creatinine ratio is a different test with different limits.

### Converting units.

urine albumin-to-creatinine ratio: A ratio in mg/mmol is multiplied by 8.84 to give mg/g: 3.4 mg/mmol is 30 mg/g and 34 mg/mmol is 300 mg/g.

## 28.4 Probable rating

The table lists every rule of this section; each is defined in full below it. The debit percentages and the decisions to decline are invented for this synthetic manual. The thresholds follow the public source named in each rule, except the edges a rule marks as this manual's own.

| Rule id    | Measure and band                                      | Probable rating  |
|------------|-------------------------------------------------------|------------------|
| UW-ALB-001 | urine albumin-to-creatinine ratio from 30 to 300 mg/g | a debit of +25 % |
| UW-ALB-002 | urine albumin-to-creatinine ratio above 300 mg/g      | a debit of +75 % |

### Readings that meet no rule.

urine albumin-to-creatinine ratio below 30 mg/g: no rule; a normal ratio.

Note on the bands. 30 and 300 mg/g are the limits of categories A2 and A3 in the cited guideline.

Rule UW-ALB-001: Albuminuria (section 28.4). Threshold: urine albumin-to-creatinine ratio from 30 to 300 mg/g. Probable rating: a debit of +25 %. If the applicant also has chronic kidney disease with eGFR from 45 to below 60 mL/min/1.73 m<sup>2</sup>, see rule UW-CKD-001. Source of the threshold: KDIGO, Clinical Practice Guideline for the Evaluation and Management of Chronic Kidney Disease (2024), the GFR categories G1 to G5 and the albuminuria categories A1 to A3.

Rule UW-ALB-002: Albuminuria (section 28.4). Threshold: urine albumin-to-creatinine ratio above 300 mg/g. Probable rating: a debit of +75 %. If the applicant also has type 2 diabetes with HbA1c from 7.0 to below 8.0 %, see rule UW-DM-002. Source of the threshold: KDIGO, Clinical Practice Guideline for the Evaluation and Management of Chronic Kidney Disease (2024), the GFR categories G1 to G5 and the albuminuria categories A1 to A3.

### **What does not change the rating.**

Taking a kidney-protecting blood pressure medicine does not change the band. A normal eGFR does not lower it.

## **28.5 Worked examples and related rules**

Example 1. An applicant aged 58, a bus driver, has albumin in the urine. The most recent clinic letter records urine albumin-to-creatinine ratio 165 mg/g. That meets rule UW-ALB-001 (urine albumin-to-creatinine ratio from 30 to 300 mg/g), so the probable rating is a debit of +25 %. If the applicant also has chronic kidney disease with eGFR from 45 to below 60 mL/min/1.73 m<sup>2</sup>, rule UW-CKD-001 would apply as well; its rating is a debit of +50 %, and the two together would give +75 %.

Example 2. Take a payroll clerk of 39 with albumin in the urine. On file, from the report on file: urine albumin-to-creatinine ratio 324 mg/g. The threshold met is that of rule UW-ALB-002 (urine albumin-to-creatinine ratio above 300 mg/g), and the probable rating is a debit of +75 %. If the applicant also has type 2 diabetes with HbA1c from 7.0 to below 8.0 %, rule UW-DM-002 would apply as well; its rating is a debit of +50 %, and the two together would give +125 %.

Readings on an edge: urine albumin-to-creatinine ratio 30 mg/g exactly, rule UW-ALB-001 (a debit of +25 %); urine albumin-to-creatinine ratio 300 mg/g exactly, rule UW-ALB-001 (a debit of +25 %).

### **Common combinations.**

With diabetes, albumin in the urine marks the start of diabetic kidney disease; both debits are added. With raised blood pressure it shows the pressure has done damage.

With a reduced eGFR, both kidney sections apply and are added.

### **Related rules.**

Under rule UW-ALB-001, if the applicant also has chronic kidney disease with eGFR from 45 to below 60 mL/min/1.73 m<sup>2</sup>, see rule UW-CKD-001 (section 27.4, Chronic kidney disease).

Under rule UW-ALB-002, if the applicant also has type 2 diabetes with HbA1c from 7.0 to below 8.0 %, see rule UW-DM-002 (section 2.4, Type 2 diabetes mellitus).

Rules of other sections that point to this one: rule UW-DM-001 (section 2.4, Type 2 diabetes mellitus); rule UW-DI-001 (section 3.4, Type 1 diabetes mellitus); rule UW-CKD-001 (section 27.4, Chronic kidney disease).

## 29 Chronic hepatitis B

### 29.1 The impairment

Chronic hepatitis B is infection with the hepatitis B virus that has lasted more than six months. Many carriers stay well for life; in others the virus slowly scars the liver and can cause liver cancer.

The amount of virus in the blood is the main guide to who is at risk and who needs antiviral treatment.

When this section applies. Hepatitis B surface antigen has been positive for more than six months.

What is measured. The rating in this section depends on HBV DNA (IU/mL). The thresholds are grounded in: American Association for the Study of Liver Diseases, Update on Prevention, Diagnosis, and Treatment of Chronic Hepatitis B: AASLD 2018 Hepatitis B Guidance (2018).

### 29.2 Key questions

#### 1. What are the latest HBV DNA level and liver enzymes?

A high viral level with raised enzymes means active liver inflammation.

#### 2. Is antiviral treatment taken?

The table is read with the level on treatment; a suppressed level on long-term treatment is a good sign.

#### 3. Has scarring been assessed by a scan or a blood index?

Scarring already present outweighs the viral level.

#### 4. Is the liver scanned every six months?

Regular scans are how liver cancer is caught early in carriers.

## 29.3 Evidence and readings

Use an HBV DNA result from the last twelve months, reported in IU/mL. Convert a result in copies/mL as shown below and record both figures.

Where cirrhosis has been diagnosed, the cirrhosis section governs and this section's rule is combined with it.

### Pitfalls with this evidence.

Antibody to the core of the virus with a negative surface antigen means past, cleared infection. It is not chronic hepatitis B and meets no rule.

Viral levels swing. One low result in an untreated carrier does not show lasting control; look for two results six months apart.

### Converting units.

HBV DNA: A result in copies/mL is divided by 5 to give IU/mL, unless the laboratory report states its own factor.

## 29.4 Probable rating

The table lists every rule of this section; each is defined in full below it. The debit percentages and the decisions to decline are invented for this synthetic manual. The thresholds follow the public source named in each rule, except the edges a rule marks as this manual's own.

| Rule id    | Measure and band                         | Probable rating   |
|------------|------------------------------------------|-------------------|
| UW-HBV-001 | HBV DNA of 2000 IU/mL or less            | a debit of +25 %  |
| UW-HBV-002 | HBV DNA above 2000 and up to 20000 IU/mL | a debit of +75 %  |
| UW-HBV-003 | HBV DNA above 20000 IU/mL                | a debit of +150 % |

**Readings that meet no rule.**

None: every reading these measures can take falls in one of the bands above.

Note on the bands. 2,000 and 20,000 IU/mL are the treatment levels in the cited guidance.

Rule UW-HBV-001: Chronic hepatitis B (section 29.4). Threshold: HBV DNA of 2000 IU/mL or less. Probable rating: a debit of +25 %. If the applicant also has HIV infection with CD4 count of 500 cells/mm3 or more, see rule UW-HIV-001. Source of the threshold: AASLD, Update on Prevention, Diagnosis, and Treatment of Chronic Hepatitis B: AASLD 2018 Hepatitis B Guidance (2018), the HBV DNA levels used to decide on treatment.

Rule UW-HBV-002: Chronic hepatitis B (section 29.4). Threshold: HBV DNA above 2000 and up to 20000 IU/mL. Probable rating: a debit of +75 %. Source of the threshold: AASLD, Update on Prevention, Diagnosis, and Treatment of Chronic Hepatitis B: AASLD 2018 Hepatitis B Guidance (2018), the HBV DNA levels used to decide on treatment.

Rule UW-HBV-003: Chronic hepatitis B (section 29.4). Threshold: HBV DNA above 20000 IU/mL. Probable rating: a debit of +150 %. If the applicant also has cirrhosis with MELD score below 15 points, see rule UW-CIR-001. Source of the threshold: AASLD, Update on Prevention, Diagnosis, and Treatment of Chronic Hepatitis B: AASLD 2018 Hepatitis B Guidance (2018), the HBV DNA levels used to decide on treatment.

**What does not change the rating.**

How the infection was acquired, and whether the e antigen is positive, do not change the band in this manual.

## 29.5 Worked examples and related rules

Example 1. The report on file for a school caretaker aged 35 with chronic hepatitis B records HBV DNA 1840 IU/mL. This is covered by rule UW-HBV-001 (HBV DNA of 2000 IU/mL or less). Probable rating: a debit of +25 %. If the applicant also has HIV infection with CD4 count of 500 cells/mm<sup>3</sup> or more, rule UW-HIV-001 would apply as well; its rating is a debit of +50 %, and the two together would give +75 %.

Example 2. An applicant aged 46, a translator, has chronic hepatitis B. The specialist's letter records HBV DNA 11000 IU/mL. That meets rule UW-HBV-002 (HBV DNA above 2000 and up to 20000 IU/mL), so the probable rating is a debit of +75 %.

Example 3. Take a surveyor of 57 with chronic hepatitis B. On file, from the attending physician's statement: HBV DNA 21600 IU/mL. The threshold met is that of rule UW-HBV-003 (HBV DNA above 20000 IU/mL), and the probable rating is a debit of +150 %. If the applicant also has cirrhosis with MELD score below 15 points, rule UW-CIR-001 would apply as well; its rating is a debit of +200 %, and the two together would give +350 %.

Readings on an edge: HBV DNA 2000 IU/mL exactly, rule UW-HBV-001 (a debit of +25 %); HBV DNA 20000 IU/mL exactly, rule UW-HBV-002 (a debit of +75 %).

### Common combinations.

Cirrhosis, once present, governs the outlook; its rule is added to the rule here. Infection with HIV or hepatitis C as well speeds liver damage; each is rated in its section.

Alcohol and hepatitis B together are far more harmful than either alone: add the alcohol debit.

**Related rules.**

Under rule UW-HBV-001, if the applicant also has HIV infection with CD4 count of 500 cells/mm<sup>3</sup> or more, see rule UW-HIV-001 (section 36.4, HIV infection).

Under rule UW-HBV-003, if the applicant also has cirrhosis with MELD score below 15 points, see rule UW-CIR-001 (section 32.4, Cirrhosis of the liver).

Rules of other sections that point to this one: rule UW-HIV-002 (section 36.4, HIV infection).

## 30 Chronic hepatitis C

### 30.1 The impairment

Hepatitis C is a blood-borne virus that, untreated, inflames the liver for decades and can end in cirrhosis. Short courses of antiviral tablets now clear it in nearly everyone treated.

Cure is confirmed when no virus is found twelve weeks after the end of treatment. The rating turns on whether that point has been reached.

When this section applies. Hepatitis C virus RNA has been detected at any time.

What is measured. The rating in this section depends on time with undetectable HCV RNA after the end of treatment (weeks). The thresholds are grounded in: American Association for the Study of Liver Diseases and Infectious Diseases Society of America, HCV Guidance: Recommendations for Testing, Managing, and Treating Hepatitis C (2023).

### 30.2 Key questions

#### 1. Has antiviral treatment been completed, and on what date?

The weeks in the table are counted from the last dose.

#### 2. What did the test twelve weeks or more after treatment show?

It is the test that confirms cure.

#### 3. How much scarring was there before treatment?

Clearing the virus stops further damage but does not remove scarring already present.

### 30.3 Evidence and readings

Count whole weeks from the end of treatment to the latest test that found no HCV RNA. An untreated applicant, or one in whom virus was found after treatment, counts as zero weeks.

Antibody to hepatitis C stays positive for life after cure and is not evidence of infection.

#### **Pitfalls with this evidence.**

A positive antibody test with RNA never detected means the infection cleared by itself. That is not chronic hepatitis C and meets no rule.

A test at the end of treatment is not the test of cure. Only a result twelve weeks or more after the last dose counts.

### 30.4 Probable rating

The table lists every rule of this section; each is defined in full below it. The debit percentages and the decisions to decline are invented for this synthetic manual. The thresholds follow the public source named in each rule, except the edges a rule marks as this manual's own.

| Rule id    | Measure and band                                                              | Probable rating   |
|------------|-------------------------------------------------------------------------------|-------------------|
| UW-HCV-001 | time with undetectable HCV RNA after the end of treatment below 12 weeks      | a debit of +100 % |
| UW-HCV-002 | time with undetectable HCV RNA after the end of treatment of 12 weeks or more | a debit of +25 %  |

#### **Readings that meet no rule.**

None: every reading these measures can take falls in one of the bands above.

Note on the bands. Twelve weeks is the point at which the cited guidance defines sustained virologic response.

Rule UW-HCV-001: Chronic hepatitis C (section 30.4). Threshold: time with undetectable HCV RNA after the end of treatment below 12 weeks. Probable rating: a debit of +100 %. If the applicant also has cirrhosis with MELD score below 15 points, see rule UW-CIR-001. Source of the threshold: AASLD-IDSA, HCV Guidance: Recommendations for Testing, Managing, and Treating Hepatitis C (2023), the definition of sustained virologic response 12 weeks after treatment.

Rule UW-HCV-002: Chronic hepatitis C (section 30.4). Threshold: time with undetectable HCV RNA after the end of treatment of 12 weeks or more. Probable rating: a debit of +25 %. If the applicant also has fatty liver disease with FIB-4 index from 1.3 to 2.67, see rule UW-FLD-002. Source of the threshold: AASLD-IDSA, HCV Guidance: Recommendations for Testing, Managing, and Treating Hepatitis C (2023), the definition of sustained virologic response 12 weeks after treatment.

### **What does not change the rating.**

The genotype of the virus, the medicines used and the route of infection do not change the band.

## **30.5 Worked examples and related rules**

Example 1. Take a joiner of 42 with hepatitis C. On file, from the specialist's letter: time with undetectable HCV RNA after the end of treatment 11 weeks. The threshold met is that of rule UW-HCV-001 (time with undetectable HCV RNA after the end of treatment below 12 weeks), and the probable rating is a debit of +100 %. If the applicant also has cirrhosis with MELD score below 15 points, rule UW-CIR-001 would apply as well; its rating is a debit of +200 %, and the two together would give +300 %.

Example 2. The attending physician's statement for a bus driver aged 53 with hepatitis C records time with undetectable HCV RNA after the end of treatment 13 weeks. This is covered by rule UW-HCV-002 (time with undetectable HCV RNA after the end of treatment of 12 weeks or more). Probable rating: a debit of +25 %. If the applicant also has fatty liver disease with FIB-4 index from 1.3 to 2.67, rule UW-FLD-002 would apply as well; its rating is a debit of +50 %, and the two together would give +75 %.

Readings on an edge: time with undetectable HCV RNA after the end of treatment 12 weeks exactly, rule UW-HCV-002 (a debit of +25 %).

### **Common combinations.**

Scarring that was present before cure remains. Where fatty liver disease has been diagnosed as well, its FIB-4 band is rated in that section and added; cirrhosis is rated in its own section.

A history of injecting drugs goes to a person. Alcohol is added as usual.

### **Related rules.**

Under rule UW-HCV-001, if the applicant also has cirrhosis with MELD score below 15 points, see rule UW-CIR-001 (section 32.4, Cirrhosis of the liver).

Under rule UW-HCV-002, if the applicant also has fatty liver disease with FIB-4 index from 1.3 to 2.67, see rule UW-FLD-002 (section 31.4, Fatty liver disease).

Rules of other sections that point to this one: rule UW-HEM-001 (section 35.4, Haemophilia).

## **31 Fatty liver disease**

### **31.1 The impairment**

Fat builds up in the liver of many people who are overweight or have diabetes. In most it stays harmless; in a minority it inflames and scars the liver over many years.

Scarring, not fat, decides the outlook. A simple index from routine blood tests sorts those at low risk of advanced scarring from those who need a scan.

When this section applies. Fatty liver has been reported on imaging or diagnosed by a physician.

What is measured. The rating in this section depends on FIB-4 index. The thresholds are grounded in: American Association for the Study of Liver Diseases, Practice Guidance on the clinical assessment and management of nonalcoholic fatty liver disease (2023).

### **31.2 Key questions**

#### **1. How was the fatty liver found?**

Most are chance findings on ultrasound; one found while investigating abnormal blood tests deserves a closer look.

#### **2. How much alcohol is taken?**

Alcohol produces the same picture and has its own section.

#### **3. Has a scan of liver stiffness been done?**

It is the usual next step when the index is not clearly low.

### **31.3 Evidence and readings**

Work out the FIB-4 index from age, AST, ALT and platelet count taken on the same day within the last twelve months, or use the figure the clinic recorded. Round to two decimal places.

The index is unreliable under the age of 35; for a younger applicant with a raised index, send the case to a person.

### **Pitfalls with this evidence.**

The index rises with age by its construction and overstates risk over 65, where the lower cut-off in practice is often taken as 2.0. The manual keeps one set of bands and sends applicants over 65 with an index from 1.3 to below 2.0 to a person.

AST and ALT taken after heavy exercise or during an acute illness distort the index.

### **Converting units.**

FIB-4 index: The index is age in years times AST, divided by the platelet count in thousands per microlitre times the square root of ALT.

## **31.4 Probable rating**

The table lists every rule of this section; each is defined in full below it. The debit percentages and the decisions to decline are invented for this synthetic manual. The thresholds follow the public source named in each rule, except the edges a rule marks as this manual's own.

| <b>Rule id</b> | <b>Measure and band</b>      | <b>Probable rating</b> |
|----------------|------------------------------|------------------------|
| UW-FLD-001     | FIB-4 index below 1.3        | no debit, +0 %         |
| UW-FLD-002     | FIB-4 index from 1.3 to 2.67 | a debit of +50 %       |
| UW-FLD-003     | FIB-4 index above 2.67       | a debit of +150 %      |

### **Readings that meet no rule.**

None: every reading these measures can take falls in one of the bands above.

Note on the bands. 1.3 and 2.67 are the cut-offs for low and high risk of advanced fibrosis in the cited guidance.

Rule UW-FLD-001: Fatty liver disease (section 31.4). Threshold: FIB-4 index below 1.3. Probable rating: no debit, +0 %. Source of the threshold: AASLD, Practice Guidance on the clinical assessment and management of nonalcoholic fatty liver disease (2023), the FIB-4 cut-offs of 1.3 and 2.67 for the risk of advanced fibrosis.

Rule UW-FLD-002: Fatty liver disease (section 31.4). Threshold: FIB-4 index from 1.3 to 2.67. Probable rating: a debit of +50 %. If the applicant also has an unusual build with body mass index from 35.0 to below 40.0 kg/m<sup>2</sup>, see rule UW-BMI-003. Source of the threshold: AASLD, Practice Guidance on the clinical assessment and management of nonalcoholic fatty liver disease (2023), the FIB-4 cut-offs of 1.3 and 2.67 for the risk of advanced fibrosis.

Rule UW-FLD-003: Fatty liver disease (section 31.4). Threshold: FIB-4 index above 2.67. Probable rating: a debit of +150 %. If the applicant also has cirrhosis with MELD score below 15 points, see rule UW-CIR-001. Source of the threshold: AASLD, Practice Guidance on the clinical assessment and management of nonalcoholic fatty liver disease (2023), the FIB-4 cut-offs of 1.3 and 2.67 for the risk of advanced fibrosis.

### **What does not change the rating.**

The amount of fat seen on the scan and mildly raised liver enzymes do not change the band.

## **31.5 Worked examples and related rules**

Example 1. An applicant aged 49, a market gardener, has fatty liver disease. The attending physician's statement records FIB-4 index 1.2. That meets rule UW-FLD-001 (FIB-4 index below 1.3), so the probable rating is no debit, +0 %.

Example 2. Take a school caretaker of 60 with fatty liver disease. On file, from the most recent clinic letter: FIB-4 index 1.98. The threshold met is that of rule UW-FLD-002 (FIB-4 index from 1.3 to 2.67), and the probable rating is a debit of +50 %. If the applicant also has an unusual build with body mass index from 35.0 to below 40.0 kg/m<sup>2</sup>, rule UW-BMI-003 would apply as well; its rating is a debit of +50 %, and the two together would give +100 %.

Example 3. The report on file for a translator aged 41 with fatty liver disease records FIB-4 index 2.88. This is covered by rule UW-FLD-003 (FIB-4 index above 2.67). Probable rating: a debit of +150 %. If the applicant also has cirrhosis with MELD score below 15 points, rule UW-CIR-001 would apply as well; its rating is a debit of +200 %, and the two together would give +350 %.

Readings on an edge: FIB-4 index 1.3 exactly, rule UW-FLD-002 (a debit of +50 %); FIB-4 index 2.67 exactly, rule UW-FLD-002 (a debit of +50 %).

### **Common combinations.**

Fatty liver disease is the liver's share of the metabolic cluster. Obesity, diabetes and raised triglycerides are each rated and added.

Where alcohol is also taken at a hazardous level, the alcohol section applies as well. A liver stiffness scan that suggests cirrhosis moves the case to that section.

### **Related rules.**

Under rule UW-FLD-002, if the applicant also has an unusual build with body mass index from 35.0 to below 40.0 kg/m<sup>2</sup>, see rule UW-BMI-003 (section 5.4, Build: underweight and obesity).

Under rule UW-FLD-003, if the applicant also has cirrhosis with MELD score below 15 points, see rule UW-CIR-001 (section 32.4, Cirrhosis of the liver).

Rules of other sections that point to this one: rule UW-ALC-001 (section 22.4, Alcohol use); rule UW-HCV-002 (section 30.4, Chronic hepatitis C).

## **32 Cirrhosis of the liver**

### **32.1 The impairment**

Cirrhosis is widespread scarring that distorts the liver and blocks the flow of blood through it. Alcohol, viral hepatitis and fatty liver disease are the usual causes.

While the liver still does its work the condition is called compensated, and years of stable life are possible. Once fluid, bleeding or confusion appear, survival is short without a transplant.

When this section applies. Cirrhosis has been diagnosed by biopsy, by a stiffness scan or by a liver specialist.

What is measured. The rating in this section depends on MELD score (points). The thresholds are grounded in: American Association for the Study of Liver Diseases and American Society of Transplantation, Evaluation for Liver Transplantation in Adults: 2013 Practice Guideline (2013).

### **32.2 Key questions**

#### **1. What caused the cirrhosis, and has the cause been removed?**

Abstinence from alcohol or cure of hepatitis C can halt the disease.

#### **2. Has there ever been fluid in the abdomen, bleeding from the gullet or confusion?**

Any one of these means decompensation, which takes the case out of the rated band.

#### **3. What are the latest bilirubin, creatinine and clotting results?**

They are the parts of the score this section rates on.

## 32.3 Evidence and readings

Use the MELD score recorded by the liver clinic in the last six months, or work it out from blood tests taken on one day.

Any past decompensation is rated as the declining rule of this section, whatever the present score.

### Pitfalls with this evidence.

Warfarin raises the clotting time and kidney disease raises creatinine, so either can inflate the score for reasons outside the liver.

A good score does not rule out past decompensation. Read the history for fluid, bleeding and confusion.

## 32.4 Probable rating

The table lists every rule of this section; each is defined in full below it. The debit percentages and the decisions to decline are invented for this synthetic manual. The thresholds follow the public source named in each rule, except the edges a rule marks as this manual's own.

| Rule id    | Measure and band                | Probable rating   |
|------------|---------------------------------|-------------------|
| UW-CIR-001 | MELD score below 15 points      | a debit of +200 % |
| UW-CIR-002 | MELD score of 15 points or more | decline           |

### Readings that meet no rule.

None: every reading these measures can take falls in one of the bands above.

Note on the bands. A score of 15 is the level at which the cited guideline advises evaluation for a transplant.

Rule UW-CIR-001: Cirrhosis of the liver (section 32.4). Threshold: MELD score below 15 points. Probable rating: a debit of +200 %. If the applicant also has a record of heavy drinking with AUDIT score from 16 to below 20 points, see rule UW-ALC-002. Source of the threshold: AASLD/AST, Evaluation for Liver Transplantation in Adults: 2013 Practice Guideline (2013), the recommendation to evaluate at a MELD score of 15 or more.

Rule UW-CIR-002: Cirrhosis of the liver (section 32.4). Threshold: MELD score of 15 points or more. Probable rating: decline. Source of the threshold: AASLD/AST, Evaluation for Liver Transplantation in Adults: 2013 Practice Guideline (2013), the recommendation to evaluate at a MELD score of 15 or more.

### **What does not change the rating.**

The cause of the cirrhosis does not change the band. A normal ultrasound does not lower it.

## **32.5 Worked examples and related rules**

Example 1. The most recent clinic letter for a pastry cook aged 56 with cirrhosis records MELD score 14 points. This is covered by rule UW-CIR-001 (MELD score below 15 points). Probable rating: a debit of +200 %. If the applicant also has a record of heavy drinking with AUDIT score from 16 to below 20 points, rule UW-ALC-002 would apply as well; its rating is a debit of +75 %, and the two together would give +275 %.

Example 2. An applicant aged 37, a joiner, has cirrhosis. The report on file records MELD score 16 points. That meets rule UW-CIR-002 (MELD score of 15 points or more), so the probable rating is decline.

Readings on an edge: MELD score 15 points exactly, rule UW-CIR-002 (decline).

**Common combinations.**

The cause is rated in its own section and added: alcohol by the AUDIT score, hepatitis B by viral level, hepatitis C by whether cure is confirmed.

Continued drinking with cirrhosis is the worst combination in this manual short of a decline; a liver cancer found on surveillance goes to a person.

**Related rules.**

Under rule UW-CIR-001, if the applicant also has a record of heavy drinking with AUDIT score from 16 to below 20 points, see rule UW-ALC-002 (section 22.4, Alcohol use).

Rules of other sections that point to this one: rule UW-ALC-003 (section 22.4, Alcohol use); rule UW-HBV-003 (section 29.4, Chronic hepatitis B); rule UW-HCV-001 (section 30.4, Chronic hepatitis C); rule UW-FLD-003 (section 31.4, Fatty liver disease).

## 33 Ulcerative colitis

### 33.1 The impairment

Ulcerative colitis is long-term inflammation of the lining of the large bowel, with bloody diarrhoea in flares and quiet spells between. Medicines keep most people well; some need the bowel removed.

Life expectancy is close to normal when the disease is quiet. Continuing inflammation brings flares, steroid courses and, over decades, a higher risk of bowel cancer.

When this section applies. Ulcerative colitis has been diagnosed at colonoscopy.

What is measured. The rating in this section depends on faecal calprotectin (mcg/g). The thresholds are grounded in: American Gastroenterological Association, Clinical Practice Guideline on the Role of Biomarkers for the Management of Ulcerative Colitis (2023).

### 33.2 Key questions

#### 1. How much of the bowel is affected?

Disease of the whole colon carries more cancer risk than disease of the rectum alone.

#### 2. How many flares needed steroids in the last two years?

Repeated steroid courses mean maintenance treatment is not holding.

#### 3. When was the last colonoscopy, and what did it show?

Surveillance looks for the early changes that come before cancer.

### 33.3 Evidence and readings

Use a faecal calprotectin from the last twelve months, taken when the applicant was not known to have a bowel infection.

An applicant whose colon has been removed is outside these bands and goes to a person.

#### **Pitfalls with this evidence.**

Anti-inflammatory painkillers and a bowel infection both raise calprotectin without a flare of colitis.

A single stool sample varies from day to day; a result near 150 mcg/g is best repeated.

#### **Converting units.**

faecal calprotectin: A result in mg/kg is the same number in mcg/g.

### 33.4 Probable rating

The table lists every rule of this section; each is defined in full below it. The debit percentages and the decisions to decline are invented for this synthetic manual. The thresholds follow the public source named in each rule, except the edges a rule marks as this manual's own.

| Rule id   | Measure and band                         | Probable rating  |
|-----------|------------------------------------------|------------------|
| UW-UC-001 | faecal calprotectin below 150 mcg/g      | a debit of +25 % |
| UW-UC-002 | faecal calprotectin of 150 mcg/g or more | a debit of +75 % |

#### **Readings that meet no rule.**

None: every reading these measures can take falls in one of the bands above.

Note on the bands. 150 mcg/g is the cut-off below which the cited guideline treats active inflammation as unlikely.

Rule UW-UC-001: Ulcerative colitis (section 33.4). Threshold: faecal calprotectin below 150 mcg/g. Probable rating: a debit of +25 %. Source of the threshold: AGA, Clinical Practice Guideline on the Role of Biomarkers for the Management of Ulcerative Colitis (2023), the faecal calprotectin cut-off of 150 mcg/g.

Rule UW-UC-002: Ulcerative colitis (section 33.4). Threshold: faecal calprotectin of 150 mcg/g or more. Probable rating: a debit of +75 %. If the applicant also has anaemia with haemoglobin from 8.0 to below 11.0 g/dL, see rule UW-ANA-001. Source of the threshold: AGA, Clinical Practice Guideline on the Role of Biomarkers for the Management of Ulcerative Colitis (2023), the faecal calprotectin cut-off of 150 mcg/g.

### **What does not change the rating.**

The medicines used, including biological medicines, and the length of bowel affected do not change the band.

## **33.5 Worked examples and related rules**

Example 1. Take a florist of 33 with ulcerative colitis. On file, from the report on file: faecal calprotectin 138 mcg/g. The threshold met is that of rule UW-UC-001 (faecal calprotectin below 150 mcg/g), and the probable rating is a debit of +25 %.

Example 2. The specialist's letter for a market gardener aged 44 with ulcerative colitis records faecal calprotectin 162 mcg/g. This is covered by rule UW-UC-002 (faecal calprotectin of 150 mcg/g or more). Probable rating: a debit of +75 %. If the applicant also has anaemia with haemoglobin from 8.0 to below 11.0 g/dL, rule UW-ANA-001 would apply as well; its rating is a debit of +50 %, and the two together would give +125 %.

Readings on an edge: faecal calprotectin 150 mcg/g exactly, rule UW-UC-002 (a debit of +75 %).

**Common combinations.**

Bleeding leads to anaemia, which is rated in its section and added.

Repeated steroid courses thin the bones.

Inflammation of the bile ducts occurs in a small minority, greatly raises the risk of cancer and goes to a person.

**Related rules.**

Under rule UW-UC-002, if the applicant also has anaemia with haemoglobin from 8.0 to below 11.0 g/dL, see rule UW-ANA-001 (section 34.4, Anaemia).

## 34 Anaemia

### 34.1 The impairment

Anaemia is a low level of haemoglobin. It is a sign rather than a disease, and its causes run from heavy periods and poor diet to kidney failure, bowel cancer and disorders of the bone marrow.

A mild, explained anaemia is of no consequence and meets no rule. Moderate anaemia is rated while its cause is confirmed; severe anaemia is postponed until it has been treated and explained.

When this section applies. A full blood count from the last six months is on file.

What is measured. The rating in this section depends on haemoglobin (g/dL). The thresholds are grounded in: World Health Organization, Haemoglobin concentrations for the diagnosis of anaemia and assessment of severity (2011).

### 34.2 Key questions

#### 1. What is the cause, and how was it established?

An unexplained anaemia in an adult must be assumed to hide something until shown otherwise.

#### 2. Has the bowel been investigated?

Iron deficiency in a man, or in a woman past the menopause, calls for a search for bleeding.

#### 3. Has the level recovered on treatment?

A normal level after iron confirms the diagnosis and ends the rating.

### 34.3 Evidence and readings

Use the haemoglobin from a full blood count in the last six months.  
Convert a result in g/L as shown below.

A haemoglobin of 11.0 g/dL or more meets no rule of this section.

#### Pitfalls with this evidence.

Haemoglobin runs lower at the end of pregnancy and for some weeks after heavy blood donation. Check for either before rating.

A carrier of thalassaemia has small red cells and a haemoglobin that sits just below the normal range for life; it usually stays at 11.0 g/dL or more and then meets no rule.

#### Converting units.

haemoglobin: A result in g/L is divided by ten to give g/dL.

### 34.4 Probable rating

The table lists every rule of this section; each is defined in full below it. The debit percentages and the decisions to decline are invented for this synthetic manual. The thresholds follow the public source named in each rule, except the edges a rule marks as this manual's own.

| Rule id    | Measure and band                        | Probable rating           |
|------------|-----------------------------------------|---------------------------|
| UW-ANA-001 | haemoglobin from 8.0 to below 11.0 g/dL | a debit of +50 %          |
| UW-ANA-002 | haemoglobin below 8.0 g/dL              | decline as a postponement |

#### Readings that meet no rule.

haemoglobin of 11.0 g/dL or more: no rule; mild anaemia or none.

Note on the bands. 8.0 and 11.0 g/dL are the limits of severe and moderate anaemia in the cited table for adults who are not pregnant.

Rule UW-ANA-001: Anaemia (section 34.4). Threshold: haemoglobin from 8.0 to below 11.0 g/dL. Probable rating: a debit of +50 %. If the applicant also has chronic kidney disease with eGFR from 30 to below 45 mL/min/1.73 m<sup>2</sup>, see rule UW-CKD-002. Source of the threshold: WHO, Haemoglobin concentrations for the diagnosis of anaemia and assessment of severity (2011), the table of haemoglobin levels for mild, moderate and severe anaemia.

Rule UW-ANA-002: Anaemia (section 34.4). Threshold: haemoglobin below 8.0 g/dL. Probable rating: decline as a postponement. This is a postponement until the anaemia has been treated and its cause is known. Source of the threshold: WHO, Haemoglobin concentrations for the diagnosis of anaemia and assessment of severity (2011), the table of haemoglobin levels for mild, moderate and severe anaemia.

### **What does not change the rating.**

The size of the red cells and the level of iron stores do not change the band, though they help explain it.

## **34.5 Worked examples and related rules**

Example 1. An applicant aged 40, a dental technician, has anaemia. The specialist's letter records haemoglobin 9.5 g/dL. That meets rule UW-ANA-001 (haemoglobin from 8.0 to below 11.0 g/dL), so the probable rating is a debit of +50 %. If the applicant also has chronic kidney disease with eGFR from 30 to below 45 mL/min/1.73 m<sup>2</sup>, rule UW-CKD-002 would apply as well; its rating is a debit of +100 %, and the two together would give +150 %.

Example 2. Take a pastry cook of 51 with anaemia. On file, from the attending physician's statement: haemoglobin 7.4 g/dL. The threshold met is that of rule UW-ANA-002 (haemoglobin below 8.0 g/dL), and the probable rating is decline as a postponement.

Readings on an edge: haemoglobin 8.0 g/dL exactly, rule UW-ANA-001 (a debit of +50 %); haemoglobin 11.0 g/dL exactly, no rule of this section.

**Common combinations.**

Anaemia is a consequence of other impairments more often than a disease in itself. With kidney disease, ulcerative colitis or rheumatoid arthritis, the debit here is added to the debit for the cause.

Anaemia with weight loss and no explanation is not rated at all until it has been investigated.

**Related rules.**

Under rule UW-ANA-001, if the applicant also has chronic kidney disease with eGFR from 30 to below 45 mL/min/1.73 m<sup>2</sup>, see rule UW-CKD-002 (section 27.4, Chronic kidney disease).

Rules of other sections that point to this one: rule UW-BMI-001 (section 5.4, Build: underweight and obesity); rule UW-CKD-003 (section 27.4, Chronic kidney disease); rule UW-UC-002 (section 33.4, Ulcerative colitis).

## 35 Haemophilia

### 35.1 The impairment

Haemophilia is an inherited shortage of clotting factor VIII or IX, almost always in men. Bleeding into joints and muscles is the hallmark, and without treatment the joints are destroyed.

Regular replacement of the missing factor has brought life expectancy close to normal. How little factor a person makes still decides how often bleeds happen and how much treatment is needed.

When this section applies. Haemophilia A or B has been diagnosed.

What is measured. The rating in this section depends on clotting factor level (IU/dL). The thresholds are grounded in: World Federation of Hemophilia, Guidelines for the Management of Hemophilia, third edition (2020).

### 35.2 Key questions

#### 1. What is the baseline factor level without treatment?

It is the figure the table is read on and it does not change through life.

#### 2. Is factor given regularly to prevent bleeds, or only when one occurs?

Regular prevention marks good care in severe disease.

#### 3. Has an inhibitor ever developed?

An antibody against the replacement factor makes bleeds much harder to treat.

#### 4. Was the applicant treated with blood products before 1992?

Older products carried hepatitis C and HIV.

### 35.3 Evidence and readings

Take the baseline factor level from the haemophilia centre's letter. A level measured shortly after an infusion is not a baseline.

A current inhibitor, or a bleed inside the skull at any time, takes the case to a person. A level of 40 IU/dL or more meets no rule of this section.

#### Pitfalls with this evidence.

Women who carry the gene can have a low factor level and are read on the same table.

A level measured during an illness or in pregnancy may be raised above baseline.

#### Converting units.

clotting factor level: A level given as a percentage of normal is the same number in IU/dL.

### 35.4 Probable rating

The table lists every rule of this section; each is defined in full below it. The debit percentages and the decisions to decline are invented for this synthetic manual. The thresholds follow the public source named in each rule, except the edges a rule marks as this manual's own.

| Rule id    | Measure and band                                 | Probable rating   |
|------------|--------------------------------------------------|-------------------|
| UW-HEM-001 | clotting factor level below 1 IU/dL              | a debit of +150 % |
| UW-HEM-002 | clotting factor level from 1 to 5 IU/dL          | a debit of +75 %  |
| UW-HEM-003 | clotting factor level above 5 and below 40 IU/dL | a debit of +25 %  |

#### Readings that meet no rule.

clotting factor level of 40 IU/dL or more: no rule; not haemophilia by factor level.

Note on the bands. 1, 5 and 40 IU/dL are the limits of severe, moderate and mild haemophilia in the cited guidelines.

Rule UW-HEM-001: Haemophilia (section 35.4). Threshold: clotting factor level below 1 IU/dL. Probable rating: a debit of +150 %. If the applicant also has hepatitis C with time with undetectable HCV RNA after the end of treatment below 12 weeks, see rule UW-HCV-001. Source of the threshold: WFH, Guidelines for the Management of Hemophilia, third edition (2020), the severity classes by clotting factor level.

Rule UW-HEM-002: Haemophilia (section 35.4). Threshold: clotting factor level from 1 to 5 IU/dL. Probable rating: a debit of +75 %. Source of the threshold: WFH, Guidelines for the Management of Hemophilia, third edition (2020), the severity classes by clotting factor level.

Rule UW-HEM-003: Haemophilia (section 35.4). Threshold: clotting factor level above 5 and below 40 IU/dL. Probable rating: a debit of +25 %. Source of the threshold: WFH, Guidelines for the Management of Hemophilia, third edition (2020), the severity classes by clotting factor level.

### **What does not change the rating.**

Whether factor VIII or factor IX is missing does not change the band. Neither does the type of replacement product.

## **35.5 Worked examples and related rules**

Example 1. The attending physician's statement for a librarian aged 47 with haemophilia records clotting factor level 0 IU/dL. This is covered by rule UW-HEM-001 (clotting factor level below 1 IU/dL). Probable rating: a debit of +150 %. If the applicant also has hepatitis C with time with undetectable HCV RNA after the end of treatment below 12 weeks, rule UW-HCV-001 would apply as well; its rating is a debit of +100 %, and the two together would give +250 %.

Example 2. An applicant aged 58, a florist, has haemophilia. The most recent clinic letter records clotting factor level 3 IU/dL. That meets rule UW-HEM-002 (clotting factor level from 1 to 5 IU/dL), so the probable rating is a debit of +75 %.

Example 3. Take a market gardener of 39 with haemophilia. On file, from the report on file: clotting factor level 22 IU/dL. The threshold met is that of rule UW-HEM-003 (clotting factor level above 5 and below 40 IU/dL), and the probable rating is a debit of +25 %.

Readings on an edge: clotting factor level 1 IU/dL exactly, rule UW-HEM-002 (a debit of +75 %); clotting factor level 5 IU/dL exactly, rule UW-HEM-002 (a debit of +75 %); clotting factor level 40 IU/dL exactly, no rule of this section.

### **Common combinations.**

People treated before blood products were made safe may carry hepatitis C or HIV. Each is rated in its own section and added.

Joint damage from past bleeds limits activity but is not rated separately.

### **Related rules.**

Under rule UW-HEM-001, if the applicant also has hepatitis C with time with undetectable HCV RNA after the end of treatment below 12 weeks, see rule UW-HCV-001 (section 30.4, Chronic hepatitis C).

## 36 HIV infection

### 36.1 The impairment

The human immunodeficiency virus attacks the CD4 cells that direct the immune system. Daily antiviral treatment stops the virus multiplying and lets the cell count recover.

People who start treatment early and take it reliably can expect a long life. The cell count shows how much immune reserve there is, and a low count means either late diagnosis or treatment that is not working.

When this section applies. HIV infection has been diagnosed.

What is measured. The rating in this section depends on CD4 count (cells/mm<sup>3</sup>). The thresholds are grounded in: Centers for Disease Control and Prevention, Revised Surveillance Case Definition for HIV Infection, United States (2014).

### 36.2 Key questions

#### 1. When was HIV diagnosed, and when did treatment start?

A long gap between the two means years of unchecked damage.

#### 2. What are the latest CD4 count and viral load?

A viral load that cannot be detected confirms that treatment is taken and working.

#### 3. What was the lowest CD4 count ever recorded?

A very low past count predicts slower and less complete recovery.

#### 4. Is there hepatitis B or C as well?

Infection with two of these viruses damages the liver faster than either alone.

### 36.3 Evidence and readings

Use the CD4 count from the last twelve months. Where the latest viral load in that period can be detected, send the case to a person, whatever the count.

Fewer than twelve months of treatment is also outside these bands.

#### **Pitfalls with this evidence.**

The CD4 count dips for weeks after an infection or a vaccination. One low figure among steady ones is noted, not rated; use the next result.

The CD4 percentage is a different figure from the count.

#### **Converting units.**

CD4 count: A count in cells per microlitre is the same number in cells/mm<sup>3</sup>.

### 36.4 Probable rating

The table lists every rule of this section; each is defined in full below it. The debit percentages and the decisions to decline are invented for this synthetic manual. The thresholds follow the public source named in each rule, except the edges a rule marks as this manual's own.

| Rule id    | Measure and band                                      | Probable rating   |
|------------|-------------------------------------------------------|-------------------|
| UW-HIV-001 | CD4 count of 500 cells/mm <sup>3</sup> or more        | a debit of +50 %  |
| UW-HIV-002 | CD4 count from 200 to below 500 cells/mm <sup>3</sup> | a debit of +100 % |
| UW-HIV-003 | CD4 count below 200 cells/mm <sup>3</sup>             | decline           |

#### **Readings that meet no rule.**

None: every reading these measures can take falls in one of the bands above.

Note on the bands. 500 and 200 cells/mm<sup>3</sup> are the limits of stages 1, 2 and 3 in the cited case definition.

Rule UW-HIV-001: HIV infection (section 36.4). Threshold: CD4 count of 500 cells/mm<sup>3</sup> or more. Probable rating: a debit of +50 %. Source of the threshold: CDC, Revised Surveillance Case Definition for HIV Infection, United States (2014), the stages of infection by CD4 lymphocyte count.

Rule UW-HIV-002: HIV infection (section 36.4). Threshold: CD4 count from 200 to below 500 cells/mm<sup>3</sup>. Probable rating: a debit of +100 %. If the applicant also has chronic hepatitis B with HBV DNA of 2000 IU/mL or less, see rule UW-HBV-001. Source of the threshold: CDC, Revised Surveillance Case Definition for HIV Infection, United States (2014), the stages of infection by CD4 lymphocyte count.

Rule UW-HIV-003: HIV infection (section 36.4). Threshold: CD4 count below 200 cells/mm<sup>3</sup>. Probable rating: decline. Source of the threshold: CDC, Revised Surveillance Case Definition for HIV Infection, United States (2014), the stages of infection by CD4 lymphocyte count.

### **What does not change the rating.**

The combination of medicines, the route of infection and the years since diagnosis do not change the band.

## **36.5 Worked examples and related rules**

Example 1. Take a surveyor of 54 with HIV infection. On file, from the most recent clinic letter: CD4 count 540 cells/mm<sup>3</sup>. The threshold met is that of rule UW-HIV-001 (CD4 count of 500 cells/mm<sup>3</sup> or more), and the probable rating is a debit of +50 %.

Example 2. The report on file for a dental technician aged 35 with HIV infection records CD4 count 350 cells/mm<sup>3</sup>. This is covered by rule UW-HIV-002 (CD4 count from 200 to below 500 cells/mm<sup>3</sup>). Probable rating: a debit of +100 %. If the applicant also has chronic hepatitis B with HBV DNA of 2000 IU/mL or less, rule UW-HBV-001 would apply as well; its rating is a debit of +25 %, and the two together would give +125 %.

Example 3. An applicant aged 46, a pastry cook, has HIV infection. The specialist's letter records CD4 count 184 cells/mm<sup>3</sup>. That meets rule UW-HIV-003 (CD4 count below 200 cells/mm<sup>3</sup>), so the probable rating is decline.

Readings on an edge: CD4 count 200 cells/mm<sup>3</sup> exactly, rule UW-HIV-002 (a debit of +100 %); CD4 count 500 cells/mm<sup>3</sup> exactly, rule UW-HIV-001 (a debit of +50 %).

### **Common combinations.**

Hepatitis B and hepatitis C share routes of infection with HIV and are each rated and added. Smoking is more common in this group and weighs more heavily; it is added.

A past AIDS-defining illness goes to a person even when the count has recovered.

### **Related rules.**

Under rule UW-HIV-002, if the applicant also has chronic hepatitis B with HBV DNA of 2000 IU/mL or less, see rule UW-HBV-001 (section 29.4, Chronic hepatitis B).

Rules of other sections that point to this one: rule UW-HBV-001 (section 29.4, Chronic hepatitis B); rule UW-MEL-003 (section 41.4, Melanoma of the skin).

## 37 Depression

### 37.1 The impairment

Depression is a lasting low mood with loss of interest, poor sleep, tiredness and, when severe, thoughts of death. It is common, and most episodes lift with talking treatment, medicine or time.

The extra mortality comes from suicide and from the neglect of physical health. Severity at the latest assessment separates the large group at small risk from the smaller group at real risk.

When this section applies. Depression has been diagnosed or treated within the last five years.

What is measured. The rating in this section depends on PHQ-9 score (points). The thresholds are grounded in: National Institute for Health and Care Excellence, Depression in adults: treatment and management, guideline NG222 (2022).

### 37.2 Key questions

#### 1. How many episodes have there been, and when was the last?

Each further episode makes another more likely.

#### 2. Has there ever been a suicide attempt or an admission to a psychiatric unit?

Either one takes the case beyond these bands to a person.

#### 3. What treatment is taken now, and who supervises it?

Care from a specialist team, rather than a family doctor, usually means a harder course.

#### 4. Is the applicant working?

Time off work is a practical measure of how disabling the episode is.

### 37.3 Evidence and readings

Use a PHQ-9 total recorded by a clinician in the last twelve months. Where several are on file, use the latest and note the highest.

With no score on file, a single past episode that ended more than five years ago with no treatment since meets no rule of this section. Anything else without a score goes to a person.

#### **Pitfalls with this evidence.**

The questionnaire measures the last two weeks. A low score during treatment shows the treatment is working, not that the tendency has gone.

Low mood after a bereavement, without a diagnosis or treatment, is not rated here.

### 37.4 Probable rating

The table lists every rule of this section; each is defined in full below it. The debit percentages and the decisions to decline are invented for this synthetic manual. The thresholds follow the public source named in each rule, except the edges a rule marks as this manual's own.

| <b>Rule id</b> | <b>Measure and band</b>          | <b>Probable rating</b> |
|----------------|----------------------------------|------------------------|
| UW-DEP-001     | PHQ-9 score below 16 points      | a debit of +25 %       |
| UW-DEP-002     | PHQ-9 score of 16 points or more | a debit of +100 %      |

#### **Readings that meet no rule.**

None: every reading these measures can take falls in one of the bands above.

Note on the bands. A score of 16 divides less severe from more severe depression in the cited guideline.

Rule UW-DEP-001: Depression (section 37.4). Threshold: PHQ-9 score below 16 points. Probable rating: a debit of +25 %. If the applicant also has a record of heavy drinking with AUDIT score from 8 to below 16 points, see rule UW-ALC-001. Source of the threshold: NICE, Depression in adults: treatment and management, guideline NG222 (2022), the definition of less severe and more severe depression by PHQ-9 score.

Rule UW-DEP-002: Depression (section 37.4). Threshold: PHQ-9 score of 16 points or more. Probable rating: a debit of +100 %. If the applicant also has long-term opioid treatment for pain with daily opioid dose below 50 MME a day, see rule UW-OPI-001. Source of the threshold: NICE, Depression in adults: treatment and management, guideline NG222 (2022), the definition of less severe and more severe depression by PHQ-9 score.

### **What does not change the rating.**

The antidepressant chosen and its dose do not change the band. Talking treatment alone is rated on the same table.

## **37.5 Worked examples and related rules**

Example 1. An applicant aged 31, a payroll clerk, has depression. The report on file records PHQ-9 score 15 points. That meets rule UW-DEP-001 (PHQ-9 score below 16 points), so the probable rating is a debit of +25 %. If the applicant also has a record of heavy drinking with AUDIT score from 8 to below 16 points, rule UW-ALC-001 would apply as well; its rating is a debit of +25 %, and the two together would give +50 %.

Example 2. Take a librarian of 42 with depression. On file, from the specialist's letter: PHQ-9 score 17 points. The threshold met is that of rule UW-DEP-002 (PHQ-9 score of 16 points or more), and the probable rating is a debit of +100 %. If the applicant also has long-term opioid treatment for pain with daily opioid dose below 50 MME a day, rule UW-OPI-001 would apply as well; its rating is a debit of +25 %, and the two together would give +125 %.

Readings on an edge: PHQ-9 score 16 points exactly, rule UW-DEP-002 (a debit of +100 %).

### **Common combinations.**

Alcohol and depression feed each other; add the alcohol debit.

Long-term opioid treatment with depression is a recognised risk for overdose and is added.

A diagnosis of bipolar disorder or psychosis is not depression for this section and goes to a person.

### **Related rules.**

Under rule UW-DEP-001, if the applicant also has a record of heavy drinking with AUDIT score from 8 to below 16 points, see rule UW-ALC-001 (section 22.4, Alcohol use).

Under rule UW-DEP-002, if the applicant also has long-term opioid treatment for pain with daily opioid dose below 50 MME a day, see rule UW-OPI-001 (section 23.4, Long-term opioid therapy).

Rules of other sections that point to this one: rule UW-ALC-002 (section 22.4, Alcohol use); rule UW-OPI-002 (section 23.4, Long-term opioid therapy).

# 38 Epilepsy

## 38.1 The impairment

Epilepsy is a tendency to repeated seizures that have no immediate cause. Medicine stops the seizures completely in about two people in three.

The risks to life are accidents during a seizure, a prolonged seizure and, rarely, sudden unexpected death. All of them fall away the longer a person stays free of seizures.

When this section applies. Epilepsy has been diagnosed.

What is measured. The rating in this section depends on time since the last seizure (years). The thresholds are grounded in: International League Against Epilepsy, A practical clinical definition of epilepsy, official report (2014).

## 38.2 Key questions

### 1. On what date was the most recent seizure of any type?

The years in the table run from that date.

### 2. What type of seizures occur, and do they happen in sleep?

Convulsive seizures at night carry the highest risk of sudden death.

### 3. How many medicines are taken, and has any been withdrawn?

Needing several medicines points to seizures that are hard to control.

### 4. Was a cause found on brain imaging?

A tumour or an old injury is an impairment to be assessed in its own right.

### 38.3 Evidence and readings

Take the date of the last seizure from the neurology letter or the attending physician's statement, and count whole years to the application.

A seizure provoked by a clear, removed cause, such as a head injury in the previous week, is not epilepsy and meets no rule here.

#### **Pitfalls with this evidence.**

Brief absences, jerks on waking and auras are seizures. Applicants often count only convulsions, so the date in the neurology letter is preferred to the date on the form.

A seizure during a planned withdrawal of medicine still restarts the count.

### 38.4 Probable rating

The table lists every rule of this section; each is defined in full below it. The debit percentages and the decisions to decline are invented for this synthetic manual. The thresholds follow the public source named in each rule, except the edges a rule marks as this manual's own.

| <b>Rule id</b> | <b>Measure and band</b>                              | <b>Probable rating</b> |
|----------------|------------------------------------------------------|------------------------|
| UW-EPI-001     | time since the last seizure below 2 years            | a debit of +100 %      |
| UW-EPI-002     | time since the last seizure from 2 to below 10 years | a debit of +50 %       |
| UW-EPI-003     | time since the last seizure of 10 years or more      | no debit, +0 %         |

#### **Readings that meet no rule.**

None: every reading these measures can take falls in one of the bands above.

Note on the bands. Ten seizure-free years is part of the cited definition of resolved epilepsy. The edge at two years is this manual's own.

Rule UW-EPI-001: Epilepsy (section 38.4). Threshold: time since the last seizure below 2 years. Probable rating: a debit of +100 %. If the applicant also has a record of heavy drinking with AUDIT score from 8 to below 16 points, see rule UW-ALC-001. The edge at 2 years is this manual's own and no guideline supplies it. The source for the measure and the other edges of this section: ILAE, A practical clinical definition of epilepsy, official report (2014), the definition of resolved epilepsy after ten seizure-free years.

Rule UW-EPI-002: Epilepsy (section 38.4). Threshold: time since the last seizure from 2 to below 10 years. Probable rating: a debit of +50 %. Source of the edge at 10 years: ILAE, A practical clinical definition of epilepsy, official report (2014), the definition of resolved epilepsy after ten seizure-free years. The edge at 2 years is this manual's own.

Rule UW-EPI-003: Epilepsy (section 38.4). Threshold: time since the last seizure of 10 years or more. Probable rating: no debit, +0 %. Source of the threshold: ILAE, A practical clinical definition of epilepsy, official report (2014), the definition of resolved epilepsy after ten seizure-free years.

### **What does not change the rating.**

The medicine taken, the result of the brain wave recording and whether a driving licence is held do not change the band.

## **38.5 Worked examples and related rules**

Example 1. The specialist's letter for a translator aged 38 with epilepsy records time since the last seizure 1 years. This is covered by rule UW-EPI-001 (time since the last seizure below 2 years). Probable rating: a debit of +100 %. If the applicant also has a record of heavy drinking with AUDIT score from 8 to below 16 points, rule UW-ALC-001 would apply as well; its rating is a debit of +25 %, and the two together would give +125 %.

Example 2. An applicant aged 49, a surveyor, has epilepsy. The attending physician's statement records time since the last seizure 6 years. That meets rule UW-EPI-002 (time since the last seizure from 2 to below 10 years), so the probable rating is a debit of +50 %.

Example 3. Take a dental technician of 60 with epilepsy. On file, from the most recent clinic letter: time since the last seizure 11 years. The threshold met is that of rule UW-EPI-003 (time since the last seizure of 10 years or more), and the probable rating is no debit, +0 %.

Readings on an edge: time since the last seizure 2 years exactly, rule UW-EPI-002 (a debit of +50 %); time since the last seizure 10 years exactly, rule UW-EPI-003 (no debit, +0 %).

### **Common combinations.**

Seizures brought on by alcohol or its withdrawal are rated here and in the alcohol section, and the debits are added.

Epilepsy after a stroke is rated in both sections. A learning disability with epilepsy goes to a person.

### **Related rules.**

Under rule UW-EPI-001, if the applicant also has a record of heavy drinking with AUDIT score from 8 to below 16 points, see rule UW-ALC-001 (section 22.4, Alcohol use).

## 39 Rheumatoid arthritis

### 39.1 The impairment

Rheumatoid arthritis is an autoimmune inflammation of the joint linings, chiefly of the hands and feet, that erodes the joints if it is not suppressed.

The inflammation is not confined to joints. Active disease over years speeds up artery disease and can involve the lungs, which is why disease activity, rather than joint damage, drives the rating.

When this section applies. Rheumatoid arthritis has been diagnosed by a rheumatologist.

What is measured. The rating in this section depends on DAS28 (points). The thresholds are grounded in: American College of Rheumatology, 2019 Update of the Recommended Rheumatoid Arthritis Disease Activity Measures (2019).

### 39.2 Key questions

#### 1. What was the disease activity score at the last two clinic visits?

Sustained low activity is the aim of treatment and the best sign.

#### 2. Which disease-modifying medicines are taken?

A biological medicine is not penalised, but it shows that standard treatment was not enough.

#### 3. Are steroid tablets taken regularly, and at what dose?

Long-term steroids bring bone thinning, diabetes and infections.

#### 4. Are the lungs involved?

Scarring of the lungs is the complication that most shortens life.

### 39.3 Evidence and readings

Use the DAS28 recorded by the rheumatology clinic in the last twelve months, whichever blood marker it was calculated with.

Lung involvement takes the case to a person.

#### Pitfalls with this evidence.

The score can be worked out with either of two blood markers, and the version using C-reactive protein runs a little lower. The manual reads both on the one table.

Some clinics record a different index with different cut-offs. Only DAS28 can be read here.

### 39.4 Probable rating

The table lists every rule of this section; each is defined in full below it. The debit percentages and the decisions to decline are invented for this synthetic manual. The thresholds follow the public source named in each rule, except the edges a rule marks as this manual's own.

| Rule id   | Measure and band                     | Probable rating   |
|-----------|--------------------------------------|-------------------|
| UW-RA-001 | DAS28 below 2.6 points               | no debit, +0 %    |
| UW-RA-002 | DAS28 from 2.6 to 3.2 points         | a debit of +25 %  |
| UW-RA-003 | DAS28 above 3.2 and up to 5.1 points | a debit of +50 %  |
| UW-RA-004 | DAS28 above 5.1 points               | a debit of +100 % |

#### Readings that meet no rule.

None: every reading these measures can take falls in one of the bands above.

Note on the bands. 2.6, 3.2 and 5.1 are the cut-offs for remission, low, moderate and high activity among the cited measures.

Rule UW-RA-001: Rheumatoid arthritis (section 39.4). Threshold: DAS28 below 2.6 points. Probable rating: no debit, +0 %. Source of the threshold: ACR, 2019 Update of the Recommended Rheumatoid Arthritis Disease Activity Measures (2019), the DAS28 cut-offs for remission, low, moderate and high disease activity.

Rule UW-RA-002: Rheumatoid arthritis (section 39.4). Threshold: DAS28 from 2.6 to 3.2 points. Probable rating: a debit of +25 %. Source of the threshold: ACR, 2019 Update of the Recommended Rheumatoid Arthritis Disease Activity Measures (2019), the DAS28 cut-offs for remission, low, moderate and high disease activity.

Rule UW-RA-003: Rheumatoid arthritis (section 39.4). Threshold: DAS28 above 3.2 and up to 5.1 points. Probable rating: a debit of +50 %. If the applicant also has osteoporosis with bone density T-score of -2.5 standard deviations or less, see rule UW-OST-001. Source of the threshold: ACR, 2019 Update of the Recommended Rheumatoid Arthritis Disease Activity Measures (2019), the DAS28 cut-offs for remission, low, moderate and high disease activity.

Rule UW-RA-004: Rheumatoid arthritis (section 39.4). Threshold: DAS28 above 5.1 points. Probable rating: a debit of +100 %. Source of the threshold: ACR, 2019 Update of the Recommended Rheumatoid Arthritis Disease Activity Measures (2019), the DAS28 cut-offs for remission, low, moderate and high disease activity.

### **What does not change the rating.**

Deformity of the hands, the presence of rheumatoid factor and joint replacements do not change the band.

## **39.5 Worked examples and related rules**

Example 1. Take a bus driver of 45 with rheumatoid arthritis. On file, from the attending physician's statement: DAS28 2.4 points. The threshold met is that of rule UW-RA-001 (DAS28 below 2.6 points), and the probable rating is no debit, +0 %.

Example 2. The most recent clinic letter for a payroll clerk aged 56 with rheumatoid arthritis records DAS28 2.9 points. This is covered by rule UW-RA-002 (DAS28 from 2.6 to 3.2 points). Probable rating: a debit of +25 %.

Example 3. An applicant aged 37, a librarian, has rheumatoid arthritis. The report on file records DAS28 4.2 points. That meets rule UW-RA-003 (DAS28 above 3.2 and up to 5.1 points), so the probable rating is a debit of +50 %. If the applicant also has osteoporosis with bone density T-score of -2.5 standard deviations or less, rule UW-OST-001 would apply as well; its rating is a debit of +25 %, and the two together would give +75 %.

Example 4. Take a florist of 48 with rheumatoid arthritis. On file, from the specialist's letter: DAS28 5.5 points. The threshold met is that of rule UW-RA-004 (DAS28 above 5.1 points), and the probable rating is a debit of +100 %.

Readings on an edge: DAS28 2.6 points exactly, rule UW-RA-002 (a debit of +25 %); DAS28 3.2 points exactly, rule UW-RA-002 (a debit of +25 %); DAS28 5.1 points exactly, rule UW-RA-003 (a debit of +50 %).

### **Common combinations.**

Years of inflammation speed artery disease: a past heart attack, raised blood pressure and smoking are added from their sections.

Steroid tablets taken for long periods thin the bones, and the osteoporosis debit is added. Anaemia of long-standing inflammation is rated in its section.

### **Related rules.**

Under rule UW-RA-003, if the applicant also has osteoporosis with bone density T-score of -2.5 standard deviations or less, see rule UW-OST-001 (section 40.4, Osteoporosis).

## 40 Osteoporosis

### 40.1 The impairment

Osteoporosis is loss of bone density that leaves bones liable to break after a minor fall. It is silent until a wrist, a vertebra or a hip fractures.

A hip fracture in later life carries a real risk of death in the following year. In applicants of working age the added mortality is small, and the single rule here reflects that.

When this section applies. A bone density scan from the last three years is on file.

What is measured. The rating in this section depends on bone density T-score (standard deviations). The thresholds are grounded in: World Health Organization, Assessment of fracture risk and its application to screening for postmenopausal osteoporosis, WHO Technical Report Series 843 (1994).

### 40.2 Key questions

#### 1. Has there been a fracture after a minor fall?

A fracture already sustained doubles the chance of another.

#### 2. Why is the bone thin?

Steroid treatment, early menopause, low weight and bowel disease are causes that have their own bearing on risk.

#### 3. Is bone-protecting treatment taken?

Treatment roughly halves the risk of spinal fracture.

### 40.3 Evidence and readings

Use the lower of the hip and spine T-scores from a bone density scan in the last three years.

A T-score above -2.5 meets no rule of this section. Two or more fractures of the spine take the case to a person.

### **Pitfalls with this evidence.**

Arthritis of the spine and a hardened aorta lying in front of it make the spine T-score read falsely high. Where hip and spine differ by more than one unit, the hip figure is the more reliable.

A Z-score compares with people of the same age and is a different figure.

## **40.4 Probable rating**

The table lists every rule of this section; each is defined in full below it. The debit percentages and the decisions to decline are invented for this synthetic manual. The thresholds follow the public source named in each rule, except the edges a rule marks as this manual's own.

| <b>Rule id</b> | <b>Measure and band</b>                                  | <b>Probable rating</b> |
|----------------|----------------------------------------------------------|------------------------|
| UW-OST-001     | bone density T-score of -2.5 standard deviations or less | a debit of +25 %       |

### **Readings that meet no rule.**

bone density T-score above -2.5 standard deviations: no rule; bone density that is normal or only reduced.

Note on the bands. A T-score of -2.5 or lower is the cited definition of osteoporosis.

Rule UW-OST-001: Osteoporosis (section 40.4). Threshold: bone density T-score of -2.5 standard deviations or less. Probable rating: a debit of +25 %. If the applicant also has an unusual build with body mass index from 16.0 to below 18.5 kg/m<sup>2</sup>, see rule UW-BMI-002. Source of the threshold: WHO, Assessment of fracture risk and its application to screening for postmenopausal osteoporosis, WHO Technical Report Series 843 (1994), the definition of osteoporosis as a T-score of -2.5 or lower.

**What does not change the rating.**

The medicine prescribed, calcium and vitamin D supplements and height lost do not change the single rule.

**40.5 Worked examples and related rules**

Example 1. An applicant aged 52, a school caretaker, has osteoporosis. The most recent clinic letter records bone density T-score -2.7 standard deviations. That meets rule UW-OST-001 (bone density T-score of -2.5 standard deviations or less), so the probable rating is a debit of +25 %. If the applicant also has an unusual build with body mass index from 16.0 to below 18.5 kg/m<sup>2</sup>, rule UW-BMI-002 would apply as well; its rating is a debit of +25 %, and the two together would give +50 %.

Readings on an edge: bone density T-score -2.5 standard deviations exactly, rule UW-OST-001 (a debit of +25 %).

**Common combinations.**

Thin bones are often the mark of something else. A body mass index below 18.5 kg/m<sup>2</sup>, long steroid treatment for rheumatoid arthritis or lung disease and heavy drinking are each rated in their own sections and added.

**Related rules.**

Under rule UW-OST-001, if the applicant also has an unusual build with body mass index from 16.0 to below 18.5 kg/m<sup>2</sup>, see rule UW-BMI-002 (section 5.4, Build: underweight and obesity).

Rules of other sections that point to this one: rule UW-RA-003 (section 39.4, Rheumatoid arthritis).

# 41 Melanoma of the skin

## 41.1 The impairment

Melanoma is a cancer of the pigment cells of the skin. Caught while thin it is cured by removal; once it has grown deep it can spread through lymph and blood years later.

The pathologist's measurement of thickness is the strongest single guide to outcome for a tumour confined to the skin. This section applies only where there was no spread to lymph nodes or beyond.

When this section applies. An invasive melanoma of the skin has been removed, with no spread to lymph nodes or beyond.

What is measured. The rating in this section depends on tumour thickness (mm). The thresholds are grounded in: American Joint Committee on Cancer, Cancer Staging Manual, eighth edition, melanoma of the skin (2017).

## 41.2 Key questions

### 1. What thickness did the pathology report give, and was the surface ulcerated?

Ulceration worsens the outlook at any thickness.

### 2. Was a sentinel lymph node sampled, and what did it show?

A positive node takes the case out of this section altogether.

### 3. On what date was the melanoma removed, and has there been a second one?

Most recurrences come within five years, and a second primary restarts the clock.

### 4. Is the applicant in regular skin follow-up?

Follow-up finds both recurrences and new melanomas early.

### 41.3 Evidence and readings

Take the tumour thickness from the pathology report, to one decimal place. A report that does not state it is not enough to rate on.

Spread to a lymph node or a distant site, or removal less than twelve months ago, takes the case to a person. Melanoma in situ, which has no thickness, meets no rule of this section.

#### **Pitfalls with this evidence.**

Thickness is measured in millimetres to one decimal place. The older level system, written in Roman numerals, is a different scale and cannot be read on this table.

A second, wider operation after the first removal is routine. It is not a recurrence.

### 41.4 Probable rating

The table lists every rule of this section; each is defined in full below it. The debit percentages and the decisions to decline are invented for this synthetic manual. The thresholds follow the public source named in each rule, except the edges a rule marks as this manual's own.

| <b>Rule id</b> | <b>Measure and band</b>                     | <b>Probable rating</b> |
|----------------|---------------------------------------------|------------------------|
| UW-MEL-001     | tumour thickness of 1.0 mm or less          | a debit of +50 %       |
| UW-MEL-002     | tumour thickness above 1.0 and up to 2.0 mm | a debit of +100 %      |
| UW-MEL-003     | tumour thickness above 2.0 and up to 4.0 mm | a debit of +200 %      |
| UW-MEL-004     | tumour thickness above 4.0 mm               | decline                |

#### **Readings that meet no rule.**

None: every reading these measures can take falls in one of the bands above.

Note on the bands. 1.0, 2.0 and 4.0 mm are the limits of categories T1 to T4 in the cited staging manual.

Rule UW-MEL-001: Melanoma of the skin (section 41.4). Threshold: tumour thickness of 1.0 mm or less. Probable rating: a debit of +50 %. Source of the threshold: AJCC, Cancer Staging Manual, eighth edition, melanoma of the skin (2017), the T categories by tumour thickness.

Rule UW-MEL-002: Melanoma of the skin (section 41.4). Threshold: tumour thickness above 1.0 and up to 2.0 mm. Probable rating: a debit of +100 %. Source of the threshold: AJCC, Cancer Staging Manual, eighth edition, melanoma of the skin (2017), the T categories by tumour thickness.

Rule UW-MEL-003: Melanoma of the skin (section 41.4). Threshold: tumour thickness above 2.0 and up to 4.0 mm. Probable rating: a debit of +200 %. If the applicant also has HIV infection with CD4 count from 200 to below 500 cells/mm<sup>3</sup>, see rule UW-HIV-002. Source of the threshold: AJCC, Cancer Staging Manual, eighth edition, melanoma of the skin (2017), the T categories by tumour thickness.

Rule UW-MEL-004: Melanoma of the skin (section 41.4). Threshold: tumour thickness above 4.0 mm. Probable rating: decline. Source of the threshold: AJCC, Cancer Staging Manual, eighth edition, melanoma of the skin (2017), the T categories by tumour thickness.

### **What does not change the rating.**

The part of the body affected and the applicant's skin type do not change the band. A clear follow-up scan does not lower it.

## **41.5 Worked examples and related rules**

Example 1. The report on file for a joiner aged 59 with a melanoma that has been removed records tumour thickness 0.9 mm. This is covered by rule UW-MEL-001 (tumour thickness of 1.0 mm or less). Probable rating: a debit of +50 %.

Example 2. An applicant aged 40, a bus driver, has a melanoma that has been removed. The specialist's letter records tumour thickness 1.5 mm. That meets rule UW-MEL-002 (tumour thickness above 1.0 and up to 2.0 mm), so the probable rating is a debit of +100 %.

Example 3. Take a payroll clerk of 51 with a melanoma that has been removed. On file, from the attending physician's statement: tumour thickness 3.0 mm. The threshold met is that of rule UW-MEL-003 (tumour thickness above 2.0 and up to 4.0 mm), and the probable rating is a debit of +200 %. If the applicant also has HIV infection with CD4 count from 200 to below 500 cells/mm<sup>3</sup>, rule UW-HIV-002 would apply as well; its rating is a debit of +100 %, and the two together would give +300 %.

Example 4. The most recent clinic letter for a librarian aged 32 with a melanoma that has been removed records tumour thickness 4.3 mm. This is covered by rule UW-MEL-004 (tumour thickness above 4.0 mm). Probable rating: decline.

Readings on an edge: tumour thickness 1.0 mm exactly, rule UW-MEL-001 (a debit of +50 %); tumour thickness 2.0 mm exactly, rule UW-MEL-002 (a debit of +100 %); tumour thickness 4.0 mm exactly, rule UW-MEL-003 (a debit of +200 %).

### **Common combinations.**

A weakened immune system makes recurrence more likely. HIV infection with a CD4 count below 500 cells/mm<sup>3</sup> is rated in its own section and added, and an organ transplant goes to a person.

A second melanoma is rated on the thicker of the two.

**Related rules.**

Under rule UW-MEL-003, if the applicant also has HIV infection with CD4 count from 200 to below 500 cells/mm<sup>3</sup>, see rule UW-HIV-002 (section 36.4, HIV infection).

## 42 Glossary

### **Abdominal aortic diameter**

The widest outer diameter of the aorta in the abdomen on ultrasound or CT. Unit: cm.

### **Apnoea-hypopnoea index**

The number of pauses and shallow breaths in each hour of sleep, from a sleep study done without treatment. Unit: events per hour.

### **Attending physician's statement**

A report on the applicant written for the insurer by the doctor who treats them.

### **AUDIT score**

The total, from 0 to 40, of the ten-question Alcohol Use Disorders Identification Test. Unit: points.

### **Band**

A range of readings of one measure that share a rating. Bands of one measure do not overlap.

### **Body mass index**

Weight in kilograms divided by the square of height in metres. Unit: kg/m<sup>2</sup>.

### **Bone density T-score**

Bone mineral density at the hip or spine compared with the average of healthy young adults, in standard deviations. Unit: standard deviations.

### **Carotid artery stenosis**

How much of the internal carotid artery's width is lost to plaque, as reported on duplex ultrasound or angiography. Unit: %.

### **CD4 count**

The number of CD4 lymphocytes in a cubic millimetre of blood, a measure of immune strength. Unit: cells/mm<sup>3</sup>.

**CHA2DS2-VASc score**

A stroke risk score for atrial fibrillation built from age, sex, heart failure, hypertension, diabetes, vascular disease and earlier stroke. Unit: points.

**Clotting factor level**

The activity of factor VIII or factor IX in the blood without treatment; 100 IU/dL is the average of healthy people. Unit: IU/dL.

**Cross-reference**

A mention of a rule outside its definition, pointing the reader to it.

**Daily opioid dose**

The total of all opioid medicines taken in a day, converted to morphine milligram equivalents (MME). Unit: MME a day.

**DAS28**

Disease Activity Score over 28 joints, built from tender and swollen joint counts, a blood marker of inflammation and the patient's own rating. Unit: points.

**Debit**

An addition to expected mortality, as a whole percentage. Debits from different rules are added.

**Decline**

The outcome in which no terms are offered on the evidence in the file.

**Diastolic blood pressure**

The lower of the two blood pressure numbers: the pressure while the heart rests between beats. Unit: mmHg.

**eGFR**

Estimated glomerular filtration rate: how much blood the kidneys clean each minute, worked out from serum creatinine. Unit: mL/min/1.73 m<sup>2</sup>.

**Faecal calprotectin**

A protein from white blood cells measured in a stool sample; it rises with inflammation of the bowel wall. Unit: mcg/g.

**Fasting triglycerides**

The main blood fat, measured after a fast of at least eight hours. Unit: mg/dL.

**FEV1**

The volume of air blown out in the first second of a forced breath, as a percentage of the value expected for age, height and sex. Unit: % of predicted.

**FEV1 after a bronchodilator**

FEV1 measured after an inhaled airway-opening medicine, so that only the fixed part of the obstruction is counted. Unit: % of predicted.

**FIB-4 index**

An estimate of liver scarring worked out from age, two liver enzymes and the platelet count.

**Haemoglobin**

The oxygen-carrying protein of red blood cells, measured in a full blood count. Unit: g/dL.

**HbA1c**

Glycated haemoglobin: the share of haemoglobin carrying glucose, which reflects average blood glucose over about three months. Unit: %.

**HBV DNA**

The amount of hepatitis B virus in the blood. Unit: IU/mL.

**Impairment**

A medical condition, finding or habit that may change expected mortality.

**LDL cholesterol**

Low-density lipoprotein cholesterol in a fasting blood sample, the fraction most closely tied to artery disease. Unit: mg/dL.

**Left ventricular ejection fraction**

The share of the blood in the left ventricle that is pumped out with each beat, usually from an echocardiogram. Unit: %.

**Lifetime smoking**

Packs of 20 cigarettes smoked a day multiplied by the number of years smoked. Unit: pack-years.

**Loading**

The sum of the debits of every rule that applies to a case.

**Maximum left ventricular wall thickness**

The thickest part of the wall of the heart's main pumping chamber, on echocardiogram or cardiac MRI. Unit: mm.

**Measure**

What a rule compares with its threshold: a reading, a score, a length of time or a status.

**MELD score**

Model for End-Stage Liver Disease: a score from bilirubin, creatinine and clotting time that predicts three-month survival in cirrhosis. Unit: points.

**NIHSS score at admission**

The National Institutes of Health Stroke Scale, from 0 to 42, scored when the patient reaches hospital; a higher score is a more severe stroke. Unit: points.

**Peak aortic jet velocity**

The fastest blood flow through the aortic valve on echocardiography; it rises as the valve narrows. Unit: m/s.

**PHQ-9 score**

The total, from 0 to 27, of a nine-question self-report of depressive symptoms over two weeks. Unit: points.

**Postponement**

A decline that invites a fresh application after a stated time or change.

**Probable rating**

The rating this manual suggests for a band. A person makes the decision.

**Reading**

One recorded value of a measure, with its date and the document it comes from.

**Resting ankle-brachial index**

Systolic pressure at the ankle divided by systolic pressure in the arm; a low value means narrowed leg arteries.

**Rule**

One band of one measure for one impairment, with its rating and its source. Each rule has its own identifier.

**Serum urate**

The level of uric acid in the blood; crystals form in joints when it stays high. Unit: mg/dL.

**Smoking status**

Whether the applicant smokes cigarettes now, used to, or never has, as the survey definitions put it.

**Standard**

The outcome for a case whose debits add up to nothing: ordinary terms.

**Systolic blood pressure**

The higher of the two blood pressure numbers: the pressure while the heart contracts. Unit: mmHg.

**Threshold**

The edge or edges of a band, taken where possible from a public guideline.

**Time since the clot**

Whole months from the diagnosis of the most recent deep vein thrombosis or pulmonary embolism to the date of the application. Unit: months.

**Time since the heart attack**

Whole months from the date of the most recent heart attack to the date of the application. Unit: months.

**Time since the last seizure**

Whole years from the most recent seizure of any kind to the date of the application. Unit: years.

**Time with undetectable HCV RNA after the end of treatment**

Whole weeks from the last dose of antiviral treatment to the most recent test that found no hepatitis C virus. Unit: weeks.

**TSH**

Thyroid-stimulating hormone; it rises when the thyroid gland makes too little hormone. Unit: mIU/L.

**Tumour thickness**

The depth of a melanoma from the skin surface to its deepest cell, measured by the pathologist. Unit: mm.

**Urine albumin-to-creatinine ratio**

Albumin in a urine sample relative to creatinine; a rise is the earliest sign of kidney damage. Unit: mg/g.

## 43 Public sources

The thresholds in this manual rest on the public guidelines below, apart from the edges that a rule marks as the manual's own. The ratings rest on nothing: they are invented.

### **American Association of Clinical Endocrinologists and American Thyroid Association (AAACE/ATA). Clinical Practice Guidelines for Hypothyroidism in Adults. 2012.**

Relied on in section 9 (Hypothyroidism: the recommendation on treating a TSH above 10 mIU/L).

### **American Association for the Study of Liver Diseases (AASLD). Practice Guidance on the clinical assessment and management of nonalcoholic fatty liver disease. 2023.**

Relied on in section 31 (Fatty liver disease: the FIB-4 cut-offs of 1.3 and 2.67 for the risk of advanced fibrosis).

### **American Association for the Study of Liver Diseases (AASLD). Update on Prevention, Diagnosis, and Treatment of Chronic Hepatitis B: AASLD 2018 Hepatitis B Guidance. 2018.**

Relied on in section 29 (Chronic hepatitis B: the HBV DNA levels used to decide on treatment).

### **American Association for the Study of Liver Diseases and Infectious Diseases Society of America (AASLD-IDSA). HCV Guidance: Recommendations for Testing, Managing, and Treating Hepatitis C. 2023.**

Relied on in section 30 (Chronic hepatitis C: the definition of sustained virologic response 12 weeks after treatment).

### **American Association for the Study of Liver Diseases and American Society of Transplantation (AASLD/AST). Evaluation for Liver Transplantation in Adults: 2013 Practice Guideline. 2013.**

Relied on in section 32 (Cirrhosis of the liver: the recommendation to evaluate at a MELD score of 15 or more).

**American Academy of Sleep Medicine (AASM). Clinical Guideline for the Evaluation, Management and Long-term Care of Obstructive Sleep Apnea in Adults. 2009.**

Relied on in section 26 (Obstructive sleep apnoea: the severity grades by apnoea-hypopnoea index).

**American College of Cardiology and American Heart Association (ACC/AHA). Guideline Focused Update on Duration of Dual Antiplatelet Therapy in Patients With Coronary Artery Disease. 2016.**

Relied on in section 11 (Myocardial infarction: the recommendations for at least 12 months of treatment after an acute coronary syndrome).

**American College of Cardiology and American Heart Association (ACC/AHA). Guideline for the Management of Patients With Valvular Heart Disease. 2020.**

Relied on in section 14 (Aortic stenosis: the stages of aortic stenosis by peak aortic jet velocity).

**American College of Cardiology and American Heart Association (ACC/AHA). Guideline for the Prevention, Detection, Evaluation, and Management of High Blood Pressure in Adults. 2017.**

Relied on in section 10 (Hypertension: the table of blood pressure categories and the section on hypertensive crises).

**American College of Cardiology, American Heart Association, American College of Clinical Pharmacy and Heart Rhythm Society (ACC/AHA/ACCP/HRS). Guideline for the Diagnosis and Management of Atrial Fibrillation. 2023.**

Relied on in section 13 (Atrial fibrillation: the recommendations on stroke risk scores and anticoagulation).

**American College of Rheumatology (ACR). 2019 Update of the Recommended Rheumatoid Arthritis Disease Activity Measures. 2019.**

Relied on in section 39 (Rheumatoid arthritis: the DAS28 cut-offs for remission, low, moderate and high disease activity).

**American College of Rheumatology (ACR). Guideline for the Management of Gout. 2020.**

Relied on in section 8 (Gout: the serum urate target for urate-lowering therapy).

**American Diabetes Association (ADA). Standards of Care in Diabetes. 2024.**

Relied on in section 2 (Type 2 diabetes mellitus: section 6, Glycemic Goals and Hypoglycemia (A1C goals for adults)); section 3 (Type 1 diabetes mellitus: section 6, Glycemic Goals and Hypoglycemia (A1C goals for adults)); section 4 (Prediabetes: section 2, Diagnosis and Classification of Diabetes (A1C criteria for prediabetes and diabetes)).

**American Gastroenterological Association (AGA). Clinical Practice Guideline on the Role of Biomarkers for the Management of Ulcerative Colitis. 2023.**

Relied on in section 33 (Ulcerative colitis: the faecal calprotectin cut-off of 150 mcg/g).

**American Heart Association and American College of Cardiology (AHA/ACC). Guideline for the Diagnosis and Treatment of Patients With Hypertrophic Cardiomyopathy. 2020.**

Relied on in section 15 (Hypertrophic cardiomyopathy: the diagnostic wall thickness and the risk markers for sudden cardiac death).

**American Heart Association and American College of Cardiology (AHA/ACC). Guideline on the Management of Blood Cholesterol. 2018.**

Relied on in section 6 (Raised LDL cholesterol: severe primary hypercholesterolemia, the risk-enhancing factors and severe hypertriglyceridemia); section 7 (Raised triglycerides: severe primary hypercholesterolemia, the risk-enhancing factors and severe hypertriglyceridemia).

**American Heart Association and American College of Cardiology (AHA/ACC). Guideline on the Management of Patients With Lower Extremity Peripheral Artery Disease. 2016.**

Relied on in section 17 (Peripheral artery disease: the interpretation of the resting ankle-brachial index).

**American Heart Association, American College of Cardiology and Heart Failure Society of America (AHA/ACC/HFSA). Guideline for the Management of Heart Failure. 2022.**

Relied on in section 12 (Heart failure: the classification of heart failure by left ventricular ejection fraction).

**American Heart Association and American Stroke Association (AHA/ASA). Guideline for the Prevention of Stroke in Patients With Stroke and Transient Ischemic Attack. 2021.**

Relied on in section 18 (Carotid artery stenosis: the recommendations on extracranial carotid stenosis).

**American Heart Association and American Stroke Association (AHA/ASA). Guidelines for the Early Management of Patients With Acute Ischemic Stroke, 2019 update. 2019.**

Relied on in section 19 (Stroke: the criteria for mechanical thrombectomy, among them an NIHSS score of 6 or more).

**American Joint Committee on Cancer (AJCC). Cancer Staging Manual, eighth edition, melanoma of the skin. 2017.**

Relied on in section 41 (Melanoma of the skin: the T categories by tumour thickness).

**Centers for Disease Control and Prevention (CDC). Guideline for Prescribing Opioids for Chronic Pain, United States. 2016.**

Relied on in section 23 (Long-term opioid therapy: the recommendation on dosage: reassess at 50, avoid 90 morphine milligram equivalents a day or more).

**Centers for Disease Control and Prevention, National Center for Health Statistics (CDC). National Health Interview Survey, adult tobacco use glossary. 2017.**

Relied on in section 21 (Tobacco use: the definitions of current smoker and former smoker).

**Centers for Disease Control and Prevention (CDC). Revised Surveillance Case Definition for HIV Infection, United States. 2014.**

Relied on in section 36 (HIV infection: the stages of infection by CD4 lymphocyte count).

**American College of Chest Physicians (CHEST). Antithrombotic Therapy for VTE Disease, second update of the CHEST guideline and expert panel report. 2021.**

Relied on in section 20 (Venous thromboembolism: the three-month treatment phase of anticoagulation).

**Global Initiative for Asthma (GINA). Global Strategy for Asthma Management and Prevention. 2023.**

Relied on in section 24 (Asthma: the risk factors for exacerbations, among them an FEV1 below 60 % of predicted).

**Global Initiative for Chronic Obstructive Lung Disease (GOLD). Global Strategy for Prevention, Diagnosis and Management of COPD. 2024.**

Relied on in section 25 (Chronic obstructive pulmonary disease: the grades of airflow obstruction by FEV1 after a bronchodilator).

**International League Against Epilepsy (ILAE). A practical clinical definition of epilepsy, official report. 2014.**

Relied on in section 38 (Epilepsy: the definition of resolved epilepsy after ten seizure-free years).

**Kidney Disease: Improving Global Outcomes (KDIGO). Clinical Practice Guideline for the Evaluation and Management of Chronic Kidney Disease. 2024.**

Relied on in section 27 (Chronic kidney disease: the GFR categories G1 to G5 and the albuminuria categories A1 to A3); section 28 (Albuminuria: the GFR categories G1 to G5 and the albuminuria categories A1 to A3).

**National Institute for Health and Care Excellence (NICE). Depression in adults: treatment and management, guideline NG222. 2022.**

Relied on in section 37 (Depression: the definition of less severe and more severe depression by PHQ-9 score).

**Society for Vascular Surgery (SVS). Practice guidelines on the care of patients with an abdominal aortic aneurysm. 2018.**

Relied on in section 16 (Abdominal aortic aneurysm: the surveillance intervals by diameter and the threshold for elective repair).

**US Preventive Services Task Force (USPSTF). Lung Cancer: Screening, recommendation statement. 2021.**

Relied on in section 21 (Tobacco use: the 20 pack-year smoking history that defines the screened group).

**World Federation of Hemophilia (WFH). Guidelines for the Management of Hemophilia, third edition. 2020.**

Relied on in section 35 (Haemophilia: the severity classes by clotting factor level).

**World Health Organization (WHO). AUDIT, the Alcohol Use Disorders Identification Test: guidelines for use in primary health care, second edition. 2001.**

Relied on in section 22 (Alcohol use: the risk zones by total AUDIT score).

**World Health Organization (WHO). Assessment of fracture risk and its application to screening for postmenopausal osteoporosis, WHO Technical Report Series 843. 1994.**

Relied on in section 40 (Osteoporosis: the definition of osteoporosis as a T-score of -2.5 or lower).

**World Health Organization (WHO). Haemoglobin concentrations for the diagnosis of anaemia and assessment of severity. 2011.**

Relied on in section 34 (Anaemia: the table of haemoglobin levels for mild, moderate and severe anaemia).

**World Health Organization (WHO). Obesity: preventing and managing the global epidemic, WHO Technical Report Series 894. 2000.**

Relied on in section 5 (Build: underweight and obesity: the classification of adults according to body mass index).